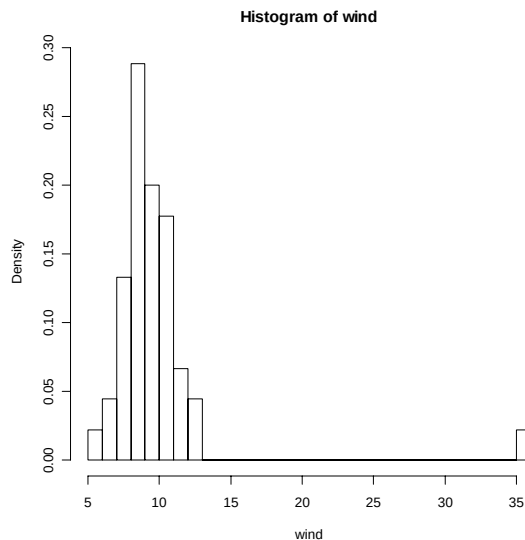
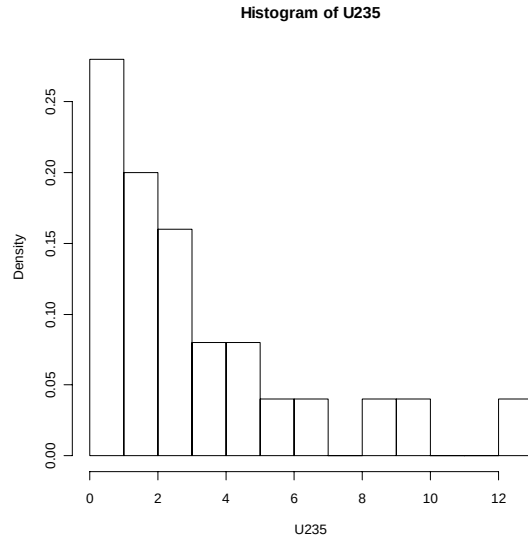


Chapter 1: What is Statistics?

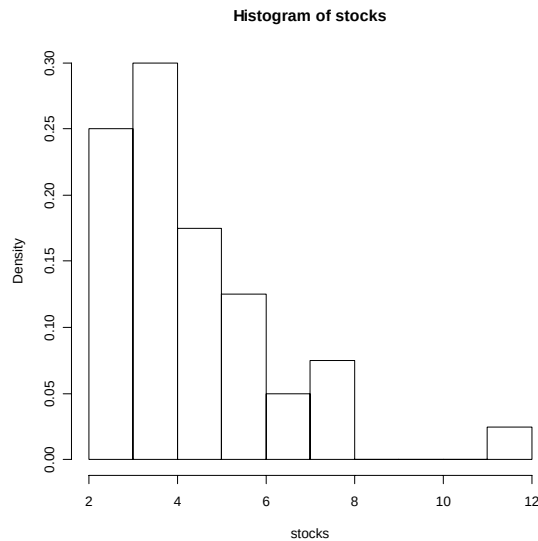
- 1.1
- a. Population: all generation X age US citizens (specifically, assign a '1' to those who want to start their own business and a '0' to those who do not, so that the population is the set of 1's and 0's). Objective: to estimate the proportion of generation X age US citizens who want to start their own business.
 - b. Population: all healthy adults in the US. Objective: to estimate the true mean body temperature
 - c. Population: single family dwelling units in the city. Objective: to estimate the true mean water consumption
 - d. Population: all tires manufactured by the company for the specific year. Objective: to estimate the proportion of tires with unsafe tread.
 - e. Population: all adult residents of the particular state. Objective: to estimate the proportion who favor a unicameral legislature.
 - f. Population: times until recurrence for all people who have had a particular disease. Objective: to estimate the true average time until recurrence.
 - g. Population: lifetime measurements for all resistors of this type. Objective: to estimate the true mean lifetime (in hours).



- 1.2
- a. This histogram is above.
 - b. Yes, it is quite windy there.
 - c. 11/45, or approx. 24.4%
 - d. it is not especially windy in the overall sample.



1.3 The histogram is above.

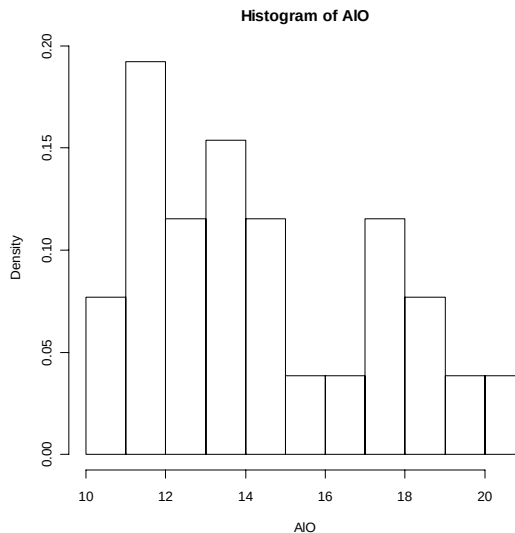


- 1.4**
- a.** The histogram is above.
 - b.** $18/40 = 45\%$
 - c.** $29/40 = 72.5\%$

- 1.5**
- a.** The categories with the largest grouping of students are 2.45 to 2.65 and 2.65 to 2.85. (both have 7 students).
 - b.** $7/30$
 - c.** $7/30 + 3/30 + 3/30 + 3/30 = 16/30$

- 1.6**
- a.** The modal category is 2 (quarts of milk). About 36% (9 people) of the 25 are in this category.
 - b.** $.2 + .12 + .04 = .36$
 - c.** Note that 8% purchased 0 while 4% purchased 5. Thus, $1 - .08 - .04 = .88$ purchased between 1 and 4 quarts.

- 1.7**
- a. There is a possibility of bimodality in the distribution.
 - b. There is a dip in heights at 68 inches.
 - c. If all of the students are roughly the same age, the bimodality could be a result of the men/women distributions.



- 1.8**
- a. The histogram is above.
 - b. The data appears to be bimodal. Llanederyn and Caldicot have lower sample values than the other two.
- 1.9**
- a. Note that $9.7 = 12 - 2.3$ and $14.3 = 12 + 2.3$. So, $(9.7, 14.3)$ should contain approximately 68% of the values.
 - b. Note that $7.4 = 12 - 2(2.3)$ and $16.6 = 12 + 2(2.3)$. So, $(7.4, 16.6)$ should contain approximately 95% of the values.
 - c. From parts (a) and (b) above, $95\% - 68\% = 27\%$ lie in both $(14.3, 16.6)$ and $(7.4, 9.7)$. By symmetry, 13.5% should lie in $(14.3, 16.6)$ so that $68\% + 13.5\% = 81.5\%$ are in $(9.7, 16.6)$
 - d. Since 5.1 and 18.9 represent three standard deviations away from the mean, the proportion outside of these limits is approximately 0.
- 1.10**
- a. $14 - 17 = -3$.
 - b. Since 68% lie within one standard deviation of the mean, 32% should lie outside. By symmetry, 16% should lie below one standard deviation from the mean.
 - c. If normally distributed, approximately 16% of people would spend less than -3 hours on the internet. Since this doesn't make sense, the population is not normal.
- 1.11**
- a. $\sum_{i=1}^n c = c + c + \dots + c = nc$.
 - b. $\sum_{i=1}^n cy_i = c(y_1 + \dots + y_n) = c \sum_{i=1}^n y_i$
 - c. $\sum_{i=1}^n (x_i + y_i) = x_1 + y_1 + x_2 + y_2 + \dots + x_n + y_n = (x_1 + x_2 + \dots + x_n) + (y_1 + y_2 + \dots + y_n)$

Using the above, the numerator of s^2 is $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) = \sum_{i=1}^n y_i^2 - 2\bar{y}\sum_{i=1}^n y_i + n\bar{y}^2$. Since $n\bar{y} = \sum_{i=1}^n y_i$, we have $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$. Let $\bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$ to get the result.

1.12 Using the data, $\sum_{i=1}^6 y_i = 14$ and $\sum_{i=1}^6 y_i^2 = 40$. So, $s^2 = (40 - 14^2/6)/5 = 1.47$. So, $s = 1.21$.

1.13 a. With $\sum_{i=1}^{45} y_i = 440.6$ and $\sum_{i=1}^{45} y_i^2 = 5067.38$, we have that $\bar{y} = 9.79$ and $s = 4.14$.

b.

k	interval	frequency	Exp. frequency
1	5.65, 13.93	44	30.6
2	1.51, 18.07	44	42.75
3	-2.63, 22.21	44	45

1.14 a. With $\sum_{i=1}^{25} y_i = 80.63$ and $\sum_{i=1}^{25} y_i^2 = 500.7459$, we have that $\bar{y} = 3.23$ and $s = 3.17$.

b.

k	interval	frequency	Exp. frequency
1	0.063, 6.397	21	17
2	-3.104, 9.564	23	23.75
3	-6.271, 12.731	25	25

1.15 a. With $\sum_{i=1}^{40} y_i = 175.48$ and $\sum_{i=1}^{40} y_i^2 = 906.4118$, we have that $\bar{y} = 4.39$ and $s = 1.87$.

b.

k	interval	frequency	Exp. frequency
1	2.52, 6.26	35	27.2
2	0.65, 8.13	39	38
3	-1.22, 10	39	40

1.16 a. Without the extreme value, $\bar{y} = 4.19$ and $s = 1.44$.

b. These counts compare more favorably:

k	interval	frequency	Exp. frequency
1	2.75, 5.63	25	26.52
2	1.31, 7.07	36	37.05
3	-0.13, 8.51	39	39

1.17 For Ex. 1.2, $\text{range}/4 = 7.35$, while $s = 4.14$. For Ex. 1.3, $\text{range}/4 = 3.04$, while $s = 3.17$. For Ex. 1.4, $\text{range}/4 = 2.32$, while $s = 1.87$.

1.18 The approximation is $(800-200)/4 = 150$.

1.19 One standard deviation below the mean is $34 - 53 = -19$. The empirical rule suggests that 16% of all measurements should lie one standard deviation below the mean. Since chloroform measurements cannot be negative, this population cannot be normally distributed.

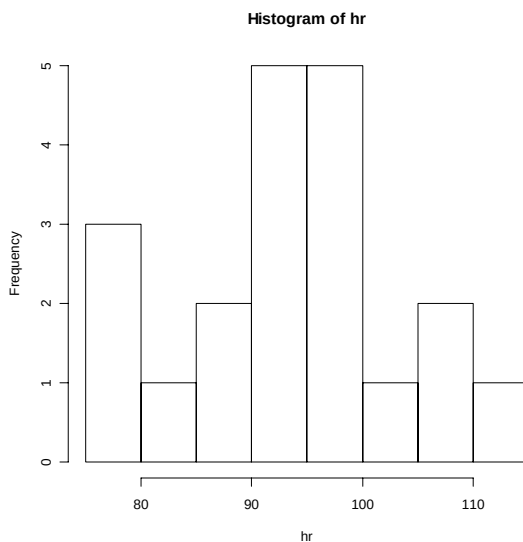
1.20 Since approximately 68% will fall between \$390 ($\$420 - \30) to \$450 ($\$420 + \30), the proportion above \$450 is approximately 16%.

1.21 (Similar to exercise 1.20) Having a gain of more than 20 pounds represents all measurements greater than one standard deviation below the mean. By the empirical rule, the proportion above this value is approximately 84%, so the manufacturer is probably correct.

1.22 (See exercise 1.11) $\sum_{i=1}^n (y_i - \bar{y}) = \sum_{i=1}^n y_i - n\bar{y} = \sum_{i=1}^n y_i - \sum_{i=1}^n y_i = 0$.

1.23 **a.** (Similar to exercise 1.20) 95 sec = 1 standard deviation above 75 sec, so this percentage is 16% by the empirical rule.
b. (35 sec., 115 sec) represents an interval of 2 standard deviations about the mean, so approximately 95%
c. 2 minutes = 120 sec = 2.5 standard deviations above the mean. This is unlikely.

1.24 **a.** $(112-78)/4 = 8.5$



b. The histogram is above.

c. With $\sum_{i=1}^{20} y_i = 1874.0$ and $\sum_{i=1}^{20} y_i^2 = 117,328.0$, we have that $\bar{y} = 93.7$ and $s = 9.55$.

d.

k	interval	frequency	Exp. frequency
1	84.1, 103.2	13	13.6
2	74.6, 112.8	20	19
3	65.0, 122.4	20	20

1.25 a. $(716-8)/4 = 177$

b. The figure is omitted.

c. With $\sum_{i=1}^{88} y_i = 18,550$ and $\sum_{i=1}^{88} y_i^2 = 6,198,356$, we have that $\bar{y} = 210.8$ and $s = 162.17$.

d.

k	interval	frequency	Exp. frequency
1	48.6, 373	63	59.84
2	-113.5, 535.1	82	83.6
3	-275.7, 697.3	87	88

1.26 For Ex. 1.12, $3/1.21 = 2.48$. For Ex. 1.24, $34/9.55 = 3.56$. For Ex. 1.25, $708/162.17 = 4.37$. The ratio increases as the sample size increases.

1.27 (64, 80) is one standard deviation about the mean, so 68% of 340 or approx. 231 scores. (56, 88) is two standard deviations about the mean, so 95% of 340 or 323 scores.

1.28 (Similar to 1.23) 13 mg/L is one standard deviation below the mean, so 16%.

1.29 If the empirical rule is assumed, approximately 95% of all bearing should lie in (2.98, 3.02) – this interval represents two standard deviations about the mean. So, approximately 5% will lie outside of this interval.

1.30 If $\mu = 0$ and $\sigma = 1.2$, we expect 34% to be between 0 and $0 + 1.2 = 1.2$. Also, approximately $95\%/2 = 47.5\%$ will lie between 0 and 2.4. So, $47.5\% - 34\% = 13.5\%$ should lie between 1.2 and 2.4.

1.31 Assuming normality, approximately 95% will lie between 40 and 80 (the standard deviation is 10). The percent below 40 is approximately 2.5% which is relatively unlikely.

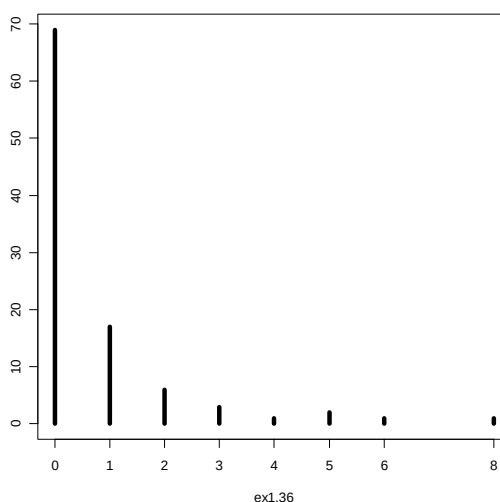
1.32 For a sample of size n , let n' denote the number of measurements that fall outside the interval $\bar{y} \pm ks$, so that $(n - n')/n$ is the fraction that falls inside the interval. To show this fraction is greater than or equal to $1 - 1/k^2$, note that

$$(n - 1)s^2 = \sum_{i \in A} (y_i - \bar{y})^2 + \sum_{i \in B} (y_i - \bar{y})^2, \text{ (both sums must be positive)}$$

where $A = \{i: |y_i - \bar{y}| \geq ks\}$ and $B = \{i: |y_i - \bar{y}| < ks\}$. We have that

$\sum_{i \in A} (y_i - \bar{y})^2 \geq \sum_{i \in A} k^2 s^2 = n' k^2 s^2$, since if i is in A , $|y_i - \bar{y}| \geq ks$ and there are n' elements in A . Thus, we have that $s^2 \geq k^2 s^2 n'/(n-1)$, or $1 \geq k^2 n'/(n-1) \geq k^2 n'/n$. Thus, $1/k^2 \geq n'/n$ or $(n - n')/n \geq 1 - 1/k^2$.

- 1.33** With $k=2$, at least $1 - 1/4 = 75\%$ should lie within 2 standard deviations of the mean. The interval is (0.5, 10.5).
- 1.34** The point 13 is $13 - 5.5 = 7.5$ units above the mean, or $7.5/2.5 = 3$ standard deviations above the mean. By Tchebysheff's theorem, at least $1 - 1/3^2 = 8/9$ will lie within 3 standard deviations of the mean. Thus, at most $1/9$ of the values will exceed 13.
- 1.35**
- a. $(172 - 108)/4 = 16$
 - b. With $\sum_{i=1}^{15} y_i = 2041$ and $\sum_{i=1}^{15} y_i^2 = 281,807$ we have that $\bar{y} = 136.1$ and $s = 17.1$
 - c. $a = 136.1 - 2(17.1) = 101.9$, $b = 136.1 + 2(17.1) = 170.3$.
 - d. There are 14 observations contained in this interval, and $14/15 = 93.3\%$. 75% is a lower bound.



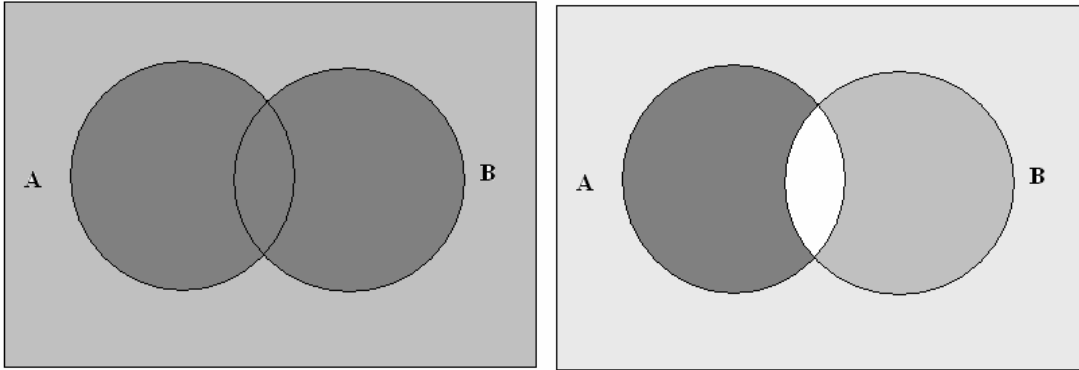
- 1.36**
- a. The histogram is above.
 - b. With $\sum_{i=1}^{100} y_i = 66$ and $\sum_{i=1}^{100} y_i^2 = 234$ we have that $\bar{y} = 0.66$ and $s = 1.39$.
 - c. Within two standard deviations: 95, within three standard deviations: 96. The calculations agree with Tchebysheff's theorem.
- 1.37** Since the lead readings must be non negative, 0 (the smallest possible value) is only 0.33 standard deviations from the mean. This indicates that the distribution is skewed.
- 1.38** By Tchebysheff's theorem, at least $3/4 = 75\%$ lie between (0, 140), at least $8/9$ lie between (0, 193), and at least $15/16$ lie between (0, 246). The lower bounds are all truncated at 0 since the measurement cannot be negative.

Chapter 2: Probability

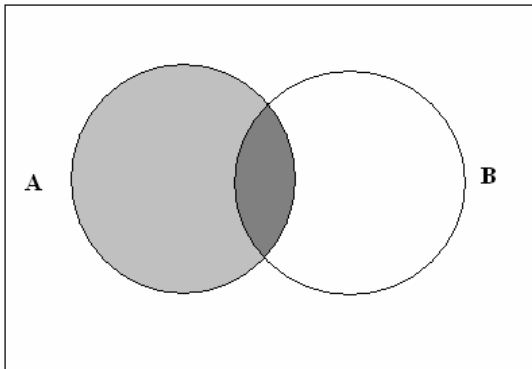
2.1 $A = \{FF\}$, $B = \{MM\}$, $C = \{MF, FM, MM\}$. Then, $A \cap B = \emptyset$, $B \cap C = \{MM\}$, $C \cap \bar{B} = \{MF, FM\}$, $A \cup B = \{FF, MM\}$, $A \cup C = S$, $B \cup C = C$.

2.2 a. $A \cap B$ b. $A \cup B$ c. $\overline{A \cup B}$ d. $(A \cap \bar{B}) \cup (\bar{A} \cap B)$

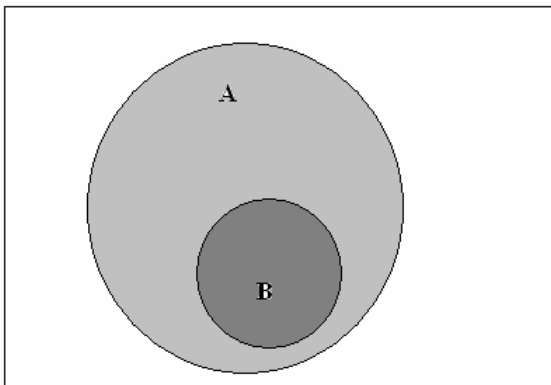
2.3



2.4 a.



b.



- 2.5** a. $(A \cap B) \cup (A \cap \bar{B}) = A \cap (B \cup \bar{B}) = A \cap S = A$.
~~b. $B \cup (A \cap \bar{B}) = (B \cap A) \cup (B \cap \bar{B}) = (B \cap A) = A$.~~
 c. $(A \cap B) \cap (A \cap \bar{B}) = A \cap (B \cap \bar{B}) = \emptyset$. The result follows from part a.
 d. $B \cap (A \cap \bar{B}) = A \cap (B \cap \bar{B}) = \emptyset$. The result follows from part b.
- 2.6** $A = \{(1,2), (2,2), (3,2), (4,2), (5,2), (6,2), (1,4), (2,4), (3,4), (4,4), (5,4), (6,4), (1,6), (2,6), (3,6), (4,6), (5,6), (6,6)\}$
 $\bar{C} = \{(2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)\}$
 $A \cap B = \{(2,2), (4,2), (6,2), (2,4), (4,4), (6,4), (2,6), (4,6), (6,6)\}$
 $A \cap \bar{B} = \{(1,2), (3,2), (5,2), (1,4), (3,4), (5,4), (1,6), (3,6), (5,6)\}$
 $\bar{A} \cup B = \text{everything but } \{(1,2), (1,4), (1,6), (3,2), (3,4), (3,6), (5,2), (5,4), (5,6)\}$
 $\bar{A} \cap C = \bar{A}$
- 2.7** $A = \{\text{two males}\} = \{M_1, M_2), (M_1, M_3), (M_2, M_3)\}$
 $B = \{\text{at least one female}\} = \{(M_1, W_1), (M_2, W_1), (M_3, W_1), (M_1, W_2), (M_2, W_2), (M_3, W_2), \{W_1, W_2\}\}$
 $\bar{B} = \{\text{no females}\} = A \quad A \cup B = S \quad A \cap B = \emptyset \quad A \cap \bar{B} = A$
- 2.8** a. $36 + 6 = 42$ b. 33 c. 18
- 2.9** $S = \{A+, B+, AB+, O+, A-, B-, AB-, O-\}$
- 2.10** a. $S = \{A, B, AB, O\}$
 b. $P(\{A\}) = 0.41, P(\{B\}) = 0.10, P(\{AB\}) = 0.04, P(\{O\}) = 0.45$.
 c. $P(\{A\} \text{ or } \{B\}) = P(\{A\}) + P(\{B\}) = 0.51$, since the events are mutually exclusive.
- 2.11** a. Since $P(S) = P(E_1) + \dots + P(E_5) = 1$, $1 = .15 + .15 + .40 + 3P(E_5)$. So, $P(E_5) = .10$ and $P(E_4) = .20$.
 b. Obviously, $P(E_3) + P(E_4) + P(E_5) = .6$. Thus, they are all equal to .2
- 2.12** a. Let $L = \{\text{left turn}\}$, $R = \{\text{right turn}\}$, $C = \{\text{continues straight}\}$.
 b. $P(\text{vehicle turns}) = P(L) + P(R) = 1/3 + 1/3 = 2/3$.
- 2.13** a. Denote the events as very likely (VL), somewhat likely (SL), unlikely (U), other (O).
 b. Not equally likely: $P(VL) = .24, P(SL) = .24, P(U) = .40, P(O) = .12$.
 c. $P(\text{at least SL}) = P(SL) + P(VL) = .48$.
- 2.14** a. $P(\text{needs glasses}) = .44 + .14 =$ 
 b. $P(\text{needs glasses but doesn't use them}) = .14$
 c. $P(\text{uses glasses}) = .44 + .02 = .46$
- 2.15** a. Since the events are M.E., $P(S) = P(E_1) + \dots + P(E_4) = 1$. So, $P(E_2) = 1 - .01 - .09 - .81 = .09$.
 b. $P(\text{at least one hit}) = P(E_1) + P(E_2) + P(E_3) = .19$.

- 2.16** **a.** $1/3$ **b.** $1/3 + 1/15 = 6/15$ **c.** $1/3 + 1/16 = 19/48$ **d.** $49/240$
- 2.17** Let B = bushing defect, SH = shaft defect.
a. $P(B) = .06 + .02 = .08$
b. $P(B \text{ or } SH) = .06 + .08 + .02 = .16$
c. $P(\text{exactly one defect}) = .06 + .08 = .14$
d. $P(\text{neither defect}) = 1 - P(B \text{ or } SH) = 1 - .16 = .84$
- 2.18** **a.** $S = \{HH, TH, HT, TT\}$
b. if the coin is fair, all events have probability .25.
c. $A = \{HT, TH\}$, $B = \{HT, TH, HH\}$
d. $P(A) = .5$, $P(B) = .75$, $P(A \cap B) = P(A) = .5$, $P(A \cup B) = P(B) = .75$, $P(\bar{A} \cup B) = 1$.
- 2.19** **a.** $(V_1, V_1), (V_1, V_2), (V_1, V_3), (V_2, V_1), (V_2, V_2), (V_2, V_3), (V_3, V_1), (V_3, V_2), (V_3, V_3)$
b. if equally likely, all have probability of $1/9$.
c. $A = \{\text{same vendor gets both}\} = \{(V_1, V_1), (V_2, V_2), (V_3, V_3)\}$
 $B = \{\text{at least one } V_2\} = \{(V_1, V_2), (V_2, V_1), (V_2, V_2), (V_2, V_3), (V_3, V_2)\}$
 So, $P(A) = 1/3$, $P(B) = 5/9$, $P(A \cup B) = 7/9$, $P(A \cap B) = 1/9$.
- 2.20** **a.** $P(G) = P(D_1) = P(D_2) = 1/3$.
b. i. The probability of selecting the good prize is $1/3$.
 ii. She will get the other dud.
 iii. She will get the good prize.
 iv. Her probability of winning is now $2/3$.
 v. The best strategy is to switch.
- 2.21** $P(A) = P((A \cap B) \cup (A \cap \bar{B})) = P(A \cap B) + P(A \cap \bar{B})$ since these are M.E. by Ex. 2.5.
- 2.22** $P(A) = P(B \cup (A \cap \bar{B})) = P(B) + P(A \cap \bar{B})$ since these are M.E. by Ex. 2.5.
- 2.23** All elements in B are in A , so that when B occurs, A must also occur. However, it is possible for A to occur and B not to occur.
- 2.24** From the relation in Ex. 2.22, $P(A \cap \bar{B}) \geq 0$, so $P(B) \leq P(A)$.
- 2.25** Unless exactly $1/2$ of all cars in the lot are Volkswagens, the claim is not true.
- 2.26** **a.** Let N_1, N_2 denote the empty cans and W_1, W_2 denote the cans filled with water. Thus, $S = \{N_1N_2, N_1W_2, N_2W_2, N_1W_1, N_2W_1, W_1W_2\}$
b. If this is merely a guess, the events are equally likely. So, $P(W_1W_2) = 1/6$.
- 2.27** **a.** $S = \{CC, CR, CL, RC, RR, RL, LC, LR, LL\}$
b. $5/9$
c. $5/9$

- 2.28** a. Denote the four candidates as A_1, A_2, A_3 , and M . Since order is not important, the outcomes are $\{A_1A_2, A_1A_3, A_1M, A_2A_3, A_2M, A_3M\}$.
 b. assuming equally likely outcomes, all have probability $1/6$.
 c. $P(\text{minority hired}) = P(A_1M) + P(A_2M) + P(A_3M) = .5$
- 2.29** a. The experiment consists of randomly selecting two jurors from a group of two women and four men.
 b. Denoting the women as $w1, w2$ and the men as $m1, m2, m3, m4$, the sample space is
- | | | | | |
|----------|----------|----------|----------|----------|
| $w1, m1$ | $w2, m1$ | $m1, m2$ | $m2, m3$ | $m3, m4$ |
| $w1, m2$ | $w2, m2$ | $m1, m3$ | $m2, m4$ | |
| $w1, m3$ | $w2, m3$ | $m1, m4$ | | |
| $w1, m4$ | $w2, m4$ | | | $w1, w2$ |
- c. $P(w1, w2) = 1/15$
- 2.30** a. Let $w1$ denote the first wine, $w2$ the second, and $w3$ the third. Each sample point is an ordered triple indicating the ranking.
 b. triples: $(w1, w2, w3), (w1, w3, w2), (w2, w1, w3), (w2, w3, w1), (w3, w1, w2), (w3, w2, w1)$
 c. For each wine, there are 4 ordered triples where it is not last. So, the probability is $2/3$.
- 2.31** a. There are four “good” systems and two “defective” systems. If two out of the six systems are chosen randomly, there are 15 possible unique pairs. Denoting the systems as $g1, g2, g3, g4, d1, d2$, the sample space is given by $S = \{g1g2, g1g3, g1g4, g1d1, g1d2, g2g3, g2g4, g2d1, g2d2, g3g4, g3d1, g3d2, g4g1, g4d1, d1d2\}$. Thus:
 $P(\text{at least one defective}) = 9/15$ $P(\text{both defective}) = P(d1d2) = 1/15$
 b. If four are defective, $P(\text{at least one defective}) = 14/15$. $P(\text{both defective}) = 6/15$.
- 2.32** a. Let “1” represent a customer seeking style 1, and “2” represent a customer seeking style 2. The sample space consists of the following 16 four-tuples:
 $1111, 1112, 1121, 1211, 2111, 1122, 1212, 2112, 1221, 2121,$
 $2211, 2221, 2212, 2122, 1222, 2222$
 b. If the styles are equally in demand, the ordering should be equally likely. So, the probability is $1/16$.
 c. $P(A) = P(1111) + P(2222) = 2/16$.
- 2.33** a. Define the events: G = family income is greater than \$43,318, N otherwise. The points are
- | | | | |
|-------------|-------------|-------------|-------------|
| $E1: GGGG$ | $E2: GGGN$ | $E3: GGNG$ | $E4: GNNG$ |
| $E5: NGGG$ | $E6: GGNN$ | $E7: GNGN$ | $E8: NGGN$ |
| $E9: GNNG$ | $E10: NGNG$ | $E11: NNGG$ | $E12: GNNN$ |
| $E13: NGNN$ | $E14: NNGN$ | $E15: NNNG$ | $E16: NNNN$ |
- b. $A = \{E1, E2, \dots, E11\}$ $B = \{E6, E7, \dots, E11\}$ $C = \{E2, E3, E4, E5\}$
 c. If $P(E) = P(N) = .5$, each element in the sample space has probability $1/16$. Thus,
 $P(A) = 11/16, P(B) = 6/16$, and $P(C) = 4/16$.

- 2.34** a. Three patients enter the hospital and randomly choose stations 1, 2, or 3 for service. Then, the sample space S contains the following 27 three-tuples:
111, 112, 113, 121, 122, 123, 131, 132, 133, 211, 212, 213, 221, 222, 223,
231, 232, 233, 311, 312, 313, 321, 322, 323, 331, 332, 333
b. $A = \{123, 132, 213, 231, 312, 321\}$
c. If the stations are selected at random, each sample point is equally likely. $P(A) = 6/27$.
- 2.35** The total number of flights is $6(7) = 42$.
- 2.36** There are $3! = 6$ orderings.
- 2.37** a. There are $6! = 720$ possible itineraries.
b. In the 720 orderings, exactly 360 have Denver before San Francisco and 360 have San Francisco before Denver. So, the probability is .5.
- 2.38** By the mn rule, $4(3)(4)(5) = 240$.
- 2.39** a. By the mn rule, there are $6(6) = 36$ possible roles.
b. Define the event $A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$. Then, $P(A) = 6/36$.
- 2.40** a. By the mn rule, the dealer must stock $5(4)(2) = 40$ autos.
b. To have each of these in every one of the eight colors, he must stock $8 \cdot 40 = 320$ autos.
- 2.41** If the first digit cannot be zero, there are 9 possible values. For the remaining six, there are 10 possible values. Thus, the total number is $9(10)(10)(10)(10)(10) = 9 \cdot 10^6$.
- 2.42** There are three *different* positions to fill using ten engineers. Then, there are $P_3^{10} = \cancel{10!/3!} = 720$ different ways to fill the positions.
- 2.43** $\binom{9}{3}\binom{6}{5}\binom{1}{1} = 504$ ways.
- 2.44** a. The number of ways the taxi needing repair can be sent to airport C is $\binom{8}{5}\binom{5}{5} = 56$.
So, the probability is $56/504 = 1/9$.
b. $3\binom{6}{2}\binom{4}{4} = 45$, so the probability that every airport receives one of the taxis requiring repair is $45/504$.
- 2.45** $\binom{17}{2 \ 7 \ 10} = 408,408$.

- 2.46** There are $\binom{10}{2}$ ways to choose two teams for the first game, $\binom{8}{2}$ for second, etc. So, there are $\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2} = \frac{10!}{2^5} = 113,400$ ways to assign the ten teams to five games.
- 2.47** There are $\binom{2n}{2}$ ways to choose two teams for the first game, $\binom{2n-2}{2}$ for second, etc. So, following Ex. 2.46, there are $\frac{2n!}{2^n}$ ways to assign $2n$ teams to n games.
- 2.48** Same answer: $\binom{8}{5} = \binom{8}{3} = 56$.
- 2.49** **a.** $\binom{130}{2} = 8385$.
b. There are $26 \cdot 26 = 676$ two-letter codes and $26(26)(26) = 17,576$ three-letter codes. Thus, 18,252 total major codes.
c. $8385 + 130 = 8515$ required.
d. Yes.
- 2.50** Two numbers, 4 and 6, are possible for each of the three digits. So, there are $2(2)(2) = 8$ potential winning three-digit numbers.
- 2.51** There are $\binom{50}{3} = 19,600$ ways to choose the 3 winners. Each of these is equally likely.
a. There are $\binom{4}{3} = 4$ ways for the organizers to win all of the prizes. The probability is $4/19600$.
b. There are $\binom{4}{2}\binom{46}{1} = 276$ ways the organizers can win two prizes and one of the other 46 people to win the third prize. So, the probability is $276/19600$.
c. $\binom{4}{1}\binom{46}{2} = 4140$. The probability is $4140/19600$.
d. $\binom{46}{3} = 15,180$. The probability is $15180/19600$.
- 2.52** The mn rule is used. The total number of experiments is $3(3)(2) = 18$.

- 2.53** a. In choosing three of the five firms, order is important. So $P_3^5 = 60$ sample points.
b. If F_3 is awarded a contract, there are $P_2^4 = 12$ ways the other contracts can be assigned. Since there are 3 possible contracts, there are $3(12) = 36$ total number of ways to award F_3 a contract. So, the probability is $36/60 = 0.6$.
- 2.54** There are $\binom{8}{4} = 70$ ways to chose four students from eight. There are $\binom{3}{2}\binom{5}{2} = 30$ ways to chose exactly 2 (of the 3) undergraduates and 2 (of the 5) graduates. ~~If each sample point is equally likely, the probability is $30/70 = 0.7$.~~
- 2.55** a. $\binom{90}{10}$ b. $\binom{20}{4}\binom{70}{6} / \binom{90}{10} = 0.111$
- 2.56** The student can solve all of the problems if the teacher selects 5 of the 6 problems that the student can do. The probability is $\binom{6}{5} / \binom{10}{5} = 0.0238$.
- 2.57** There are $\binom{52}{2} = 1326$ ways to draw two cards from the deck. The probability is $4 \cdot 12 / 1326 = 0.0362$.
- 2.58** There are $\binom{52}{5} = 2,598,960$ ways to draw five cards from the deck.
a. There are $\binom{4}{3}\binom{4}{2} = 24$ ways to draw three Aces and two Kings. So, the probability is $24/2598960$.
b. There are $13(12) = 156$ types of “full house” hands. From part a. above there are 24 different ways each type of full house hand can be made. So, the probability is $156 \cdot 24 / 2598960 = 0.00144$.
- 2.59** There are $\binom{52}{5} = 2,598,960$ ways to draw five cards from the deck.
a. $\binom{4}{1}\binom{4}{1}\binom{4}{1}\binom{4}{1}\binom{4}{1} = 4^5 = 1024$. So, the probability is $1024/2598960 = 0.000394$.
b. There are 9 different types of “straight” hands. So, the probability is $9(4^5)/2598960 = 0.00355$. Note that this also includes “straight flush” and “royal straight flush” hands.
- 2.60** a. $\frac{365(364)(363)\cdots(365-n+1)}{365^n}$ b. With $n = 23$, $1 - \frac{365(364)\cdots(343)}{365^{23}} = 0.507$.

2.61 a. $\frac{364(364)(364)\cdots(364)}{365^n} = \frac{364^n}{365^n}$. b. With $n = 253$, $1 - \left(\frac{364}{365}\right)^{253} = 0.5005$.

2.62 The number of ways to divide the 9 motors into 3 groups of size 3 is $\left(\frac{9!}{3! 3! 3!}\right) = 1680$. If both motors from a particular supplier are assigned to the first line, there are only 7 motors to be assigned: one to line 1 and three to lines 2 and 3. This can be done $\left(\frac{7!}{1! 3! 3!}\right) = 140$ ways. Thus, $140/1680 = 0.0833$.

2.63 There are $\binom{8}{5} = 56$ sample points in the experiment, and only one of which results in choosing five women. So, the probability is $1/56$.

2.64 $6!\left(\frac{1}{6}\right)^6 = 5/324$.

2.65 $5!\left(\frac{2}{6}\right)^6\left(\frac{1}{6}\right)^4 = 5/162$.

2.66 a. After assigning an ethnic group member to each type of job, there are 16 laborers remaining for the other jobs. Let n_a be the number of ways that one ethnic group can be assigned to each type of job. Then:

$$n_a = \binom{4}{1 \ 1 \ 1 \ 1} \binom{16}{5 \ 3 \ 4 \ 4}. \text{ The probability is } n_a/N = 0.1238.$$

b. It doesn't matter how the ethnic group members are assigned to jobs type 1, 2, and 3. Let n_a be the number of ways that no ethnic member gets assigned to a type 4 job. Then:

$$n_a = \binom{4}{0} \binom{16}{5}. \text{ The probability is } \binom{4}{0} \binom{16}{5} / \binom{20}{5} = 0.2817.$$

2.67 As shown in Example 2.13, $N = 10^7$.

a. Let A be the event that all orders go to different vendors. Then, A contains $n_a = 10(9)(8)\cdots(4) = 604,800$ sample points. Thus, $P(A) = 604,800/10^7 = 0.0605$.

b. The 2 orders assigned to Distributor I can be chosen from the 7 in $\binom{7}{2} = 21$ ways.

The 3 orders assigned to Distributor II can be chosen from the remaining 5 in $\binom{5}{3} = 10$ ways. The final 2 orders can be assigned to the remaining 8 distributors in 8^2 ways. Thus, there are $21(10)(8^2) = 13,440$ possibilities so the probability is $13440/10^7 = 0.001344$.

- c. Let A be the event that Distributors I, II, and III get exactly 2, 3, and 1 order(s) respectively. Then, there is one remaining unassigned order. Thus, A contains

$$\binom{7}{2}\binom{5}{3}\binom{2}{1}7 = 2940 \text{ sample points and } P(A) = 2940/10^7 = 0.00029.$$

- 2.68** a. $\binom{n}{n} = \frac{n!}{n!(n-n)!} = 1$. There is only one way to choose all of the items.
 b. $\binom{n}{0} = \frac{n!}{0!(n-0)!} = 1$. There is only one way to choose none of the items.
 c. $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)!(n-(n-r))!} = \binom{n}{n-r}$. There are the same number of ways to choose r out of n objects as there are to choose $n-r$ out of n objects.
 d. $2^n = (1+1)^n = \sum_{i=1}^n \binom{n}{i} 1^{n-i} 1^i = \sum_{i=1}^n \binom{n}{i}$.

2.69 $\binom{n}{k} + \binom{n}{k-1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} = \frac{n!(n-k+1)}{k!(n-k+1)!} + \frac{n!k}{k!(n-k+1)!} = \frac{(n+1)!}{k!(n+1-k)!}$

- 2.70** From Theorem 2.3, let $y_1 = y_2 = \dots = y_k = 1$.

- 2.71** a. $P(A|B) = .1/.3 = 1/3$. b. $P(B|A) = .1/.5 = 1/5$.
 c. $P(A|A \cup B) = .5/(.5+.3-.1) = 5/7$ d. $P(A|A \cap B) = 1$, since A has occurred.
 e. $P(A \cap B|A \cup B) = .1/(.5+.3-.1) = 1/7$.

- 2.72** Note that $P(A) = 0.6$ and $P(A|M) = .24/.4 = 0.6$. So, A and M are independent. Similarly, $P(\bar{A}|F) = .24/.6 = 0.4 = P(\bar{A})$, so $\bar{A}$ and F are independent.

- 2.73** a. $P(\text{at least one R}) = P(\text{Red}) = 3/4$. b. $P(\text{at least one r}) = 3/4$.
 c. $P(\text{one r} | \text{Red}) = .5/.75 = 2/3$.

- 2.74** a. $P(A) = 0.61$, $P(D) = .30$. $P(A \cap D) = .20$. Dependent.
 b. $P(B) = 0.30$, $P(D) = .30$. $P(B \cap D) = 0.09$. Independent.
 c. $P(C) = 0.09$, $P(D) = .30$. $P(C \cap D) = 0.01$. Dependent.

2.75 a. Given the first two cards drawn are spades, there are 11 spades left in the deck. Thus,

the probability is $\frac{\binom{11}{3}}{\binom{50}{3}} = 0.0084$. Note: this is also equal to $P(S_3S_4S_5|S_1S_2)$.

b. Given the first three cards drawn are spades, there are 10 spades left in the deck. Thus,

the probability is $\frac{\binom{10}{2}}{\binom{49}{2}} = 0.0383$. Note: this is also equal to $P(S_4S_5|S_1S_2S_3)$.

c. Given the first four cards drawn are spades, there are 9 spades left in the deck. Thus,

the probability is $\frac{\binom{9}{1}}{\binom{48}{1}} = 0.1875$. Note: this is also equal to $P(S_5|S_1S_2S_3S_4)$

2.76 Define the events: U : job is unsatisfactory A : plumber A does the job

a. $P(U|A) = P(A \cap U)/P(A) = P(A|U)P(U)/P(A) = .5 \cdot .1/.4 = 0.125$

b. From part a. above, $1 - P(U|A) = 0.875$.

2.77 **a.** 0.40 **b.** 0.37 **c.** 0.10 **d.** $0.40 + 0.37 - 0.10 = 0.67$
e. $1 - 0.4 = 0.6$ **f.** $1 - 0.67 = 0.33$ **g.** $1 - 0.10 = 0.90$
h. $.1/.37 = 0.27$ **i.** $1/.4 = 0.25$

2.78 1. Assume $P(A|B) = P(A)$. Then:

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B). \quad P(B|A) = P(B \cap A)/P(A) = P(A)P(B)/P(A) = P(B).$$

2. Assume $P(B|A) = P(B)$. Then:

$$P(A \cap B) = P(B|A)P(A) = P(B)P(A). \quad P(A|B) = P(A \cap B)/P(B) = P(A)P(B)/P(B) = P(A).$$

3. Assume $P(A \cap B) = P(B)P(A)$. The results follow from above.

2.79 If A and B are M.E., $P(A \cap B) = 0$. But, $P(A)P(B) > 0$. So they are not independent.

2.80 If $A \subset B$, $P(A \cap B) = P(A) \neq P(A)P(B)$, unless $B = S$ (in which case $P(B) = 1$). 

2.81 Given $P(A) < P(A|B) = P(A \cap B)/P(B) = P(B|A)P(A)/P(B)$, solve for $P(B|A)$ in the inequality.

2.82 $P(B|A) = P(B \cap A)/P(A) = P(A)/P(A) = 1$
 $P(A|B) = P(A \cap B)/P(B) = P(A)/P(B)$.

2.83 $P(A|A \cup B) = P(A)/P(A \cup B) = \frac{P(A)}{P(A) + P(B)}$, since A and B are M.E. events.

2.84 Note that if $P(A_2 \cap A_3) = 0$, then $P(A_1 \cap A_2 \cap A_3)$ also equals 0. The result follows from Theorem 2.6.

2.85 $P(A|\bar{B}) = P(A \cap \bar{B})/P(\bar{B}) = \frac{P(\bar{B}|A)P(A)}{P(\bar{B})} = \frac{[1 - P(B|A)]P(A)}{P(\bar{B})} = \frac{[1 - P(B)]P(A)}{P(\bar{B})} = \frac{P(\bar{B})P(A)}{P(\bar{B})} = P(A)$. So, A and $\bar{B}$ are independent.

$P(\bar{B}|\bar{A}) = P(\bar{B} \cap \bar{A})/P(\bar{A}) = \frac{P(\bar{A}|\bar{B})P(\bar{B})}{P(\bar{A})} = \frac{[1 - P(A|\bar{B})]P(\bar{B})}{P(\bar{A})}$. From the above, A and $\bar{B}$ are independent. So $P(\bar{B}|\bar{A}) = \frac{[1 - P(A)]P(\bar{B})}{P(\bar{A})} = \frac{P(\bar{A})P(\bar{B})}{P(\bar{A})} = P(\bar{B})$. So, $\bar{A}$ and $\bar{B}$ are independent

2.86 **a.** No. It follows from $P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq 1$.

b. $P(A \cap B) \geq 0.5$

c. No.

d. $P(A \cap B) \leq 0.70$.

2.87 **a.** $P(A) + P(B) - 1$.

b. the smaller of $P(A)$ and $P(B)$.

2.88 **a.** Yes.

b. 0, since they could be disjoint.

c. No, since $P(A \cap B)$ cannot be larger than either of $P(A)$ or $P(B)$.

d. $0.3 = P(A)$.

2.89 **a.** 0, since they could be disjoint.

b. the smaller of $P(A)$ and $P(B)$.

2.90 **a.** $(1/50)(1/50) = 0.0004$.

b. $P(\text{at least one injury}) = 1 - P(\text{no injuries in 50 jumps}) = 1 - (49/50)^{50} = 0.636$. Your friend is not correct.

2.91 If A and B are M.E., $P(A \cup B) = P(A) + P(B)$. This value is greater than 1 if $P(A) = 0.4$ and $P(B) = 0.7$. So they cannot be M.E. It is possible if $P(A) = 0.4$ and $P(B) = 0.3$.

2.92 **a.** The three tests are independent. So, the probability in question is $(.05)^3 = 0.000125$.

b. $P(\text{at least one mistake}) = 1 - P(\text{no mistakes}) = 1 - (.95)^3 = 0.143$.

2.93 Let H denote a hit and let M denote a miss. Then, she wins the game in three trials with the events HHH, HHM, and MHH. If she begins with her right hand, the probability she wins the game, assuming independence, is $(.7)(.4)(.7) + (.7)(.4)(.3) + (.3)(.4)(.7) = 0.364$.

2.94 Define the events A: device A detects smoke B: device B detects smoke

a. $P(A \cup B) = .95 + .90 - .88 = 0.97$.

b. $P(\text{smoke is undetected}) = 1 - P(A \cup B) = 1 - 0.97 = 0.03$.

2.95 Part a is found using the Addition Rule. Parts b and c use DeMorgan's Laws.

a. $0.2 + 0.3 - 0.4 = 0.1$

b. $1 - 0.1 = 0.9$

c. $1 - 0.4 = 0.6$

d. $P(\bar{A} | B) = \frac{P(\bar{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = \frac{.3 - .1}{.3} = 2/3$.

2.96 ~~Using the results of Ex. 2.95:~~

a. $0.5 + 0.2 - (0.5)(0.2) = 0.6$.

b. $1 - 0.6 = 0.4$.

c. $1 - 0.1 = 0.9$.

2.97 a. $P(\text{current flows}) = 1 - P(\text{all three relays are open}) = 1 - (.1)^3 = 0.999$.

b. Let A be the event that current flows and B be the event that relay 1 closed properly. Then, $P(B|A) = P(B \cap A)/P(A) = P(B)/P(A) = .9/.999 = 0.9009$. Note that $B \subset A$.

2.98 Series system: $P(\text{both relays are closed}) = (.9)(.9) = 0.81$

Parallel system: $P(\text{at least one relay is closed}) = .9 + .9 - .81 = 0.99$.

2.99 Given that $P(\overline{A \cup B}) = a$, $P(B) = b$, and that A and B are independent. Thus $P(A \cup B) = 1 - a$ and $P(B \cap A) = bP(A)$. Thus, $P(A \cup B) = P(A) + b - bP(A) = 1 - a$. Solve for $P(A)$.

2.100
$$P(A \cup B | C) = \frac{P((A \cup B) \cap C)}{P(C)} = \frac{P((A \cap C) \cup (B \cap C))}{P(C)} =$$

$$\frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} = P(A|C) + P(B|C) - P(A \cap B|C).$$

2.101 Let A be the event the item gets past the first inspector and B the event it gets past the second inspector. Then, $P(A) = 0.1$ and $P(B|A) = 0.5$. Then $P(A \cap B) = .1(.5) = 0.05$.

2.102 Define the events: I: disease I ~~us~~ contracted II: disease II is contracted. Then, $P(I) = 0.1$, $P(II) = 0.15$, and $P(I \cap II) = 0.03$.

a. $P(I \cup II) = .1 + .15 - .03 = 0.22$

b. $P(I \cap II | I \cup II) = .03/.22 = 3/22$.

2.103 Assume that the two state lotteries are independent.

a. $P(666 \text{ in CT} | 666 \text{ in PA}) = P(666 \text{ in CT}) = 0.001$

b. $P(666 \text{ in CT} \cap 666 \text{ in PA}) = P(666 \text{ in CT})P(666 \text{ in PA}) = .001(1/8) = 0.000125.$

2.104 By DeMorgan's Law, $P(A \cap B) = 1 - P(\overline{A \cap B}) = 1 - P(\overline{A} \cup \overline{B})$. Since $P(\overline{A} \cup \overline{B}) \leq P(\overline{A}) + P(\overline{B})$, $P(A \cap B) \geq 1 - P(\overline{A}) - P(\overline{B})$.

2.105 $P(\text{landing safely on both jumps}) \geq 1 - 0.05 - 0.05 = 0.90.$

2.106 Note that it must be also true that $P(\overline{A}) = P(\overline{B})$. Using the result in Ex. 2.104,

$$P(A \cap B) \geq 1 - 2P(\overline{A}) \geq 0.98, \text{ so } P(A) \geq 0.99.$$

2.107 (Answers vary) Consider flipping a coin twice. Define the events:

A: observe at least one tail B: observe two heads or two tails C: observe two heads

2.108 Let U and V be two events. Then, by Ex. 2.104, $P(U \cap V) \geq 1 - P(\overline{U}) - P(\overline{V})$. Let $U = A \cap B$ and $V = C$. Then, $P(A \cap B \cap C) \geq 1 - P(\overline{A \cap B}) - P(\overline{C})$. Apply Ex. 2.104 to $P(\overline{A \cap B})$ to obtain the result.

2.109 This is similar to Ex. 2.106. Apply Ex. 2.108: $0.95 \leq 1 - P(\overline{A}) - P(\overline{B}) - P(\overline{C}) \leq P(A \cap B \cap C)$. Since the events have the same probability, $0.95 \leq 1 - 3P(\overline{A})$. Thus, $P(A) \geq 0.9833$.

2.110 Define the events:

I: item is from line I II: item is from line II N: item is not defective

Then, $P(N) = P(N \cap (I \cup II)) = P(N \cap I) + P(N \cap II) = .92(.4) + .90(.6) = 0.908.$

2.111 Define the following events:

A: buyer sees the magazine ad

B: buyer sees the TV ad

C: buyer purchases the product

The following are known: $P(A) = .02$, $P(B) = .20$, $P(A \cap B) = .01$. Thus $P(A \cup B) = .21$.

Also, $P(\text{buyer sees no ad}) = P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B) = 1 - 0.21 = 0.79$. Finally, it is known that $P(C | A \cup B) = 0.1$ and $P(C | \overline{A} \cap \overline{B}) = 1/3$. So, we can find $P(C)$ as

$$P(C) = P(C \cap (A \cup B)) + P(C \cap (\overline{A} \cap \overline{B})) = (1/3)(.21) + (.1)(.79) = 0.149.$$

2.112 a. $P(\text{aircraft undetected}) = P(\text{all three fail to detect}) = (.02)(.02)(.02) = (.02)^3$.

b. $P(\text{all three detect aircraft}) = (.98)^3$.

2.113 By independence, $(.98)(.98)(.98)(.02) = (.98)^3(.02)$.

2.114 Let $T = \{\text{detects truth}\}$ and $L = \{\text{detects lie}\}$. The sample space is TT, TL, LT, LL . Since one suspect is guilty, assume the guilty suspect is questioned first:

- a.** $P(LL) = .95(.10) = 0.095$ **b.** $P(LT) = .95(.9) = 0.885$
b. $P(TL) = .05(.10) = 0.005$ **d.** $1 - (.05)(.90) = 0.955$

2.115 By independence, $(.75)(.75)(.75)(.75) = (.75)^4$.

2.116 By the complement rule, $P(\text{system works}) = 1 - P(\text{system fails}) = 1 - (.01)^3$.

2.117 a. From the description of the problem, there is a 50% chance a car will be rejected. To find the probability that three out of four will be rejected (i.e. the drivers chose team 2), note that there are $\binom{4}{3} = 4$ ways that three of the four cars are evaluated by team 2. Each one has probability $(.5)(.5)(.5)(.5)$ of occurring, so the probability is $4(.5)^4 = 0.25$.

b. The probability that all four pass (i.e. all four are evaluated by team 1) is $(.5)^4 = 1/16$.

2.118 If the victim is to be saved, a proper donor must be found within eight minutes. The patient will be saved if the proper donor is found on the 1st, 2nd, 3rd, or 4th try. But, if the donor is found on the 2nd try, that implies he/she wasn't found on the 1st try. So, the probability of saving the patient is found by, letting $A = \{\text{correct donor is found}\}$:

$$P(\text{save}) = P(A) + P(\bar{A}A) + P(\bar{A}\bar{A}A) + P(\bar{A}\bar{A}\bar{A}A).$$

By independence, this is $.4 + .6(.4) + (.6)^2(.4) + (.6)^3(.4) = 0.8704$

2.119 a. Define the events: A : obtain a sum of 3 B : do not obtain a sum of 3 or 7
 Since there are 36 possible rolls, $P(A) = 2/36$ and $P(B) = 28/36$. Obtaining a sum of 3 before a sum of 7 can happen on the 1st roll, the 2nd roll, the 3rd roll, etc. Using the events above, we can write these as $A, BA, BBA, BBBA$, etc. The probability of obtaining a sum of 3 before a sum of 7 is given by $P(A) + P(B)P(A) + [P(B)]^2P(A) + [P(B)]^3P(A) + \dots$. (Here, we are using the fact that the rolls are independent.) This is an infinite sum, and it follows as a geometric series. Thus, $2/36 + (28/36)(2/36) + (28/36)^2(2/36) + \dots = 1/4$.

b. Similar to part a. Define C : obtain a sum of 4 D : do not obtain a sum of 4 or 7
 Then, $P(C) = 3/36$ and $P(D) = 27/36$. The probability of obtaining a 4 before a 7 is $1/3$.

2.120 Denote the events G : good refrigerator D : defective refrigerator

a. If the last defective refrigerator is found on the 4th test, this means the first defective refrigerator was found on the 1st, 2nd, or 3rd test. So, the possibilities are $DGGD, GDGD$, and $GGDD$. So, the probability is $(\frac{2}{6})(\frac{4}{5})(\frac{3}{4})\frac{1}{3}$. The probabilities associated with the other two events are identical to the first. So, the desired probability is $3(\frac{2}{6})(\frac{4}{5})(\frac{3}{4})\frac{1}{3} = \frac{1}{5}$.

b. Here, the second defective refrigerator must be found on the 2nd, 3rd, or 4th test.

Define: A_1 : second defective found on 2nd test
 A_2 : second defective found on 3rd test
 A_3 : second defective found on 4th test

Clearly, $P(A_1) = \left(\frac{2}{6}\right)\left(\frac{1}{5}\right) = \frac{1}{15}$. Also, $P(A_3) = \frac{1}{5}$ from part a. Note that $A_2 = \{DGD, GDD\}$.
Thus, $P(A_2) = 2\left(\frac{2}{6}\right)\left(\frac{4}{5}\right)\left(\frac{1}{4}\right) = \frac{2}{15}$. So, $P(A_1) + P(A_2) + P(A_3) = 2/5$.

c. Define: B_1 : second defective found on 3rd test
 B_2 : second defective found on 4th test

Clearly, $P(B_1) = 1/4$ and $P(B_2) = (3/4)(1/3) = 1/4$. So, $P(B_1) + P(B_2) = 1/2$.

2.121 a. $1/n$

b. $\frac{n-1}{n} \cdot \frac{1}{n-1} = 1/n$. $\frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdot \frac{1}{n-2} = 1/n$.

c. $P(\text{gain access}) = P(\text{first try}) + P(\text{second try}) + P(\text{third try}) = 3/7$.

2.122 Applet exercise (answers vary).

2.123 Applet exercise (answers vary).

2.124 Define the events for the voter: D : democrat R : republican F : favors issue

$$P(D|F) = \frac{P(F|D)P(D)}{P(F|D)P(D) + P(F|R)P(R)} = \frac{.7(.6)}{.7(.6) + .3(.4)} = 7/9$$

2.125 Define the events for the person: D : has the disease H : test indicates the disease

Thus, $P(H|D) = .9$, $P(\bar{H}|\bar{D}) = .9$, $P(D) = .01$, and $P(\bar{D}) = .99$. Thus,

$$P(D|H) = \frac{P(H|D)P(D)}{P(H|D)P(D) + P(H|\bar{D})P(\bar{D})} = 1/12.$$

2.126 a. $(.95*.01)/(.95*.01 + .1*.99) = 0.08756$.

b. $.99*.01/(.99*.01 + .1*.99) = 1/11$. 

c. Only a small percentage of the population has the disease.

d. If the specificity is .99, the positive predictive value is .5.

e. The sensitivity and specificity should be as close to 1 as possible.

2.127 a. $.9*.4/(.9*.4 + .1*.6) = 0.857$.

b. A larger proportion of the population has the disease, so the numerator and denominator values are closer.

c. No; if the sensitivity is 1, the largest value for the positive predictive value is .8696.

d. Yes, by increasing the specificity.

e. The specificity is more important with tests used for rare diseases.

2.128 a. Let $P(A|B) = P(A|\bar{B}) = p$. By the Law of Total Probability,

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}) = p(P(B) + P(\bar{B})) = p.$$

Thus, A and B are independent.

b. $P(A) = P(A|C)P(C) + P(A|\bar{C})P(\bar{C}) > P(B|C)P(C) + P(B|\bar{C})P(\bar{C}) = P(B)$.

- 2.129** Define the events: P : positive response M : male respondent F : female respondent
 $P(P|F) = .7$, $P(P|M) = .4$, $P(M) = .25$. Using Bayes' rule,

$$P(M | \bar{P}) = \frac{P(\bar{P} | M)P(M)}{P(\bar{P} | M)P(M) + P(\bar{P} | F)P(F)} = \frac{.6(.25)}{.6(.25) + .3(.75)} = 0.4.$$

- 2.130** Define the events: C : contract lung cancer S : worked in a shipyard
 Thus, $P(S|C) = .22$, $P(S | \bar{C}) = .14$, and $P(C) = .0004$. Using Bayes' rule,

$$P(C | S) = \frac{P(S | C)P(C)}{P(S | C)P(C) + P(S | \bar{C})P(\bar{C})} = \frac{.22(.0004)}{.22(.0004) + .14(.9996)} = 0.0006.$$

- 2.131** The additive law of probability gives that $P(A \Delta B) = P(A \cap \bar{B}) + P(\bar{A} \cap B)$. Also, A and B can be written as the union of two disjoint sets: $A = (A \cap \bar{B}) \cup (A \cap B)$ and $B = (\bar{A} \cap B) \cup (A \cap B)$. Thus, $P(A \cap \bar{B}) = P(A) - P(A \cap B)$ and $P(\bar{A} \cap B) = P(B) - P(A \cap B)$. Thus, $P(A \Delta B) = P(A) + P(B) - 2P(A \cap B)$.

- 2.132** For $i = 1, 2, 3$, let F_i represent the event that the plane is found in region i and N_i be the complement. Also R_i is the event the plane is in region i . Then $P(F_i | R_i) = 1 - \alpha_i$ and $P(R_i) = 1/3$ for all i . Then,

$$\begin{aligned} \text{a. } P(R_1 | N_1) &= \frac{P(N_1 | R_1)P(R_1)}{P(N_1 | R_1)P(R_1) + P(N_1 | R_2)P(R_2) + P(N_1 | R_3)P(R_3)} = \frac{\alpha_1 \frac{1}{3}}{\alpha_1 \frac{1}{3} + \frac{1}{3} + \frac{1}{3}} \\ &= \frac{\alpha_1}{\alpha_1 + 2}. \end{aligned}$$

$$\text{b. Similarly, } P(R_2 | N_1) = \frac{1}{\alpha_1 + 2} \text{ and } \text{c. } P(R_3 | N_1) = \frac{1}{\alpha_1 + 2}.$$

- 2.133** Define the events: G : student guesses C : student is correct

$$P(\bar{G} | C) = \frac{P(C | \bar{G})P(\bar{G})}{P(C | \bar{G})P(\bar{G}) + P(C | G)P(G)} = \frac{1(.8)}{1(.8) + .25(.2)} = 0.9412.$$

- 2.134** Define F as "failure to learn. Then, $P(F|A) = .2$, $P(F|B) = .1$, $P(A) = .7$, $P(B) = .3$. By Bayes' rule, $P(A|F) = 14/17$.

- 2.135** Let M = major airline, P = private airline, C = commercial airline, B = travel for business

- $P(B) = P(B|M)P(M) + P(B|P)P(P) + P(B|C)P(C) = .6(.5) + .3(.6) + .1(.9) = 0.57$.
- $P(B \cap P) = P(B|P)P(P) = .3(.6) = 0.18$.
- $P(P|B) = P(B \cap P)/P(B) = .18/.57 = 0.3158$.
- $P(B|C) = 0.90$.

- 2.136** Let A = woman's name is selected from list 1, B = woman's name is selected from list 2.
 Thus, $P(A) = 5/7$, $P(\bar{B} | A) = 2/3$, $P(\bar{B} | \bar{A}) = 7/9$.

$$P(A | \bar{B}) = \frac{P(\bar{B} | A)P(A)}{P(\bar{B} | A)P(A) + P(\bar{B} | \bar{A})P(\bar{A})} = \frac{\frac{2}{3}(\frac{5}{7})}{\frac{2}{3}(\frac{5}{7}) + \frac{7}{9}(\frac{2}{7})} = \frac{30}{44}.$$

$A_i =$ both balls selected from bowl i are white. Then $\bigcup A_i = A$.

a. $P(A) = \sum P(A_i | B_i)P(B_i) = \frac{1}{5}[0 + \frac{2}{5}(\frac{1}{4}) + \frac{3}{5}(\frac{2}{4}) + \frac{4}{5}(\frac{3}{4}) + 1] = 2/5.$

b. Using Bayes' rule, $P(B_3|A) = \frac{\frac{3}{50}}{\frac{2}{50}} = 3/20$.

C_k : obtain a sum of k before obtaining a 7

Now, $P(A) = \sum_{i=1}^{12} P(A \cap B_i)$. We have that $P(A \cap B_2) = P(A \cap B_3) = P(A \cap B_{12}) = 0$.

Also, $P(A \cap B_7) = P(B_7) = \frac{6}{36}$, $P(A \cap B_{11}) = P(B_{11}) = \frac{2}{36}$.

Now, $P(A \cap B_4) = P(C_4 \cap B_7) = P(C_4)P(B_7) = \frac{1}{3}(\frac{3}{36}) = \frac{3}{36}$ (using independence Ex. 119).

Similarly, $P(C_5) = P(C_9) = \frac{4}{10}$, $P(C_6) = P(C_8) = \frac{5}{11}$, and $P(C_{10}) = \frac{3}{9}$.

Thus, $P(A \cap B_5) = P(A \cap B_9) = \frac{2}{45}$, $P(A \cap B_6) = P(A \cap B_8) = \frac{25}{396}$, $P(A \cap B_{10}) = \frac{1}{36}$.

Putting all of this together, $P(A) = 0.493$.

2.139 From Ex. 1.112, $P(Y=0) = (.02)^3$ and $P(Y=3) = (.98)^3$. The event $Y=1$ are the events FDF , DFF , and FFD , each having probability $(.02)^2(.98)$. So, $P(Y=1) = 3(.02)^2(.98)$. Similarly, $P(Y=2) = 3(.02)(.98)^2$.

2.140 The total number of ways to select 3 from 6 refrigerators is $\binom{6}{3} = 20$. The total number

of ways to select y defectives and $3 - y$ nondefectives is $\binom{2}{y} \binom{4}{3-y}$, $y = 0, 1, 2$. So,

$$P(Y=0) = \frac{\binom{2}{0}\binom{4}{3}}{20} = 4/20, P(Y=1) = 4/20, \text{ and } P(Y=2) = 1/20.$$

$$DGGGD \quad GDGGD \quad GGDGD \quad GGGDD$$

Each of these possibilities has probability $1/15$, so that $P(Y = 5) = 4/15$. By the complement rule, $P(Y = 6) = 5/15$.

2.142 Each position has probability $1/4$, so every ordering of two positions (from two spins) has probability $1/16$. The values for Y are 2, 3. $P(Y = 2) = \binom{4}{2} \frac{1}{16} = 3/4$. So, $P(Y = 3) = 1/4$.

2.143 Since $P(B) = P(B \cap A) + P(B \cap \bar{A})$, $1 = \frac{P(B \cap A)}{P(B)} + \frac{P(B \cap \bar{A})}{P(B)} = P(A|B) + P(\bar{A}|B)$.

2.144 a. $S = \{16 \text{ possibilities of drawing } 0 \text{ to } 4 \text{ of the sample points}\}$

b. $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 1 + 4 + 6 + 4 + 1 = 16 = 2^4$.

c. $A \cup B = \{E_1, E_2, E_3, E_4\}$, $A \cap B = \{E_2\}$, $\bar{A} \cap \bar{B} = \emptyset$, $\bar{A} \cup B = \{E_2, E_4\}$.

2.145 All 18 orderings are possible, so the total number of orderings is $18!$

2.146 There are $\binom{52}{5}$ ways to draw 5 cards from the deck. For each suit, there are $\binom{13}{5}$ ways to select 5 cards. Since there are 4 suits, the probability is $4 \binom{13}{5} / \binom{52}{5} = 0.00248$.

2.147 The gambler will have a full house if he is dealt {two kings} or {an ace and a king} (there are 47 cards remaining in the deck, two of which are aces and three are kings).

The probabilities of these two events are $\binom{3}{2} / \binom{47}{2}$ and $\binom{3}{1} \binom{2}{1} / \binom{47}{2}$, respectively.

So, the probability of a full house is $\binom{3}{2} / \binom{47}{2} + \binom{3}{1} \binom{2}{1} / \binom{47}{2} = 0.0083$.

2.148 Note that $\binom{12}{4} = 495$. $P(\text{each supplier has at least one component tested})$ is given by

$$\frac{\binom{3}{2} \binom{4}{1} \binom{5}{1} + \binom{3}{1} \binom{4}{2} \binom{5}{1} + \binom{3}{1} \binom{4}{1} \binom{5}{2}}{495} = 270/495 = 0.545.$$

2.149 Let A be the event that the person has symptom A and define B similarly. Then

a. $P(\overline{A \cup B}) = P(\bar{A} \cap \bar{B}) = 0.4$

b. $P(A \cup B) = 1 - P(\overline{A \cup B}) = 0.6$.

c. $P(A \cap B | B) = P(A \cap B) / P(B) = .1/.4 = 0.25$

2.150 $P(Y = 0) = 0.4$, $P(Y = 1) = 0.2 + 0.3 = 0.5$, $P(Y = 2) = 0.1$.

2.151 The probability that team A wins in 5 games is $p^4(1 - p)$ and the probability that team B wins in 5 games is $p(1 - p)^4$. Since there are 4 ways that each team can win in 5 games, the probability is $4[p^4(1 - p) + p(1 - p)^4]$.

2.152 Let R denote the event that the specimen turns red and N denote the event that the specimen contains nitrates.

a. $P(R) = P(R | N)P(N) + P(R | \bar{N})P(\bar{N}) = .95(.3) + .1(.7) = 0.355$.

b. Using Bayes' rule, $P(N|R) = .95(.3)/.355 = 0.803$.

2.153 Using Bayes' rule,

$$P(I_1 | H) = \frac{P(H | I_1)P(I_1)}{P(H | I_1)P(I_1) + P(H | I_2)P(I_2) + P(H | I_3)P(I_3)} = 0.313.$$

2.154 Let Y = the number of pairs chosen. Then, the possible values are 0, 1, and 2.

a. There are $\binom{10}{4} = 210$ ways to choose 4 socks from 10 and there are $\binom{5}{4} 2^4 = 80$ ways to pick 4 non-matching socks. So, $P(Y = 0) = 80/210$.

b. Generalizing the above, the probability is $\binom{n}{2r} 2^{2r} / \binom{2n}{2r}$.

2.155 a. $P(A) = .25 + .1 + .05 + .1 = .5$

b. $P(A \cap B) = .1 + .05 = 0.15$.

c. 0.10

d. Using the result from Ex. 2.80, $\frac{.25 + .25 - .15}{.4} = 0.875$.

2.156 a. i. $1 - 5686/97900 = 0.942$ ii. $(97900 - 43354)/97900 = 0.557$
ii. $10560/14113 = 0.748$ iv. $(646+375+568)/11533 = 0.138$

b. If the US population in 2002 was known, this could be used to divide into the total number of deaths in 2002 to give a probability.

2.157 Let D denote death due to lung cancer and S denote being a smoker. Thus:

$$P(D) = P(D | S)P(S) + P(D | \bar{S})P(\bar{S}) = 10P(D | \bar{S})(.2) + P(D | \bar{S})(.8) = 0.006. \text{ Thus, } P(D | S) = 0.021.$$

- 2.158** Let W denote the event that the first ball is white and B denote the event that the second ball is black. Then:

$$P(W | B) = \frac{P(B | W)P(W)}{P(B | W)P(W) + P(B | \bar{W})P(\bar{W})} = \frac{\frac{b}{w+b+n} \left(\frac{w}{w+b} \right)}{\frac{b}{w+b+n} \left(\frac{w}{w+b} \right) + \frac{b+n}{w+b+n} \left(\frac{b}{w+b} \right)} = \frac{w}{w+b+n}$$

- 2.159** Note that $S = S \cup \emptyset$, and S and $\emptyset$ are disjoint. So, $1 = P(S) = P(S) + P(\emptyset)$ and therefore $P(\emptyset) = 0$.

- 2.160** There are 10 nondefective and 2 defective tubes that have been drawn from the machine, and number of distinct arrangements is $\binom{12}{2} = 66$.

- a. The probability of observing the specific arrangement is $1/66$.
- b. There are two such arrangements that consist of “runs.” In addition to what was given in part (a), the other is $DDNNNNNNNNNN$. Thus, the probability of two runs is $2/66 = 1/33$.

- 2.161** We must find $P(R \leq 3) = P(R = 3) + P(R = 2)$, since the minimum value for R is 2. If the two D 's occurs on consecutive trials (but not in positions 1 and 2 or 11 and 12), there are 9 such arrangements. The only other case is a defective in position 1 and 12, so that (combining with Ex. 2.160 with $R = 2$), there are 12 possibilities. So, $P(R \leq 3) = 12/66$.

- 2.162** There are $9!$ ways for the attendant to park the cars. There are $3!$ ways to park the expensive cars together and there are 7 ways the expensive cars can be next to each other in the 9 spaces. So, the probability is $7(3!)/9! = 1/12$.

- 2.163** Let A be the event that current flows in design A and let B be defined similarly. Design A will function if (1 or 2) & (3 or 4) operate. Design B will function if (1 & 3) or (2 & 4) operate. Denote the event $R_i = \{\text{relay } i \text{ operates properly}\}$, $i = 1, 2, 3, 4$. So, using independence and the addition rule,

$$P(A) = (R_1 \cup R_2) \cap (R_3 \cup R_4) = (.9 + .9 - .9^2)(.9 + .9 - .9^2) = 0.9801.$$

$$P(B) = (R_1 \cap R_3) \cup (R_2 \cap R_4) = .9^2 + .9^2 - (.9^2)^2 = .9639.$$

So, design A has the higher probability.

- 2.164** Using the notation from Ex. 2.163, $P(R_1 \cap R_4 | A) = P(R_1 \cap R_4 \cap A) / P(A)$.

Note that $R_1 \cap R_4 \cap A = R_1 \cap R_4$, since the event $R_1 \cap R_4$ represents a path for the current to flow. The probability of this above event is $.9^2 = .81$, and the conditional probability is in question is $.81/.9801 = 0.8264$.

- 2.165** Using the notation from Ex. 2.163, $P(R_1 \cap R_4 | B) = P(R_1 \cap R_4 \cap B) / P(B)$.

$R_1 \cap R_4 \cap B = (R_1 \cap R_4) \cap (R_1 \cap R_3) \cup (R_2 \cap R_4) = (R_1 \cap R_4 \cap R_3) \cup (R_2 \cap R_4)$. The probability of the above event is $.9^3 + .9^2 - .9^4 = 0.8829$. So, the conditional probability in question is $.8829/.9639 = 0.916$.

2.166 There are $\binom{8}{4} = 70$ ways to choose the tires. If the best tire the customer has is ranked #3, the other three tires are from ranks 4, 5, 6, 7, 8. There are $\binom{5}{3} = 10$ ways to select three tires from these five, so that the probability is $10/70 = 1/7$.

2.167 If $Y = 1$, the customer chose the best tire. There are $\binom{7}{3} = 35$ ways to choose the remaining tires, so $P(Y = 1) = 35/70 = .5$.

If $Y = 2$, the customer chose the second best tire. There are $\binom{6}{3} = 20$ ways to choose the remaining tires, so $P(Y = 2) = 20/70 = 2/7$. Using the same logic, $P(Y = 4) = 4/70$ and so $P(Y = 5) = 1/70$.

2.168 a. The two other tires picked by the customer must have ranks 4, 5, or 6. So, there are $\binom{3}{2} = 3$ ways to do this. So, the probability is $3/70$.

b. There are four ways the range can be 4: #1 to #5, #2 to #6, #3 to #7, and #4 to #8. Each has probability $3/70$ (as found in part **a**). So, $P(R = 4) = 12/70$.

c. Similar to parts **a** and **b**, $P(R = 3) = 5/70$, $P(R = 5) = 18/70$, $P(R = 6) = 20/70$, and $P(R = 7) = 15/70$.

2.169 a. For each beer drinker, there are $4! = 24$ ways to rank the beers. So there are $24^3 = 13,824$ total sample points.

b. In order to achieve a combined score of 4 or less, the given beer may receive at most one score of two and the rest being one. Consider brand A. If a beer drinker assigns a one to A there are still $3! = 6$ ways to rank the other brands. So, there are 6^3 ways for brand A to be assigned all ones. Similarly, brand A can be assigned two ones and one two in $3(3!)^3$ ways. Thus, some beer may earn a total rank less than or equal to four in $4[6^3 + 3(3!)^3] = 3456$ ways. So, the probability is $3456/13824 = 0.25$.

2.170 There are $\binom{7}{3} = 35$ ways to select three names from seven. If the first name on the list is included, the other two names can be picked $\binom{6}{2} = 15$ ways. So, the probability is $15/35 = 3/7$.

2.171 It is stated that the probability that Skylab will hit someone is (unconditionally) $1/150$, without regard to where that person lives. If one wants to know the probability condition on living in a certain area, it is not possible to determine.

2.172 Only $P(A|B) + P(\bar{A}|B) = 1$ is true for any events A and B .

2.173 Define the events: D : item is defective C : item goes through inspection
Thus $P(D) = .1$, $P(C|D) = .6$, and $P(C|\bar{D}) = .2$. Thus,

$$P(D|C) = \frac{P(C|D)P(D)}{P(C|D)P(D) + P(C|\bar{D})P(\bar{D})} = .25.$$

2.174 Let A = athlete disqualified previously B = athlete disqualified next term
Then, we know $P(B|\bar{A}) = .15$, $P(B|A) = .5$, $P(A) = .3$. To find $P(B)$, use the law of total probability: $P(B) = .3(.5) + .7(.15) = 0.255$.

2.175 Note that $P(A) = P(B) = P(C) = .5$. But, $P(A \cap B \cap C) = P(HH) = .25 \neq (.5)^3$. So, they are not mutually independent.

2.176 a. $P[(A \cup B) \cap C] = P[(A \cap C) \cup (B \cap C)] = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$
 $= P(A)P(C) + P(B)P(C) - P(A)P(B)P(C) = [P(A) + P(B) - P(A)P(B)]P(C)$
 $= P(A \cap B)P(C)$.

b. Similar to part a above.

2.177 a. $P(\text{no injury in 50 jumps}) = (49/50)^{50} = 0.364$.

b. $P(\text{at least one injury in 50 jumps}) = 1 - P(\text{no injury in 50 jumps}) = 0.636$.

c. $P(\text{no injury in } n \text{ jumps}) = (49/50)^n \geq 0.60$, so n is at most 25.

2.178 Define the events: E : person is exposed to the flu F : person gets the flu
Consider two employees, one of who is inoculated and one not. The probability of interest is the probability that at least one contracts the flu. Consider the complement:

$P(\text{at least one gets the flu}) = 1 - P(\text{neither employee gets the flu})$.

For the inoculated employee: $P(\bar{F}) = P(\bar{F} \cap E) + P(\bar{F} \cap \bar{E}) = .8(.6) + 1(.4) = 0.88$.

For the non-inoculated employee: $P(\bar{F}) = P(\bar{F} \cap E) + P(\bar{F} \cap \bar{E}) = .1(.6) + 1(.4) = 0.46$.

So, $P(\text{at least one gets the flu}) = 1 - .88(.46) = 0.5952$

2.179 a. The gamblers break even if each win three times and lose three times. Considering the possible sequences of “wins” and “losses”, there are $\binom{6}{3} = 20$ possible orderings. Since each has probability $(\frac{1}{2})^6$, the probability of breaking even is $20(\frac{1}{2})^6 = 0.3125$.

b. In order for this event to occur, the gambler Jones must have \$11 at trial 9 and must win on trial 10. So, in the nine remaining trials, seven “wins” and two “losses” must be placed. So, there are $\binom{9}{2} = 36$ ways to do this. However, this includes cases where

Jones would win before the 10th trial. Now, Jones can only win the game on an even trial (since he must gain \$6). Included in the 36 possibilities, there are three ways Jones could win on trial 6: *WWWWWWLL*, *WWWWWWLLW*, *WWWWWWLWL*, and there are six ways Jones could win on trial 8: *LWWWWWWWL*, *WLWWWWWWL*, *WWLWWWWWL*, *WWWLWWWWL*, *WWWWLWWWL*, *WWWWLWWLWL*. So, these nine cases must be removed from the 36. So, the probability is $27\left(\frac{1}{2}\right)^{10}$.

2.180 a. If the patrolman starts in the center of the 16x16 square grid, there are 4^8 possible paths to take. Only four of these will result in reaching the boundary. Since all possible paths are equally likely, the probability is $4/4^8 = 1/4^7$.

b. Assume the patrolman begins by walking north. There are nine possible paths that will bring him back to the starting point: *NNSS*, *NSNS*, *NSSN*, *NESW*, *NWSE*, *NWES*, *NEWS*, *NSEW*, *NSWE*. By symmetry, there are nine possible paths for each of north, south, east, and west as the starting direction. Thus, there are 36 paths in total that result in returning to the starting point. So, the probability is $36/4^8 = 9/4^7$.

2.181 We will represent the n balls as 0's and create the N boxes by placing bars (|) between the 0's. For example if there are 6 balls and 4 boxes, the arrangement

0|00||000

represents one ball in box 1, two balls in box 2, no balls in box 3, and three balls in box 4. Note that six 0's were needed but only 3 bars. In general, n 0's and $N - 1$ bars are needed to

represent each possible placement of n balls in N boxes. Thus, there are $\binom{N+n-1}{N-1}$

ways to arrange the 0's and bars. Now, if no two bars are placed next to each other, no box will be empty. So, the $N - 1$ bars must be placed in the $n - 1$ spaces between the 0's.

The total number of ways to do this is $\binom{n-1}{N-1}$, so that the probability is as given in the problem.

Chapter 3: Discrete Random Variables and Their Probability Distributions

3.1 $P(Y = 0) = P(\text{no impurities}) = .2$, $P(Y = 1) = P(\text{exactly one impurity}) = .7$, $P(Y = 2) = .1$.

3.2 We know that $P(HH) = P(TT) = P(HT) = P(TH) = 0.25$. So, $P(Y = -1) = .5$, $P(Y = 1) = .25 = P(Y = 2)$.

3.3 $p(2) = P(DD) = 1/6$, $p(3) = P(DGD) + P(GDD) = 2(2/4)(2/3)(1/2) = 2/6$, $p(4) = P(GGDD) + P(DGGD) + P(GDGD) = 3(2/4)(1/3)(2/2) = 1/2$.

3.4 Define the events: A: ~~valve~~ 1 fails B: valve 2 fails C: valve 3 fails
 $P(Y = 2) = P(\bar{A} \cap \bar{B} \cap \bar{C}) = .8^3 = 0.512$
 $P(Y = 0) = P(A \cap (B \cup C)) = P(A)P(B \cup C) = .2(.2 + .2 - .2^2) = 0.072$.
Thus, $P(Y = 1) = 1 - .512 - .072 = 0.416$.

3.5 There are $3! = 6$ possible ways to assign the words to the pictures. Of these, one is a perfect match, three have one match, and two have zero matches. Thus,
 $p(0) = 2/6$, $p(1) = 3/6$, $p(3) = 1/6$.

3.6 There are $\binom{5}{2} = 10$ sample points, and all are equally likely: (1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5).
a. $p(2) = .1$, $p(3) = .2$, $p(4) = .3$, $p(5) = .4$.

b. $p(3) = .1$, $p(4) = .1$, $p(5) = .2$, $p(6) = .2$, $p(7) = .2$, $p(8) = .1$, $p(9) = .1$.

3.7 There are $3^3 = 27$ ways to place the three balls into the three bowls. Let $Y = \#$ of empty bowls. Then:
 $p(0) = P(\text{no bowls are empty}) = \frac{3!}{27} = \frac{6}{27}$
 $p(2) = P(2 \text{ bowls are empty}) = \frac{3}{27}$
 $p(1) = P(1 \text{ bowl is empty}) = 1 - \frac{6}{27} - \frac{3}{27} = \frac{18}{27}$.

3.8 Note that the number of cells cannot be odd.
 $p(0) = P(\text{no cells in the next generation}) = P(\text{the first cell dies or the first cell splits and both die}) = .1 + .9(.1)(.1) = 0.109$
 $p(4) = P(\text{four cells in the next generation}) = P(\text{the first cell splits and both created cells split}) = .9(.9)(.9) = 0.729$.
 $p(2) = 1 - .109 - .729 = 0.162$.

3.9 The random variable Y takes on values 0, 1, 2, and 3.
a. Let E denote an error on a single entry and let N denote no error. There are 8 sample points: EEE , EEN , ENE , NEE , ENN , NEN , NNE , NNN . With $P(E) = .05$ and $P(N) = .95$ and assuming independence:
 $P(Y = 3) = (.05)^3 = 0.000125$ $P(Y = 2) = 3(.05)^2(.95) = 0.007125$
 $P(Y = 1) = 3(.05)(.95)^2 = 0.135375$ $P(Y = 0) = (.95)^3 = 0.857375$.

- b. The graph is omitted.
c. $P(Y > 1) = P(Y = 2) + P(Y = 3) = 0.00725$.

- 3.10** Denote R as the event a rental occurs on a given day and N denotes no rental. Thus, the sequence of interest is $RR, RNR, RNNR, RNNNR, \dots$. Consider the position immediately following the first R : it is filled by an R with probability .2 and by an N with probability .8. Thus, $P(Y = 0) = .2$, $P(Y = 1) = .8(.2) = .16$, $P(Y = 2) = .128$, $\dots$. In general,
$$P(Y = y) = .2(.8)^y, y = 0, 1, 2, \dots$$

- 3.11** There is a $1/3$ chance a person has O^+ blood and $2/3$ they do not. Similarly, there is a $1/15$ chance a person has O^- blood and $14/15$ chance they do not. Assuming the donors are randomly selected, if $X = \#$ of O^+ blood donors and $Y = \#$ of O^- blood donors, the probability distributions are

	0	1	2	3
$p(x)$	$(2/3)^3 = 8/27$	$3(2/3)^2(1/3) = 12/27$	$3(2/3)(1/3)^2 = 6/27$	$(1/3)^3 = 1/27$
$p(y)$	$2744/3375$	$192/375$	$16/375$	$1/3375$

Note that $Z = X + Y = \#$ will type O blood. The probability a donor will have type O blood is $1/3 + 1/15 = 6/15 = 2/5$. The probability distribution for Z is

	0	1	2	3
$p(z)$	$(2/5)^3 = 27/125$	$2(2/5)^2(3/5) = 54/125$	$2(2/5)(3/5)^2 = 36/125$	$(3/5)^3 = 27/125$

- 3.12** $E(Y) = 1(.4) + 2(.3) + 3(.2) + 4(.1) = 2.0$
 $E(1/Y) = 1(.4) + 1/2(.3) + 1/3(.2) + 1/4(.1) = 0.6417$
 $E(Y^2 - 1) = E(Y^2) - 1 = [1(.4) + 2^2(.3) + 3^2(.2) + 4^2(.1)] - 1 = 5 - 1 = 4$.
 $V(Y) = E(Y^2) - [E(Y)]^2 = 5 - 2^2 = 1$.

- 3.13** $E(Y) = -1(1/2) + 1(1/4) + 2(1/4) = 1/4$
 $E(Y^2) = (-1)^2(1/2) + 1^2(1/4) + 2^2(1/4) = 7/4$
 $V(Y) = 7/4 - (1/4)^2 = 27/16$.
 Let $C = \text{cost of play}$, then the net winnings is $Y - C$. If $E(Y - C) = 0$, $C = 1/4$.

- 3.14** a. $\mu = E(Y) = 3(.03) + 4(.05) + 5(.07) + \dots + 13(.01) = 7.9$
 b. $\sigma^2 = V(Y) = E(Y^2) - [E(Y)]^2 = 3^2(.03) + 4^2(.05) + 5^2(.07) + \dots + 13^2(.01) - 7.9^2 = 67.14 - 62.41 = 4.73$. So, $\sigma = 2.17$.
 c. $(\mu - 2\sigma, \mu + 2\sigma) = (3.56, 12.24)$. So, $P(3.56 < Y < 12.24) = P(4 \leq Y \leq 12) = .05 + .07 + .10 + .14 + .20 + .18 + .12 + .07 + .03 = 0.96$.

- 3.15** a. $p(0) = P(Y = 0) = (.48)^3 = .1106$, $p(1) = P(Y = 1) = 3(.48)^2(.52) = .3594$, $p(2) = P(Y = 2) = 3(.48)(.52)^2 = .3894$, $p(3) = P(Y = 3) = (.52)^3 = .1406$.
 b. The graph is omitted.
 c. $P(Y = 1) = .3594$.

d. $\mu = E(Y) = 0(.1106) + 1(.3594) + 2(.3894) + 3(.1406) = 1.56$,
 $\sigma^2 = V(Y) = E(Y^2) - [E(Y)]^2 = 0^2(.1106) + 1^2(.3594) + 2^2(.3894) + 3^2(.1406) - 1.56^2 = 3.1824 - 2.4336 = .7488$. So, $\sigma = 0.8653$.

e. $(\mu - 2\sigma, \mu + 2\sigma) = (-.1706, 3.2906)$. So, $P(-.1706 < Y < 3.2906) = P(0 \leq Y \leq 3) = 1$.

3.16 As shown in Ex. 2.121, $P(Y = y) = 1/n$ for $y = 1, 2, \dots, n$. Thus, $E(Y) = \frac{1}{n} \sum_{y=1}^n y = \frac{n+1}{2}$.

$$E(Y^2) = \frac{1}{n} \sum_{y=1}^n y^2 = \frac{(n+1)(2n+1)}{6}. \text{ So, } V(Y) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}.$$

3.17 $\mu = E(Y) = 0(6/27) + 1(18/27) + 2(3/27) = 24/27 = .889$
 $\sigma^2 = V(Y) = E(Y^2) - [E(Y)]^2 = 0^2(6/27) + 1^2(18/27) + 2^2(3/27) - (24/27)^2 = 30/27 - 576/729 = .321$. So, $\sigma = 0.567$

For $(\mu - 2\sigma, \mu + 2\sigma) = (-.245, 2.023)$. So, $P(-.245 < Y < 2.023) = P(0 \leq Y \leq 2) = 1$.

3.18 $\mu = E(Y) = 0(.109) + 2(.162) + 4(.729) = 3.24$.

3.19 Let P be a random variable that represents the company's profit. Then, $P = C - 15$ with probability $98/100$ and $P = C - 15 - 1000$ with probability $2/100$. Then,
 $E(P) = (C - 15)(98/100) + (C - 15 - 1000)(2/100) = 50$. Thus, $C = \$85$.

3.20 With probability .3 the volume is $8(10)(30) = 2400$. With probability .7 the volume is $8*10*40 = 3200$. Then, the mean is $.3(2400) + .7(3200) = 2960$.

3.21 Note that $E(N) = E(8\pi R^2) = 8\pi E(R^2)$. So, $E(R^2) = 21^2(.05) + 22^2(.20) + \dots + 26^2(.05) = 549.1$. Therefore $E(N) = 8\pi(549.1) = 13,800.388$.

3.22 Note that $p(y) = P(Y = y) = 1/6$ for $y = 1, 2, \dots, 6$. ~~This is similar to Ex. 3.16 with $n = 6$.~~
 So, $E(Y) = 3.5$ and $V(Y) = 2.9167$.

3.23 Define G to be the gain to a person in drawing one card. The possible values for G are \$15, \$5, or \$-4 with probabilities $\frac{1}{13}$, $2/13$, and $9/13$ respectively. So,
 $E(G) = 15(\frac{1}{13}) + 5(2/13) - 4(9/13) = 4/13$ (roughly \$.31).

3.24 The probability distribution for $Y =$ number of bottles with serious flaws is:

$p(y)$	0	1	2
y	.81	.18	.01

Thus, $E(Y) = 0(.81) + 1(.18) + 2(.01) = 0.20$ and $V(Y) = 0^2(.81) + 1^2(.18) + 2^2(.01) - (.20)^2 = 0.18$.

3.25 Let $X_1 = \#$ of contracts assigned to firm 1; $X_2 = \#$ of contracts assigned to firm 2. The sample space for the experiment is $\{(I,I), (I,II), (I,III), (II,I), (II,II), (II,III), (III,I), (III,II), (III,III)\}$, each with probability $1/9$. So, the probability distributions for X_1 and X_2 are:

x_1	0	1	2
$p(x_1)$	4/9	4/9	1/9

x_2	0	1	2
$p(x_2)$	4/9	4/9	1/9

Thus, $E(X_1) = E(X_2) = 2/3$. The expected profit for the owner of both firms is given by $90000(2/3 + 2/3) = \$120,000$.

3.26 The random variable Y = daily sales can have values \$0, \$50,000 and \$100,000.

If $Y = 0$, either the salesperson contacted only one customer and failed to make a sale or the salesperson contacted two customers and failed to make both sales. Thus $P(Y = 0) = 1/3(9/10) + 2/3(9/10)(9/10) = 252/300$.

If $Y = 2$, the salesperson contacted to customers and made both sales. So, $P(Y = 2) = 2/3(1/10)(1/10) = 2/300$.

Therefore, $P(Y = 1) = 1 - 252/300 - 2/300 = 46/300$.

Then, $E(Y) = 0(252/300) + 50000(46/300) + 100000(2/300) = 25000/3$ (or \$8333.33).

$V(Y) = 380,561,111$ and $\sigma = \$19,507.98$.

3.27 Let Y = the payout on an individual policy. Then, $P(Y = 85,000) = .001$, $P(Y = 42,500) = .01$, and $P(Y = 0) = .989$. Let C represent the premium the insurance company charges. Then, the company's net gain/loss is given by $C - Y$. If $E(C - Y) = 0$, $E(Y) = C$. Thus, $E(Y) = 85000(.001) + 42500(.01) + 0(.989) = 510 = C$.

3.28 Using the probability distribution found in Ex. 3.3, $E(Y) = 2(1/6) + 3(2/6) + 4(3/6) = 20/6$. The cost for testing and repairing is given by $2Y + 4$. So, $E(2Y + 4) = 2(20/6) + 4 = 64/6$.

3.29
$$\sum_{k=1}^{\infty} P(Y \geq k) = \sum_{k=1}^{\infty} \sum_{j=k}^{\infty} P(Y=j) = \sum_{k=1}^{\infty} \sum_{j=k}^{\infty} p(j) = \sum_{j=1}^{\infty} \sum_{k=1}^j p(j) = \sum_{j=1}^{\infty} j \cdot p(j) = \sum_{y=1}^{\infty} y \cdot p(y) = E(Y).$$

3.30 a. The mean of X will be larger than the mean of Y .

b. $E(X) = E(Y + 1) = E(Y) + 1 = \mu + 1$.

c. The variances of X and Y will be the same (the addition of 1 doesn't affect variability).

d. $V(X) = E[(X - E(X))^2] = E[(Y + 1 - \mu - 1)^2] = E[(Y - \mu)^2] = \sigma^2$.

3.31 a. The mean of W will be larger than the mean of Y if $\mu > 0$. If $\mu < 0$, the mean of W will be smaller than μ . If $\mu = 0$, the mean of W will equal μ .

b. $E(W) = E(2Y) = 2E(Y) = 2\mu$.

c. The variance of W will be larger than σ^2 , since the spread of values of W has increased.

d. $V(W) = E[(W - E(W))^2] = E[(2Y - 2\mu)^2] = 4E[(Y - \mu)^2] = 4\sigma^2$.

3.32 a. The mean of W will be smaller than the mean of Y if $\mu > 0$. If $\mu < 0$, the mean of W will be larger than μ . If $\mu = 0$, the mean of W will equal μ .

b. $E(W) = E(Y/10) = (.1)E(Y) = (.1)\mu$.

c. The variance of W will be smaller than σ^2 , since the spread of values of W has decreased.

d. $V(W) = E[(W - E(W))^2] = E[(.1Y - .1\mu)^2] = (.01)E[(Y - \mu)^2] = (.01)\sigma^2$.

- 3.33** a. $E(aY + b) = E(aY) + E(b) = aE(Y) + b = a\mu + b$.
 b. $V(aY + b) = E[(aY + b - a\mu - b)^2] = E[(aY - a\mu)^2] = a^2 E[(Y - \mu)^2] = a^2 \sigma^2$.
- 3.34** The mean cost is $E(10Y) = 10E(Y) = 10[0(.1) + 1(.5) + 2(.4)] = \13 . Since $V(Y) = .41$, $V(10Y) = 100V(Y) = 100(.41) = 41$.
- 3.35** With $B = SS \cup FS$, $P(B) = P(SS) + P(FS) = \frac{2000}{5000} \left(\frac{1999}{4999} \right) + \frac{3000}{5000} \left(\frac{2000}{4999} \right) = 0.4$
 $P(B|\text{first trial success}) = \frac{1999}{4999} = 0.3999$, which is not very different from the above.
- 3.36** a. The random variable Y does not have a binomial distribution. The days are not independent.
 b. This is not a binomial experiment. The number of trials is not fixed.
- 3.37** a. Not a binomial random variable.
 b. Not a binomial random variable.
 c. Binomial with $n = 100$, $p =$ proportion of high school students who scored above 1026.
 d. Not a binomial random variable (not discrete).
 e. Not binomial, since the sample was not selected among all female HS grads.
- 3.38** Note that Y is binomial with $n = 4$, $p = 1/3 = P(\text{judge chooses formula } B)$.
 a. $p(y) = \binom{4}{y} \left(\frac{1}{3} \right)^y \left(\frac{2}{3} \right)^{4-y}$, $y = 0, 1, 2, 3, 4$.
 b. $P(Y \geq 3) = p(3) + p(4) = 8/81 + 1/81 = 9/81 = 1/9$.
 c. $E(Y) = 4(1/3) = 4/3$.
 d. $V(Y) = 4(1/3)(2/3) = 8/9$
- 3.39** Let $Y = \#$ of components failing in less than 1000 hours. Then, Y is binomial with $n = 4$ and $p = .2$.
 a. $P(Y = 2) = \binom{4}{2} .2^2 (.8)^2 = 0.1536$.
 b. The system will operate if 0, 1, or 2 components fail in less than 1000 hours. So,
 $P(\text{system operates}) = .4096 + .4096 + .1536 = .9728$.
- 3.40** Let $Y = \#$ that recover from stomach disease. Then, Y is binomial with $n = 20$ and $p = .8$. To find these probabilities, Table 1 in Appendix III will be used. 
 $P(Y \geq 10) = 1 - P(Y \leq 9) = 1 - .001 = .999$.
 $P(14 \leq Y \leq 18) = P(Y \leq 18) - P(Y \leq 13) = .931 - .087 = .844$
 $P(Y \leq 16) = .589$.
- 3.41** Let $Y = \#$ of correct answers. Then, Y is binomial with $n = 15$ and $p = .2$. Using Table 1 in Appendix III, $P(Y \geq 10) = 1 - P(Y \leq 9) = 1 - 1.000 = 0.000$ (to three decimal places).

3.42 a. If one answer can be eliminated on every problem, then, Y is binomial with $n = 15$ and $p = .25$. Then, $P(Y \geq 10) = 1 - P(Y \leq 9) = 1 - 1.000 = 0.000$ (to three decimal places).

b. If two answers can be (correctly) eliminated on every problem, then, Y is binomial with $n = 15$ and $p = 1/3$. Then, $P(Y \geq 10) = 1 - P(Y \leq 9) = 0.0085$.

3.43 Let $Y = \#$ of qualifying subscribers. Then, Y is binomial with $n = 5$ and $p = .7$.

a. $P(Y = 5) = .7^5 = .1681$

b. $P(Y \geq 4) = P(Y = 4) + P(Y = 5) = 5(.7^4)(.3) + .7^5 = .3601 + .1681 = 0.5282$.

3.44 Let $Y = \#$ of successful operations. Then Y is binomial with $n = 5$.

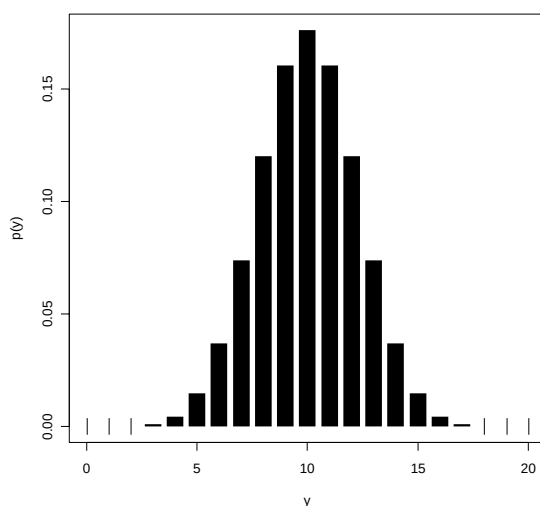
a. With $p = .8$, $P(Y = 5) = .8^5 = 0.328$.

b. With $p = .6$, $P(Y = 4) = 5(.6^4)(.4) = 0.259$.

c. With $p = .3$, $P(Y < 2) = P(Y = 1) + P(Y = 0) = 0.528$.

3.45 Note that Y is binomial with $n = 3$ and $p = .8$. The alarm will function if $Y = 1, 2$, or 3 . Thus, $P(Y \geq 1) = 1 - P(Y = 0) = 1 - .008 = 0.992$.

3.46 When $p = .5$, the distribution is symmetric. When $p < .5$, the distribution is skewed to the left. When $p > .5$, the distribution is skewed to the right.



3.47 The graph is above.

3.48 a. Let $Y = \#$ of sets that detect the missile. Then, Y has a binomial distribution with $n = 5$ and $p = .9$. Then, $P(Y = 4) = 5(.9)^4(.1) = 0.32805$ and

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - 5(.9)^4(.1) = 0.32805.$$

b. With n radar sets, the probability of at least one detection is $1 - (.1)^n$. If $1 - (.1)^n = .999$, $n = 3$.

3.49 Let $Y = \#$ of housewives preferring brand A. Thus, Y is binomial with $n = 15$ and $p = .5$.

a. Using the Appendix, $P(Y \geq 10) = 1 - P(Y \leq 9) = 1 - .849 = 0.151$.

b. $P(10 \text{ or more prefer A or B}) = P(6 \leq Y \leq 9) = 0.302$.

- 3.50** The only way team A can win in exactly 5 games is to win 3 in the first 4 games and then win the 5th game. Let $Y = \#$ of games team A wins in the first 4 games. Thus, Y has a binomial distribution with $n = 4$. Thus, the desired probability is given by

$$\begin{aligned} P(\text{Team A wins in 5 games}) &= P(Y = 3)P(\text{Team A wins game 5}) \\ &= \binom{4}{3} p^3 (1-p) p = 4p^4 (1-p). \end{aligned}$$

- 3.51 a.** $P(\text{at least one 6 in four rolls}) = 1 - P(\text{no 6's in four rolls}) = 1 - (5/6)^4 = 0.51775$.

b. Note that in a single toss of two dice, $P(\text{double 6}) = 1/36$. Then:
 $P(\text{at least one double 6 in twenty-four rolls}) = 1 - P(\text{no double 6's in twenty-four rolls}) = 1 - (35/36)^{24} = 0.4914$.

- 3.52** Let $Y = \#$ that are tasters. Then, Y is binomial with $n = 20$ and $p = .7$.

a. $P(Y \geq 17) = 1 - P(Y \leq 16) = 0.107$.

b. $P(Y < 15) = P(Y \leq 14) = 0.584$.

- 3.53** There is a 25% chance the offspring of the parents will develop the disease. Then, $Y = \#$ of offspring that develop the disease is binomial with $n = 3$ and $p = .25$.

a. $P(Y = 3) = (.25)^3 = 0.015625$.

b. $P(Y = 1) = 3(.25)(.75)^2 = 0.421875$

c. Since the pregnancies are mutually independent, the probability is simply 25%.

- 3.54 a. and b.** follow from simple substitution

c. the classifications of "success" and "failure" are arbitrary.

$$\begin{aligned} \mathbf{3.55} \quad E\{Y(Y-1)(Y-2)\} &= \sum_{y=0}^n \frac{y(y-1)(y-2)n!}{y!(n-y)!} p^y (1-p)^{n-y} = \sum_{y=3}^n \frac{n(n-1)(n-2)(n-3)!}{(y-3)!(n-3-(y-3))!} p^y (1-p)^{n-y} \\ &= n(n-1)(n-2)p^3 \sum_{z=0}^{n-3} \binom{n-3}{z} p^z (1-p)^{n-3-z} = n(n-1)(n-2)p^3. \end{aligned}$$

Equating this to $E(Y^3) - 3E(Y^2) + 2E(Y)$, it is found that

$$E(Y^3) = 3n(n-1)p^2 - n(n-1)(n-2)p^3 + np.$$

- 3.56** Using expression for the mean and variance of $Y = \#$ of successful explorations, a binomial random variable with $n = 10$ and $p = .1$, $E(Y) = 10(.1) = 1$, and $V(Y) = 10(.1)(.9) = 0.9$.

- 3.57** If $Y = \#$ of successful explorations, then $10 - Y$ is the number of unsuccessful explorations. Hence, the cost C is given by $C = 20,000 + 30,000Y + 15,000(10 - Y)$. Therefore, $E(C) = 20,000 + 30,000(1) + 15,000(10 - 1) = \$185,000$.

- 3.58** If Y is binomial with $n = 4$ and $p = .1$, $E(Y) = .4$ and $V(Y) = .36$. Thus, $E(Y^2) = .36 + (.4)^2 = 0.52$. Therefore, $E(C) = 3(.52) + (.36) + 2 = 3.96$.

- 3.59** If $Y = \#$ of defective motors, then Y is binomial with $n = 10$ and $p = .08$. Then, $E(Y) = .8$. The seller's expected next gain is $\$1000 - \$200E(Y) = \$840$.
- 3.60** Let $Y = \#$ of fish that survive. Then, Y is binomial with $n = 20$ and $p = .8$.
- $P(Y = 14) = .109$.
 - $P(Y \geq 10) = .999$.
 - $P(Y \leq 16) = .589$.
 - $\mu = 20(.8) = 16$, $\sigma^2 = 20(.8)(.2) = 3.2$.
- 3.61** Let $Y = \#$ with Rh⁺ blood. Then, Y is binomial with $n = 5$ and $p = .8$
- $1 - P(Y = 5) = .672$.
 - $P(Y \leq 4) = .672$.
 - We need n for which $P(Y \geq 5) = 1 - P(Y \leq 4) > .9$. The smallest n is 8.
- 3.62**
- Assume independence of the three inspection events.
 - Let $Y = \#$ of plane with wing cracks that are detected. Then, Y is binomial with $n = 3$ and $p = .9(.8)(.5) = .36$. Then, $P(Y \geq 1) = 1 - P(Y = 0) = 0.737856$.
- 3.63**
- Found by pulling in the formula for $p(y)$ and $p(y - 1)$ and simplifying.
 - Note that $P(Y < 3) = P(Y \leq 2) = P(Y = 2) + P(Y = 1) + P(Y = 0)$. Now, $P(Y = 0) = (.96)^{90} = .0254$. Then, $P(Y = 1) = \frac{(90-1+1)!.04}{1!.96}(.0254) = .0952$ and $P(Y = 2) = \frac{(90-2+1)!.04}{2!.96}(.0952) = .1765$. Thus, $P(Y < 3) = .0254 + .0952 + .1765 = 0.2971$
 - $\frac{(n-y+1)}{yq} > 1$ is equivalent to $(n+1)p - yp > yq$ is equivalent to $(n+1)p > y$. The others are similar.
 - Since for $y \leq (n+1)p$, then $p(y) \geq p(y-1) > p(y-2) > \dots$. Also, for $y \geq (n+1)p$, then $p(y) \geq p(y+1) > p(y+2) > \dots$. It is clear that $p(y)$ is maximized when y is as close to $(n+1)p$ as possible.
- 3.64** To maximize the probability distribution as a function of p , consider taking the natural log (since $\ln()$ is a strictly increasing function, it will not change the maximum). By taking the first derivative of $\ln[p(y_0)]$ and setting it equal to 0, the maximum is found to be y_0/n .
- 3.65**
- $E(Y/n) = E(Y)/n = np/n = p$.
 - $V(Y/n) = V(Y)/n^2 = npq/n^2 = pq/n$. This quantity goes to zero as n goes to infinity.
- 3.66**
- $\sum_{y=1}^{\infty} q^{y-1} p = p \sum_{x=0}^{\infty} q^x = p \frac{1}{1-q} = 1$ (infinite sum of a geometric series)
 - $\frac{q^{y-1} p}{q^{y-2} p} = q$. The event $Y = 1$ has the highest probability for all p , $0 < p < 1$.

3.67 $(.7)^4(.3) = 0.07203.$

3.68 $1/(.30) = 3.33.$

3.69 Y is geometric with $p = 1 - .41 = .59$. Thus, $p(y) = (.41)^{y-1}(.59), y = 1, 2, \dots$

3.70 Let $Y = \#$ of holes drilled until a productive well is found.

a. $P(Y = 3) = (.8)^2(.2) = .128$

b. $P(Y > 10) = P(\text{first 10 are not productive}) = (.8)^{10} = .107.$

3.71 a. $P(Y > a) = \sum_{y=a+1}^{\infty} q^{y-1} p = q^a \sum_{x=1}^{\infty} q^{x-1} p = q^a.$

b. From part a, $P(Y > a + b | Y > a) = \frac{P(Y > a + b, Y > a)}{P(Y > a)} = \frac{P(Y > a + b)}{P(Y > a)} = \frac{q^{a+b}}{q^a} = q^b.$

c. The results in the past are not relevant to a future outcome (independent trials).

3.72 Let $Y = \#$ of tosses until the first head. $P(Y \geq 12 | Y > 10) = P(Y > 11 | Y > 10) = 1/2.$

3.73 Let $Y = \#$ of accounts audited until the first with substantial errors is found.

a. $P(Y = 3) = .1^2(.9) = .009.$

b. $P(Y \geq 3) = P(Y > 2) = .1^2 = .01.$

3.74 $\mu = 1/.9 = 1.1, \sigma = \sqrt{\frac{1-.9}{.9^2}} = .35$

3.75 Let $Y = \#$ of one second intervals until the first arrival, so that $p = .1$

a. $P(Y = 3) = (.9)^2(.1) = .081.$

b. $P(Y \geq 3) = P(Y > 2) = .9^2 = .81.$

3.76 $P(Y > y_0) = (.7)^{y_0} \geq .1$. Thus, $y_0 \leq \frac{\ln(.1)}{\ln(.7)} = 6.46$, so $y_0 \leq 6$.

3.77 $P(Y = 1, 3, 5, \dots) = P(Y = 1) + P(Y = 3) + P(Y = 5) + \dots = p + q^2p + q^4p + \dots =$
 $p[1 + q^2 + q^4 + \dots] = p \frac{1}{1 - q^2}.$ (Sum an infinite geometric series in $(q^2)^k$.)

3.78 a. $(.4)^4(.6) = .01536.$

b. $(.4)^4 = .0256.$

3.79 Let $Y = \#$ of people questioned before a “yes” answer is given. Then, Y has a geometric distribution with $p = P(\text{yes}) = P(\text{smoker and “yes”}) + P(\text{nonsmoker and “yes”}) = .3(.2) + 0 = .06$. Thus, $p(y) = .06(.94)^{y-1}, y = 1, 2, \dots$

- 3.80** Let $Y = \#$ of tosses until the first 6 appears, so Y has a geometric distribution. Using the result from Ex. 3.77,

$$P(\text{B tosses first 6}) = P(Y = 2, 4, 6, \dots) = 1 - P(Y = 1, 3, 5, \dots) = 1 - p \frac{1}{1 - q^2}.$$

Since $p = 1/6$, $P(\text{B tosses first 6}) = 5/11$. Then,

$$P(Y = 4 \mid \text{B tosses the first 6}) = \frac{P(Y = 4)}{5/11} = \frac{(5/6)^2(1/6)}{5/11} = 275/1296.$$

- 3.81** With $p = 1/2$, then $\mu = 1/(1/2) = 2$.

- 3.82** With $p = .2$, then $\mu = 1/(.2) = 5$. The 5th attempt is the expected first successful well.

- 3.83** Let $Y = \#$ of trials until the correct password is picked. Then, Y has a geometric distribution with $p = 1/n$. $P(Y = 6) = \frac{1}{n} \left(\frac{n-1}{n} \right)^5$.

- 3.84** $E(Y) = n$, $V(Y) = (1 - \frac{1}{n})n^2 = n(n-1)$.

- 3.85** Note that $\frac{d^2}{dq^2} q^y = y(y-1)q^{y-2}$. Thus, $\frac{d^2}{dq^2} \sum_{y=2}^{\infty} q^y = \sum_{y=2}^{\infty} y(y-1)q^{y-2}$. Thus,

$$E[Y(Y-1)] = \sum_{y=1}^{\infty} y(y-1)q^{y-1} = pq \sum_{y=1}^{\infty} y(y-1)q^{y-2} = pq \frac{d^2}{dq^2} \sum_{y=2}^{\infty} q^y = pq \frac{d^2}{dq^2} \left\{ \frac{1}{1-q} - 1 - q \right\} = \frac{2pq}{(1-q)^3} = \frac{2q}{p^2}.$$

Use this with $V(Y) = E[Y(Y-1)] + E(Y) - [E(Y)]^2$.

- 3.86** $P(Y = y_0) = q^{y_0-1} p$. Like Ex. 3.64, maximize this probability by first taking the natural log.

- 3.87** $E(1/Y) = \sum_{y=1}^{\infty} \frac{1}{y} (1-p)^{y-1} p = \frac{p}{1-p} \sum_{y=1}^{\infty} \frac{(1-p)^y}{y} = -\frac{p \ln(p)}{1-p}.$

- 3.88** $P(Y^* = y) = P(Y = y+1) = q^{y+1-1} p = q^y p, y = 0, 1, 2, \dots$

- 3.89** $E(Y^*) = E(Y) - 1 = \frac{1}{p} - 1$. $V(Y^*) = V(Y - 1) = V(Y)$.

- 3.90** Let $Y = \#$ of employees tested until three positives are found. Then, Y is negative binomial with $r = 3$ and $p = .4$. $P(Y = 10) = \binom{9}{2} .4^3 (.6)^7 = .06$.

- 3.91** The total cost is given by $20Y$. So, $E(20Y) = 20E(Y) = 20 \cdot \frac{3}{.4} = \150 . Similarly, $V(20Y) = 400V(Y) = 4500$.

3.92 Let $Y = \#$ of trials until this first non-defective engine is found. Then, Y is geometric with $p = .9$. $P(Y = 2) = .9(.1) = .09$.

3.93 From Ex. 3.92:

a. $P(Y = 5) = \binom{4}{2} (.9)^3 (.1)^2 = .04374$.

b. $P(Y \leq 5) = P(Y = 3) + P(Y = 4) + P(Y = 5) = .729 + .2187 + .04374 = .99144$.

3.94 a. $\mu = 1/(.9) = 1.11$, $\sigma^2 = (.1)/(.9)^2 = .1234$.

b. $\mu = 3/(.9) = 3.33$, $\sigma^2 = 3(.1)/(.9)^2 = .3704$.

3.95 From Ex. 3.92 (and the memory-less property of the geometric distribution),
 $P(Y \geq 4 | Y > 2) = P(Y > 3 | Y > 2) = P(Y > 1) = 1 - P(Y = 0) = .1$.

3.96 a. Let $Y = \#$ of attempts until you complete your call. Thus, Y is geometric with $p = .4$. Thus, $P(Y = 1) = .4$, $P(Y = 2) = (.6).4 = .24$, $P(Y = 3) = (.6)^2 .4 = .144$.

b. Let $Y = \#$ of attempts until both calls are completed. Thus, Y is negative binomial with $r = 2$ and $p = .4$. Thus, $P(Y = 4) = 3(.4)^2 (.6)^2 = .1728$.

3.97 a. Geometric probability calculation: $(.8)^2 (.2) = .128$.

b. Negative binomial probability calculation: $\binom{6}{2} (.2)^3 (.8)^4 = .049$.

c. The trials are independent and the probability of success is the same from trial to trial.

d. $\mu = 3/.2 = 15$, $\sigma^2 = 3(.8)/(.04) = 60$.

3.98 a. $\frac{p(y)}{p(y-1)} = \frac{\frac{(y-1)!}{(r-1)!(y-r)!} p^r q^{y-r}}{\frac{(y-2)!}{(r-1)!(y-1-r)!} p^r q^{y-1-r}} = \frac{y-1}{y-r} q$

b. If $\frac{y-1}{y-r} q > 1$, then $yq - q > y - r$ or equivalently $\frac{r-q}{1-q} > y$. The 2nd result is similar.

c. If $r = 7$, $p = .5 = q$, then $\frac{r-q}{1-q} = \frac{7-.5}{1-.5} = 13 > y$.

3.99 Define a random variable $X = y$ trials before the first success, $y = r - 1, r, r + 1, \dots$. Then, $X = Y - 1$, where Y has the negative binomial distribution with parameters r and p . Thus, $p(x) = \frac{y!}{(r-1)!(y-r+1)!} p^r q^{y+1-r}$, $y = r - 1, r, r + 1, \dots$.

3.100 a. $P(Y^* = y) = P(Y = y + r) = \binom{y+r-1}{r-1} p^r q^{y+r-r} = \binom{y+r-1}{r-1} p^r q^y$, $y = 0, 1, 2, \dots$.

b. $E(Y^*) = E(Y) - r = r/p - r = r/q$, $V(Y^*) = V(Y - r) = V(Y)$.

3.101 a. Note that $P(Y = 11) = \binom{10}{4} p^5 (1-p)^6$. Like Ex. 3.64 and 3.86, maximize this

probability by first taking the natural log. The maximum is 5/11.

b. In general, the maximum is r/y_0 .

3.102 Let $Y = \#$ of green marbles chosen in three draws. Then, $P(Y = 3) = \frac{\binom{5}{3}}{\binom{10}{3}} = 1/12$.

3.103 Use the hypergeometric probability distribution with $N = 10$, $r = 4$, $n = 5$. $P(Y = 0) = \frac{1}{42}$.

3.104 Define the events: A : 1st four selected packets contain cocaine
 B : 2nd two selected packets do not contain cocaine

Then, the desired probability is $P(A \cap B) = P(B|A)P(A)$. So,

$$P(A) = \frac{\binom{15}{4}}{\binom{20}{4}} = .2817 \text{ and } P(B|A) = \frac{\binom{5}{2}}{\binom{16}{2}} = .0833. \text{ Thus,}$$

$$P(A \cap B) = .2817(.0833) = 0.0235.$$

3.105 a. The random variable Y follows a hypergeometric distribution. The probability of being chosen on a trial is dependent on the outcome of previous trials.

$$\mathbf{b.} \ P(Y \geq 2) = P(Y = 2) + P(Y = 3) = \frac{\binom{5}{2}\binom{3}{1}}{\binom{8}{3}} + \frac{\binom{5}{3}}{\binom{8}{3}} = .5357 + .1786 = 0.7143.$$

$$\mathbf{c.} \ \mu = 3(5/8) = 1.875, \sigma^2 = 3(5/8)(3/8)(5/7) = .5022, \text{ so } \sigma = .7087.$$

3.106 Using the results from Ex.103, $E(50Y) = 50E(Y) = 50[5(\frac{4}{10})] = \100 . Furthermore, $V(50Y) = 2500V(Y) = 2500[5(\frac{4}{10})(\frac{6}{10})(\frac{5}{9})] = 1666.67$.

3.107 The random variable Y follows a hypergeometric distribution with $N = 6$, $n = 2$, and $r = 4$.

3.108 Use the fact that $P(\text{at least one is defective}) = 1 - P(\text{none are defective})$. Then, we require $P(\text{none are defective}) \leq .2$. If $n = 8$,

$$P(\text{none are defective}) = \left(\frac{17}{20}\right)\left(\frac{16}{19}\right)\left(\frac{15}{18}\right)\left(\frac{14}{17}\right)\left(\frac{13}{16}\right)\left(\frac{12}{15}\right)\left(\frac{11}{14}\right)\left(\frac{10}{13}\right) = 0.193.$$

3.109 Let $Y = \#$ of treated seeds selected.

$$\mathbf{a.} \ P(Y = 4) = \frac{\binom{5}{4}\binom{5}{0}}{\binom{10}{4}} = .0238$$

$$\mathbf{b.} \ P(Y \leq 3) = 1 - P(Y = 4) = 1 - \frac{\binom{5}{4}\binom{5}{0}}{\binom{10}{4}} = 1 - .0238 = .9762.$$

c. same answer as part (b) above.

$$3.110 \quad \mathbf{a.} \quad P(Y=1) = \frac{\binom{4}{2}\binom{2}{1}}{\binom{6}{3}} = .6.$$

$$\mathbf{b.} \quad P(Y \geq 1) = p(1) + p(2) = \frac{\binom{4}{2}\binom{2}{1}}{\binom{6}{3}} + \frac{\binom{4}{1}\binom{2}{2}}{\binom{6}{3}} = .8$$

$$\mathbf{c.} \quad P(Y \leq 1) = p(0) + p(1) = .8.$$

$$3.111 \quad \mathbf{a.} \quad \text{The probability function for } Y \text{ is } p(y) = \frac{\binom{2}{y}\binom{8}{3-y}}{\binom{10}{3}}, y = 0, 1, 2. \text{ In tabular form, this is}$$

y	0	1	2
$p(y)$	14/30	14/30	2/30

$$\mathbf{b.} \quad \text{The probability function for } Y \text{ is } p(y) = \frac{\binom{4}{y}\binom{6}{3-y}}{\binom{10}{3}}, y = 0, 1, 2, 3. \text{ In tabular form, this is}$$

y	0	1	2	3
$p(y)$	5/30	15/30	9/30	1/30

3.112 Let $Y = \#$ of malfunctioning copiers selected. Then, Y is hypergeometric with probability function

$$p(y) = \frac{\binom{3}{y}\binom{5}{4-y}}{\binom{8}{4}}, y = 0, 1, 2, 3.$$

$$\mathbf{a.} \quad P(Y=0) = p(0) = 1/14.$$

$$\mathbf{b.} \quad P(Y \geq 1) = 1 - P(Y=0) = 13/14.$$

3.113 The probability of an event as rare or rarer than one observed can be calculated according to the hypergeometric distribution. Let $Y = \#$ of black members. Then, Y is

hypergeometric and $P(Y \leq 1) = \frac{\binom{8}{3}\binom{12}{5}}{\binom{20}{6}} + \frac{\binom{8}{0}\binom{12}{6}}{\binom{20}{6}} = .187$. This is nearly 20%, so it is not unlikely.

$$3.114 \quad \mu = 6(8)/20 = 2.4, \sigma^2 = 6(8/20)(12/20)(14/19) = 1.061.$$

3.115 The probability distribution for Y is given by

y	0	1	2
$p(y)$	1/5	3/5	1/5

3.116 (Answers vary, but with $n=100$, the relative frequencies should be close to the probabilities in the table above.)

3.117 Let $Y = \#$ of improperly drilled gearboxes. Then, Y is hypergeometric with $N = 20$, $n = 5$, and $r = 2$.

a. $P(Y = 0) = .553$

b. The random variable T , the total time, is given by $T = 10Y + (5 - Y) = 9Y + 5$. Thus,
 $E(T) = 9E(Y) + 5 = 9[5(2/20)] + 5 = 9.5$.
 $V(T) = 81V(Y) = 81(.355) = 28.755$, $\sigma = 5.362$.

3.118 Let $Y = \#$ of aces in the hand. Then, $P(Y = 4 | Y \geq 3) = \frac{P(Y = 4)}{P(Y = 3) + P(Y = 4)}$. Note that Y

is a hypergeometric random variable. So, $P(Y = 3) = \frac{\binom{4}{3} \binom{48}{2}}{\binom{52}{5}} = .001736$ and

$$P(Y = 4) = \frac{\binom{4}{4} \binom{48}{1}}{\binom{52}{5}} = .00001847. \text{ Thus, } P(Y = 4 | Y \geq 3) = .0105.$$

3.119 Let the event $A = 2^{\text{nd}}$ king is dealt on 5^{th} card. The four possible outcomes for this event are $\{KNNNK, NKNNK, NNKNN, NNNKK\}$, where K denotes a king and N denotes a non-king. Each of these outcomes has probability: $\left(\frac{4}{52}\right)\left(\frac{48}{51}\right)\left(\frac{47}{50}\right)\left(\frac{46}{49}\right)\left(\frac{3}{48}\right)$. Then, the desired probability is $P(A) = 4\left(\frac{4}{52}\right)\left(\frac{48}{51}\right)\left(\frac{47}{50}\right)\left(\frac{46}{49}\right)\left(\frac{3}{48}\right) = .016$.

3.120 There are N animals in this population. After taking a sample of k animals, making and releasing them, there are $N - k$ unmarked animals. We then choose a second sample of size 3 from the N animals. There are $\binom{N}{3}$ ways of choosing this second sample and

there are $\binom{N-k}{2} \binom{k}{1}$ ways of finding exactly one of the originally marked animals. For $k = 4$, the probability of finding just one marked animal is

$$P(Y = 1) = \frac{\binom{N-4}{2} \binom{4}{1}}{\binom{N}{3}} = \frac{12(N-4)(N-5)}{N(N-1)(N-2)}.$$

Calculating this for various values of N , we find that the probability is largest for $N = 11$ or $N = 12$ (the same probability is found: .503).

3.121 a. $P(Y = 4) = \frac{2^4}{4!} e^{-2} = .090$.

b. $P(Y \geq 4) = 1 - P(Y \leq 3) = 1 - .857 = .143$ (using Table 3, Appendix III).

c. $P(Y < 4) = P(Y \leq 3) = .857$.

d. $P(Y \geq 4 | Y \geq 2) = \frac{P(Y \geq 4)}{P(Y \geq 2)} = .143/.594 = .241$

- 3.122** Let $Y = \#$ of customers that arrive during the hour. Then, Y is Poisson with $\lambda = 7$.
- $P(Y \leq 3) = .0818$.
 - $P(Y \geq 2) = .9927$.
 - $P(Y = 5) = .1277$
- 3.123** If $p(0) = p(1)$, $e^{-\lambda} = \lambda e^{-\lambda}$. Thus, $\lambda = 1$. Therefore, $p(2) = \frac{1^2}{2!} e^{-1} = .1839$.
- 3.124** Using Table 3 in Appendix III, we find that if Y is Poisson with $\lambda = 6.6$, $P(Y \leq 2) = .04$. Using this value of λ , $P(Y > 5) = 1 - P(Y \leq 5) = 1 - .355 = .645$.
- 3.125** Let $S = \text{total service time} = 10Y$. From Ex. 3.122, Y is Poisson with $\lambda = 7$. Therefore, $E(S) = 10E(Y) = 70$ and $V(S) = 100V(Y) = 700$. Also, $P(S > 150) = P(Y > 15) = 1 - P(Y \leq 15) = 1 - .998 = .002$, and unlikely event.
- 3.126** **a.** Let $Y = \#$ of customers that arrive in a given two-hour time. Then, Y has a Poisson distribution with $\lambda = 2(7) = 14$ and $P(Y = 2) = \frac{14^2}{2!} e^{-14}$.
- b.** The same answer as in part a. is found.
- 3.127** Let $Y = \#$ of typing errors per page. Then, Y is Poisson with $\lambda = 4$ and $P(Y \leq 4) = .6288$.
- 3.128** Note that over a one-minute period, $Y = \#$ of cars that arrive at the toll booth is Poisson with $\lambda = 80/60 = 4/3$. Then, $P(Y \geq 1) = 1 - P(Y = 0) = 1 - e^{-4/3} = .7364$.
- 3.129** Following the above exercise, suppose the phone call is of length t , where t is in minutes. Then, $Y = \#$ of cars that arrive at the toll booth is Poisson with $\lambda = 4t/3$. Then, we must find the value of t such that
- $$P(Y = 0) = 1 - e^{-4t/3} \geq .4.$$
- Therefore, $t \leq -\frac{3}{4} \ln(.6) = .383$ minutes, or about $.383(60) = 23$ seconds.
- 3.130** Define: $Y_1 = \#$ of cars through entrance I, $Y_2 = \#$ of cars through entrance II. Thus, Y_1 is Poisson with $\lambda = 3$ and Y_2 is Poisson with $\lambda = 4$.
- Then, $P(\text{three cars arrive}) = P(Y_1 = 0, Y_2 = 3) + P(Y_1 = 1, Y_2 = 2) + P(Y_1 = 2, Y_2 = 1) + P(Y_1 = 3, Y_2 = 0)$.
- By independence, $P(\text{three cars arrive}) = P(Y_1 = 0)P(Y_2 = 3) + P(Y_1 = 1)P(Y_2 = 2) + P(Y_1 = 2)P(Y_2 = 1) + P(Y_1 = 3)P(Y_2 = 0)$.
- Using Poisson probabilities, this is equal to 0.0521
- 3.131** Let the random variable $Y = \#$ of knots in the wood. Then, Y has a Poisson distribution with $\lambda = 1.5$ and $P(Y \leq 1) = .5578$.
- 3.132** Let the random variable $Y = \#$ of cars entering the tunnel in a two-minute period. Then, Y has a Poisson distribution with $\lambda = 1$ and $P(Y > 3) = 1 - P(Y \leq 3) = 0.01899$.

3.133 Let $X = \#$ of two-minute intervals with more than three cars. Therefore, X is binomial with $n = 10$ and $p = .01899$ and $P(X \geq 1) = 1 - P(X = 0) = 1 - (1 - .01899)^{10} = .1745$.

3.134 The probabilities are similar, even with a fairly small n .

y	$p(y)$, exact binomial	$p(y)$, Poisson approximation
0	.358	.368
1	.378	.368
2	.189	.184
3	.059	.061
4	.013	.015

3.135 Using the Poisson approximation, $\lambda \approx np = 100(.03) = 3$, so $P(Y \geq 1) = 1 - P(Y = 0) = .9524$.

3.136 Let $Y = \#$ of *E. coli* cases observed this year. Then, Y has an approximate Poisson distribution with $\lambda \approx 2.4$.

a. $P(Y \geq 5) = 1 - P(Y \leq 4) = 1 - .904 = .096$.

b. $P(Y > 5) = 1 - P(Y \leq 5) = 1 - .964 = .036$. Since there is a small probability associated with this event, the rate probably has changed.

3.137 Using the Poisson approximation to the binomial with $\lambda \approx np = 30(.2) = 6$. Then, $P(Y \leq 3) = .1512$.

3.138 $E[Y(Y-1)] = \sum_{y=0}^{\infty} \frac{y(y-1)\lambda^y e^{-\lambda}}{y!} = \lambda^2 \sum_{y=0}^{\infty} \frac{y(y-1)\lambda^{y-2} e^{-\lambda}}{y!}$. Using the substitution $z = y - 2$, it is found that $E[Y(Y-1)] = \lambda^2$. Use this with $V(Y) = E[Y(Y-1)] + E(Y) - [E(Y)]^2 = \lambda$.

3.139 Note that if Y is Poisson with $\lambda = 2$, $E(Y) = 2$ and $E(Y^2) = V(Y) + [E(Y)]^2 = 2 + 4 = 6$. So, $E(X) = 50 - 2E(Y) - E(Y^2) = 50 - 2(2) - 6 = 40$.

3.140 Since Y is Poisson with $\lambda = 2$, $E(C) = E\left[100\left(\frac{1}{2}\right)^Y\right] = \sum_{y=0}^{\infty} \frac{100\left(\frac{1}{2}\right)^y e^{-2}}{y!} = 100e^{-1} \sum_{y=1}^{\infty} \frac{1^y e^{-1}}{y!} = 100e^{-1}$.

3.141 Similar to Ex. 3.139: $E(R) = E(1600 - 50Y^2) = 1600 - 50(6) = \1300 .

3.142 a. $\frac{p(y)}{p(y-1)} = \frac{\frac{\lambda^y e^{-\lambda}}{y!}}{\frac{\lambda^{y-1} e^{-\lambda}}{(y-1)!}} = \frac{\lambda}{y}, y = 1, 2, \dots$

b. Note that if $\lambda > y$, $p(y) > p(y-1)$. If $\lambda > y$, $p(y) > p(y-1)$. If $\lambda = y$ for some integer y , $p(y) = p(y-1)$.

c. Note that for λ a non-integer, part b. implies that for $y-1 < y < \lambda$,
 $p(y-1) < p(y) > p(y+1)$.

Hence, $p(y)$ is maximized for $y =$ largest integer less than λ . If λ is an integer, then $p(y)$ is maximized at both values $\lambda - 1$ and λ .

3.143 Since λ is a non-integer, $p(y)$ is maximized at $y = 5$.

3.144 Observe that with $\lambda = 6$, $p(5) = \frac{6^5 e^{-6}}{5!} = .1606$, $p(6) = \frac{6^6 e^{-6}}{6!} = .1606$.

3.145 Using the binomial theorem, $m(t) = E(e^{ty}) = \sum_{y=0}^n \binom{n}{y} (pe^t)^y q^{n-y} = (pe^t + q)^n$.

3.146 $\frac{d}{dt} m(t) = n(pe^t + q)^{n-1} pe^t$. At $t = 0$, this is $np = E(Y)$.

$\frac{d^2}{dt^2} m(t) = n(n-1)(pe^t + q)^{n-2} (pe^t)^2 + n(pe^t + q)^{n-1} pe^t$. At $t = 0$, this is $np^2(n-1) + np$.
Thus, $V(Y) = np^2(n-1) + np - (np)^2 = np(1-p)$.

3.147 The moment-generating function is $m(t) = E(e^{ty}) = \sum_{y=1}^n pe^{ty} q^{y-1} = pe^t \sum_{y=0}^{\infty} (qe^t)^y = \frac{pe^t}{1-qe^t}$.

3.148 $\frac{d}{dt} m(t) = \frac{pe^t}{(1-qe^t)^2}$. At $t = 0$, this is $1/p = E(Y)$.

$\frac{d^2}{dt^2} m(t) = \frac{(1-qe^t)^2 pe^t - 2pe^t(1-qe^t)(-qe^t)}{(1-qe^t)^4}$. At $t = 0$, this is $(1+q)/p^2$.
Thus, $V(Y) = (1+q)/p^2 - (1/p)^2 = q/p^2$.

3.149 This is the moment-generating function for the binomial with $n = 3$ and $p = .6$.

3.150 This is the moment-generating function for the geometric with $p = .3$.

3.151 This is the moment-generating function for the binomial with $n = 10$ and $p = .7$, so $P(Y \leq 5) = .1503$.

3.152 This is the moment-generating function for the Poisson with $\lambda = 6$. So, $\mu = 6$ and $\sigma = \sqrt{6} \approx 2.45$. So, $P(|Y - \mu| \leq 2\sigma) = P(\mu - 2\sigma \leq Y \leq \mu + 2\sigma) = P(1.1 \leq Y \leq 10.9) = P(2 \leq Y \leq 10) = .940$.

3.153 a. Binomial with $n = 5$, $p = 1$

b. If $m(t)$ is multiplied top and bottom by $1/2$, this is a geometric mgf with $p = 1/2$.

c. Poisson with $\lambda = 2$.

3.154 a. Binomial mean and variance: $\mu = 1.667$, $\sigma^2 = 1.111$.

b. Geometric mean and variance: $\mu = 2$, $\sigma^2 = 2$.

c. Poisson mean and variance: $\mu = 2$, $\sigma^2 = 2$.

3.155 Differentiate to find the necessary moments:

- a. $E(Y) = 7/3$.
- b. $V(Y) = E(Y^2) - [E(Y)]^2 = 6 - (7/3)^2 = 5/9$.
- c. Since $m(t) = E(e^{tY})$, Y can only take on values 1, 2, and 3 with probabilities 1/6, 2/6, and 3/6.

3.156 a. $m(0) = E(e^{0Y}) = E(1) = 1$.

b. $m_W(t) = E(e^{tW}) = E(e^{t3Y}) = E(e^{(3t)Y}) = m(3t)$.

c. $m_X(t) = E(e^{tX}) = E(e^{t(Y-2)}) = E(e^{-2t} e^{tY}) = e^{-2t} m(t)$.

3.157 a. From part b. in Ex. 3.156, the results follow from differentiating to find the necessary moments.

b. From part c. in Ex. 3.156, the results follow from differentiating to find the necessary moments.

3.158 The mgf for W is $m_W(t) = E(e^{tW}) = E(e^{t(aY+b)}) = E(e^{bt} e^{(at)Y}) = e^{bt} m(at)$.

3.159 From Ex. 3.158, the results follow from differentiating the mgf of W to find the necessary moments.

3.160 a. $E(Y^*) = E(n - Y) = n - E(Y) = n - np = n(1 - p) = nq$. $V(Y^*) = V(n - Y) = V(Y) = npq$.

b. $m_{Y^*}(t) = E(e^{tY^*}) = E(e^{t(n-Y)}) = E(e^{nt} e^{(-t)Y}) = e^{nt} m(-t) = (pe^t + q)^n$.

c. Based on the moment-generating function, Y^* has a binomial distribution.

d. The random variable $Y^* = \#$ of failures.

e. The classification of “success” and “failure” in the Bernoulli trial is arbitrary.

3.161 $m_{Y^*}(t) = E(e^{tY^*}) = E(e^{t(Y-1)}) = E(e^{-t} e^{tY}) = e^{-t} m(t) = \frac{p}{1-qe^t}$.

3.162 Note that $r^{(1)}(t) = \frac{m^{(1)}(t)}{m(t)}$, $r^{(2)}(t) = \frac{m^{(2)}(t)m(t) - [m^{(1)}(t)]^2}{(m(t))^2}$. Then, $r^{(1)}(0) = \frac{m^{(1)}(0)}{m(0)} = \frac{E(Y)}{1} = \mu$.

$$r^{(2)}(0) = \frac{m^{(2)}(0)m(0) - [m^{(1)}(0)]^2}{(m(0))^2} = \frac{E(Y^2) - [E(Y)]^2}{1} = \sigma^2.$$

3.163 Note that $r(t) = 5(e^t - 1)$. Then, $r^{(1)}(t) = 5e^t$ and $r^{(2)}(t) = 5e^t$. So, $r^{(1)}(0) = 5 = \mu = \lambda$ and $r^{(2)}(0) = 5e^0 = \sigma^2 = \lambda$.

3.164 For the binomial, $P(t) = E(t^Y) = \sum_{y=0}^n \binom{n}{y} (pt)^y q^{n-y} = (q + pt)^n$. Differentiating with respect to t , $\left. \frac{d}{dt} P(t) \right|_{t=1} = np(q + pt)^{n-1} \Big|_{t=1} = np$.

3.165 For the Poisson, $P(t) = E(t^Y) = \sum_{y=0}^{\infty} \frac{\lambda^y e^{-\lambda} t^y}{y!} = \frac{e^{-\lambda}}{e^{-\lambda t}} \sum_{y=0}^{\infty} \frac{(t\lambda)^y e^{-\lambda t}}{y!} = e^{\lambda(t-1)}$. Differentiating with respect to t , $E(Y) = \left. \frac{d}{dt} P(t) \right|_{t=1} = \lambda e^{\lambda(t-1)} \Big|_{t=1} = \lambda$ and $\left. \frac{d^2}{dt^2} P(t) \right|_{t=1} = \lambda^2 e^{\lambda(t-1)} \Big|_{t=1} = \lambda^2 = E[Y(Y-1)] = E(Y^2) - E(Y)$. Thus, $V(Y) = \lambda$.

3.166 $E[Y(Y-1)(Y-2)] = \left. \frac{d^3}{dt^3} P(t) \right|_{t=1} = \lambda^3 e^{\lambda(t-1)} \Big|_{t=1} = \lambda^3 = E(Y^3) - 3E(Y^2) + 2E(Y)$. Therefore, $E(Y^3) = \lambda^3 + 3(\lambda^2 + \lambda) - 2\lambda = \lambda^3 + 3\lambda^2 + \lambda$.

3.167 a. The value 6 lies $(11-6)/3 = 5/3$ standard deviations below the mean. Similarly, the value 16 lies $(16-11)/3 = 5/3$ standard deviations above the mean. By Tchebysheff's theorem, at least $1 - 1/(5/3)^2 = 64\%$ of the distribution lies in the interval 6 to 16.

b. By Tchebysheff's theorem, $.09 = 1/k^2$, so $k = 10/3$. Since $\sigma = 3$, $k\sigma = \frac{10}{3} \cdot 3 = 10 = C$.

3.168 Note that Y has a binomial distribution with $n = 100$ and $p = 1/5 = .2$

a. $E(Y) = 100(.2) = 20$.

b. $V(Y) = 100(.2)(.8) = 16$, so $\sigma = 4$.

c. The intervals are $20 \pm 2(4)$ or $(12, 28)$, $20 + 3(4)$ or $(8, 32)$.

d. By Tchebysheff's theorem, $1 - 1/3^2$ or approximately 89% of the time the number of correct answers will lie in the interval $(8, 32)$. Since a passing score of 50 is far from this range, receiving a passing score is very unlikely.

3.169 a. $E(Y) = -1(1/18) + 0(16/18) + 1(1/18) = 0$. $E(Y^2) = 1(1/18) + 0(16/18) + 1(1/18) = 2/18 = 1/9$. Thus, $V(Y) = 1/9$ and $\sigma = 1/3$.

b. $P(|Y - 0| \geq 1) = P(Y = -1) + P(Y = 1) = 1/18 + 1/18 = 2/18 = 1/9$. According to Tchebysheff's theorem, an upper bound for this probability is $1/3^2 = 1/9$.

c. Example: let X have probability distribution $p(-1) = 1/8$, $p(0) = 6/8$, $p(1) = 1/8$. Then, $E(X) = 0$ and $V(X) = 1/4$.

d. For a specified k , assign probabilities to the points -1 , 0 , and 1 as $p(-1) = p(1) = \frac{1}{2k^2}$ and $p(0) = 1 - \frac{1}{k}$.

3.170 Similar to Ex. 3.167: the interval $(.48, .52)$ represents two standard deviations about the mean. Thus, the lower bound for this interval is $1 - 1/4 = 3/4$. The expected number of coins is $400(3/4) = 300$.

3.171 Using Tchebysheff's theorem, $5/9 = 1 - 1/k^2$, so $k = 3/2$. The interval is $100 \pm (3/2)10$, or 85 to 115.

3.172 From Ex. 3.115, $E(Y) = 1$ and $V(Y) = .4$. Thus, $\sigma = .63$. The interval of interest is $1 \pm 2(.63)$, or $(-.26, 2.26)$. Since Y can only take on values 0, 1, or 2, 100% of the values will lie in the interval. According to Tchebysheff's theorem, the lower bound for this probability is 75%.

3.173 a. The binomial probabilities are $p(0) = 1/8$, $p(1) = 3/8$, $p(2) = 3/8$, $p(3) = 1/8$.

b. The graph represents a symmetric distribution.

c. $E(Y) = 3(1/2) = 1.5$, $V(Y) = 3(1/2)(1/2) = .75$. Thus, $\sigma = .866$.

d. For *one* standard deviation about the mean:  $1.5 \pm .866$ or $(.634, 2.366)$
This traps the values 1 and 2, which represents $7/8$ or 87.5% of the probability. This is consistent with the empirical rule.

For *two* standard deviations about the mean: $1.5 \pm 2(.866)$ or $(-.232, 3.232)$
This traps the values 0, 1, and 2, which represents 100% of the probability. This is consistent with both the empirical rule and Tchebysheff's theorem.

3.174 a. (Similar to Ex. 3.173) the binomial probabilities are $p(0) = .729$, $p(1) = .243$, $p(2) = .027$, $p(3) = .001$.

b. The graph represents a skewed distribution.

c. $E(Y) = 3(.1) = .3$, $V(Y) = 3(.1)(.9) = .27$. Thus, $\sigma = .520$.

d. For *one* standard deviation about the mean: $.3 \pm .520$ or $(-.220, .820)$
This traps the value 1, which represents 24.3% of the probability. This is not consistent with the empirical rule.

For *two* standard deviations about the mean: $.3 \pm 2(.520)$ or $(-.740, 1.34)$
This traps the values 0 and 1, which represents 97.2% of the probability. This is consistent with both the empirical rule and Tchebysheff's theorem.

3.175 a. The expected value is $120(.32) = 38.4$

b. The standard deviation is $\sqrt{120(.32)(.68)} = 5.11$.

c. It is quite likely, since 40 is close to the mean 38.4 (less than .32 standard deviations away).

3.176 Let Y represent the number of students in the sample who favor banning clothes that display gang symbols. If the teenagers are actually equally split, then $E(Y) = 549(.5) = 274.5$ and $V(Y) = 549(.5)(.5) = 137.25$. Now, $Y/549$ represents the proportion in the sample who favor banning clothes that display gang symbols, so $E(Y/549) = .5$ and $V(Y/549) = .5(.5)/549 = .000455$. Then, by Tchebysheff's theorem,

$$P(Y/549 \geq .85) \leq P(|Y/549 - .5| \geq .35) \leq 1/k^2,$$

where k is given by $k\sigma = .35$. From above, $\sigma = .02134$ so $k = 16.4$ and $1/(16.4)^2 = .0037$. This is a very unlikely result. It is also unlikely using the empirical rule. We assumed that the sample was selected randomly from the population.

3.177 For $C = 50 + 3Y$, $E(C) = 50 + 3(10) = \$80$ and $V(C) = 9(10) = 90$, so that $\sigma = 9.487$. Using Tchebysheff's theorem with $k = 2$, we have $P(|Y - 80| < 2(9.487)) \geq .75$, so that the required interval is $(80 - 2(9.487), 80 + 2(9.487))$ or $(61.03, 98.97)$.

3.178 Using the binomial, $E(Y) = 1000(.1) = 100$ and $V(Y) = 1000(.1)(.9) = 90$. Using the result that at least 75% of the values will fall within two standard deviation of the mean, the interval can be constructed as $100 \pm 2\sqrt{90}$, or $(81, 119)$.

3.179 Using Tchebysheff's theorem, observe that

$$P(Y \geq \mu + k\sigma) = P(Y - \mu \geq k\sigma) \leq P(|Y - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Therefore, to find $P(Y \geq 350) \leq 1/k^2$, we solve $150 + k(67.081) = 350$, so $k = 2.98$. Thus, $P(Y \geq 350) \leq 1/(2.98)^2 = .1126$, which is not highly unlikely.

3.180 Number of combinations = $26(26)(10)(10)(10)(10) = 6,760,000$. Thus, $E(\text{winnings}) = 100,000(1/6,760,000) + 50,000(2/6,760,000) + 1000(10/6,760,000) = \$.031$, which is much less than the price of the stamp.

3.181 Note that $P(\text{acceptance}) = P(\text{observe no defectives}) = \binom{5}{0} p^0 q^5$. Thus:

$p = \text{Fraction defective}$	$P(\text{acceptance})$
0	1
.10	.5905
.30	.1681
.50	.0312
1.0	0

3.182 OC curves can be constructed using points given in the tables below.

a. Similar to Ex. 3.181: $P(\text{acceptance}) = \binom{10}{0} p^0 q^{10}$. Thus,

p	0	.05	.10	.30	.50	1
$P(\text{acceptance})$	1	.599	.349	.028	.001	0

b. Here, $P(\text{acceptance}) = \binom{10}{0} p^0 q^{10} + \binom{10}{1} p^1 q^9$. Thus,

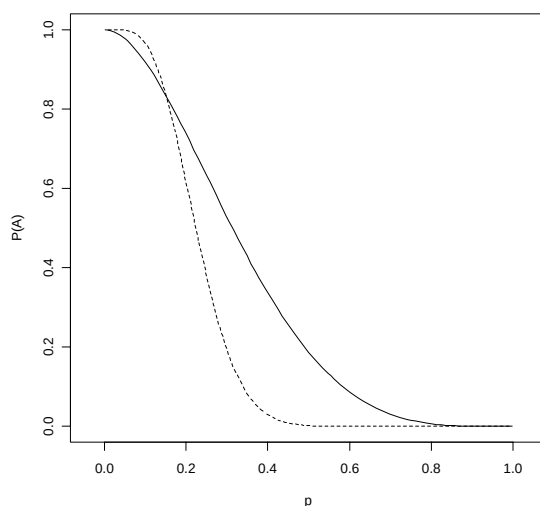
p	0	.05	.10	.30	.50	1
$P(\text{acceptance})$	1	.914	.736	.149	.01	0

c. Here, $P(\text{acceptance}) = \binom{10}{0} p^0 q^{10} + \binom{10}{1} p^1 q^9 + \binom{10}{2} p^2 q^8$. Thus,

p	0	.05	.10	.30	.50	1
$P(\text{acceptance})$	1	.988	.930	.383	.055	0

3.183 Graph the two OC curves with $n = 5$ and $a = 1$ in the first case and $n = 25$ and $a = 5$ in the second case.

- By graphing the OC curves, it is seen that if the defectives fraction ranges from $p = 0$ to $p = .10$, the seller would want the probability of accepting in this interval to be as high as possible. So, he would choose the second plan.
- If the buyer wishes to be protected against accepting lots with a defective fraction greater than .3, he would want the probability of acceptance (when $p > .3$) to be as small as possible. Thus, he would also choose the second plan.



The above graph illustrates the two OC curves. The solid line represents the first case and the dashed line represents the second case.

3.184 Let $Y = \#$ in the sample who favor garbage collect by contract to a private company.

Then, Y is binomial with $n = 25$.

- If $p = .80$, $P(Y \geq 22) = 1 - P(Y \leq 21) = 1 - .766 = .234$,
- If $p = .80$, $P(Y = 22) = .1358$.
- There is not strong evidence to show that the commissioner is incorrect.

3.185 Let $Y = \#$ of students who choose the numbers 4, 5, or 6. Then, Y is binomial with $n = 20$ and $p = 3/10$.

- $P(Y \geq 8) = 1 - P(Y \leq 7) = 1 - .7723 = .2277$.
- Given the result in part a, it is not an unlikely occurrence for 8 students to choose 4, 5 or 6.

3.186 The total cost incurred is $W = 30Y$. Then,

$$E(W) = 30E(Y) = 30(1/.3) = 100, \quad V(W) = 900V(Y) = 900(.7/.3^2) = 7000.$$

Using the empirical rule, we can construct a interval of three standard deviations about the mean: $100 \pm 3\sqrt{7000}$, or (151, 351).

3.187 Let $Y = \#$ of rolls until the player stops. Then, Y is geometric with $p = 5/6$.

a. $P(Y = 3) = (1/6)^2(5/6) = .023.$

b. $E(Y) = 6/5 = 1.2.$

c. Let $X =$ amount paid to player. Then, $X = 2^{Y-1}$.

$$E(X) = E(2^{Y-1}) = \sum_{y=1}^{\infty} 2^{y-1} q^{y-1} p = p \sum_{x=0}^{\infty} (2q)^x = \frac{p}{1-2q}, \text{ since } 2q < 1. \text{ With } p = 5/6,$$

this is \$1.25.

3.188 The result follows from $P(Y > 1 | Y \geq 1) = \frac{P(Y > 1)}{P(Y \geq 1)} = \frac{P(Y \geq 2)}{P(Y \geq 1)} = \frac{1 - P(Y = 1) - P(Y = 0)}{1 - P(Y = 0)}.$

3.189 The random variable $Y = \#$ of failures in 10,000 starts is binomial with $n = 10,000$ and $p = .00001$. Thus, $P(Y \geq 1) = 1 - P(Y = 0) = 1 - (.9999)^{10000} = .09516$.
Poisson approximation: $1 - e^{-1} = .09516$.

3.190 Answers vary, but with $n = 100$, $\bar{y}$ should be quite close to $\mu = 1$.

3.191 Answers vary, but with $n = 100$, s^2 should be quite close to $\sigma^2 = .4$.

3.192 Note that $p(1) = p(2) = \dots p(6) = 1/6$. From Ex. 3.22, $\mu = 3.5$ and $\sigma^2 = 2.9167$. The interval constructed of two standard deviations about the mean is (.08, 6.92) which contains 100% of the possible values for Y .

3.193 Let $Y_1 = \#$ of defectives from line I, Y_2 is defined similarly. Then, both Y_1 and Y_2 are binomial with $n = 5$ and defective probability p . In addition, $Y_1 + Y_2$ is also binomial with $n = 10$ and defective probability p . Thus,

$$P(Y_1 = 2 | Y_1 + Y_2 = 4) = \frac{P(Y_1 = 2)P(Y_2 = 2)}{P(Y_1 + Y_2 = 4)} = \frac{\binom{5}{2}p^2q^3\binom{5}{2}p^2q^3}{\binom{10}{4}p^4q^6} = \frac{\binom{5}{2}\binom{5}{2}}{\binom{10}{4}} = 0.476.$$

Notice that the probability does not depend on p .

3.194 The possible outcomes of interest are:

WLLLLLLLLL, LWLLLLLLLLL, LLWLLLLLLLLL

So the desired probability is $.1(.9)^{10} + .9(.1)(.9)^9 + (.9)^2(.1)(.9)^8 = 3(.1)(.9)^{10} = .104$.

- 3.195** Let $Y = \#$ of imperfections in one-square yard of weave. Then, Y is Poisson with $\lambda = 4$.
a. $P(Y \geq 1) = 1 - P(Y = 0) = 1 - e^{-4} = .982$.
b. Let $W = \#$ of imperfections in three-square yards of weave. Then, W is Poisson with $\lambda = 12$. $P(W \geq 1) = 1 - P(W = 0) = 1 - e^{-12}$.

- 3.196** For an 8-square yard bolt, let $X = \#$ of imperfections so that X is Poisson with $\lambda = 32$.
 Thus, $C = 10X$ is the cost to repair the weave and
 $E(C) = 10E(X) = \$320$ and $V(C) = 100V(X) = 3200$.

- 3.197 a.** Let $Y = \#$ of samples with at least one bacteria colony. Then, Y is binomial with $n = 4$ and $p = P(\text{at least one bacteria colony}) = 1 - P(\text{no bacteria colonies}) = 1 - e^{-2} = .865$ (by the Poisson). Thus, $P(Y \geq 1) = 1 - P(Y = 0) = 1 - (.135)^4 = .9997$.
b. Following the above, we require $1 - (.135)^n = .95$ or $(.135)^n = .05$. Solving for n , we have $n = \frac{\ln(.05)}{\ln(.135)} = 1.496$, so take $n = 2$.

- 3.198** Let $Y = \#$ of neighbors for a seedling within an area of size A . Thus, Y is Poisson with $\lambda = A*d$, where for this problem $d = 4$ per square meter.
a. Note that “within 1 meter” denotes an area $A = \pi(1 \text{ m})^2 = \pi \text{ m}^2$. Thus, $P(Y = 0) = e^{-4\pi}$.
b. “Within 2 meters” denotes an area $A = \pi(2 \text{ m})^2 = 4\pi \text{ m}^2$. Thus,

$$P(Y \leq 3) = P(Y = 0) + P(Y = 1) + P(Y = 2) + P(Y = 3) = \sum_{y=0}^3 \frac{(16\pi)^y e^{-16\pi}}{y!}.$$

- 3.199 a.** Using the binomial model with $n = 1000$ and $p = 30/100,000$, let $\lambda \approx np = 1000(30/100000) = .300$ for the Poisson approximation.
b. Let $Y = \#$ of cases of IDD. $P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1) = 1 - .963 = .037$.

- 3.200** Note that

$$(q + pe^t)^n = \left[q + p\left(1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots\right) \right]^n = \left[1 + pt + p\frac{t^2}{2!} + p\frac{t^3}{3!} + \cdots \right]^n.$$

Expanding the above multinomial (but only showing the first three terms gives

$$(q + pe^t)^n = 1^n + (np)t1^{n-1} + \left[n(n-1)p^2 + np \right] \frac{t^2}{2!} 1^{n-2} + \cdots$$

The coefficients agree with the first and second moments for the binomial distribution.

- 3.201** From Ex. 103 and 106, we have that $\mu = 100$ and $\sigma = \sqrt{1666.67} = 40.825$. Using an interval of two standard deviations about the mean, we obtain $100 \pm 2(40.825)$ or $(18.35, 181.65)$

- 3.202** Let $W = \#$ of drivers who wish to park and $W' = \#$ of cars, which is Poisson with mean λ .
a. Observe that

$$\begin{aligned} P(W = k) &= \sum_{n=k}^{\infty} P(W = k | W' = n) P(W' = n) = \sum_{n=k}^{\infty} \frac{n!}{k!(n-k)!} p^k q^{n-k} e^{-\lambda} \frac{\lambda^n}{n!} \\ &= \lambda^k e^{-\lambda} \left(\frac{p}{k!} \right) \sum_{n=k}^{\infty} \frac{q^{n-k}}{(n-k)!} \lambda^{n-k} = \frac{(\lambda p)^k}{k!} \sum_{j=0}^{\infty} \frac{q^j}{j!} \lambda^j = \frac{(\lambda p)^k}{k!} e^{-\lambda} e^{q\lambda} \\ &= \frac{(\lambda p)^k}{k!} e^{-\lambda p}, \quad k = 0, 1, 2, \dots \end{aligned}$$

Thus, $P(W = 0) = e^{-\lambda p}$.

- b.** This is a Poisson distribution with mean λp .

- 3.203** Note that $Y(t)$ has a negative binomial distribution with parameters $r = k$, $p = e^{-\lambda t}$.

a. $E[Y(t)] = ke^{\lambda t}$, $V[Y(t)] = \frac{k(1-e^{-\lambda t})}{e^{-2\lambda t}} = k(e^{2\lambda t} - e^{\lambda t})$.

b. With $k = 2$, $\lambda = .1$, $E[Y(5)] = 3.2974$, $V[Y(5)] = 2.139$.

- 3.204** Let $Y = \#$ of left-turning vehicles arriving while the light is red. Then, Y is binomial with $n = 5$ and $p = .2$. Thus, $P(Y \leq 3) = .993$.

- 3.205** One solution: let $Y = \#$ of tosses until 3 sixes occur. Therefore, Y is negative binomial

where $r = 3$ and $p = 1/6$. Then, $P(Y = 9) = \binom{8}{2} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^6 = .0434127$. Note that this

probability contains all events where a six occurs on the 9th toss. Multiplying the above probability by 1/6 gives the probability of observing 4 sixes in 10 trials, where a six occurs on the 9th and 10th trial: $(.0424127)(1/6) = .007235$.

- 3.206** Let Y represent the gain to the insurance company for a particular insured driver and let P be the premium charged to the driver. Given the information, the probability distribution for Y is given by:

y	$p(y)$
P	.85
$P - 2400$	.15(.80) = .12
$P - 7200$	.15(.12) = .018
$P - 12,000$	.15(.08) = .012

If the expected gain is 0 (breakeven), then:

$$E(Y) = P(.85) + (P - 2400).12 + (P - 7200).018 + (P - 12000).012 = 0, \text{ so } P = \$561.60.$$

- 3.207** Use the Poisson distribution with $\lambda = 5$.

a. $p(2) = .084$, $P(Y \leq 2) = .125$.

b. $P(Y > 10) = 1 - P(Y \leq 10) = 1 - .986 = .014$, which represents an unlikely event.

3.208 If the general public was split 50–50 in the issue, then $Y = \#$ of people in favor of the proposition is binomial with $n = 2000$ and $p = .5$. Thus,

$$E(Y) = 2000(.5) = 1000 \text{ and } V(Y) = 2000(.5)(.5) = 500.$$

Since $\sigma = \sqrt{500} = 22.36$, observe that 1373 is $(1373 - 1000)/22.36 = 16.68$ standard deviations above the mean. Such a value is unlikely.

3.209 Let $Y = \#$ of contracts necessary to obtain the third sale. Then, Y is negative binomial with $r = 3$, $p = .3$. So, $P(Y < 5) = P(Y = 3) + P(Y = 4) = .3^3 + 3(.3)^3(.7) = .0837$.

3.210 In Example 3.22, $\lambda = \mu = 3$ and $\sigma^2 = 3$ and that $\sigma = \sqrt{3} = 1.732$. Thus, $P(|Y - 3| \leq 2(1.732)) = P(-.464 \leq Y \leq 6.464) = P(Y \leq 6) = .966$. This is consistent with the empirical rule (approximately 95%).

3.211 There are three scenarios:

- if she stocks two items, both will sell with probability 1. So, her profit is \$.40.
- if she stocks three items, two will sell with probability .1 (a loss of .60) and three will sell with probability .9. Thus, her expected profit is $(-.60).1 + .60(.9) = $.48$.
- if she stocks four items, two will sell with probability .1 (a loss of 1.60), three will sell with probability .4 (a loss of .40), and four will sell with probability .5 (a gain of .80). Thus, her expected profit is $(-1.60).1 + (-.40).4 + (.80).5 = $.08$

So, to maximize her expected profit, stock three items.

3.212 Note that: $\frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}} = \frac{n!}{y!(n-y)!} \left[\left(\frac{r}{N} \right) \left(\frac{r-1}{N-1} \right) \left(\frac{r-2}{N-2} \right) \cdots \left(\frac{r-y+1}{N-y+1} \right) \right] \times \left[\left(\frac{N-r}{N-y} \right) \left(\frac{N-r-1}{N-y-1} \right) \left(\frac{N-r-2}{N-y-2} \right) \cdots \left(\frac{N-r-n+y+1}{N-n+1} \right) \right]$. In the

first bracketed part, each quotient in parentheses has a limiting value of p . There are y such quotients. In the second bracketed part, each quotient in parentheses has a limiting value of $1 - p = q$. There are $n - y$ such quotients. Thus,

$$\frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}} \rightarrow \binom{n}{y} p^y q^{n-y} \text{ as } N \rightarrow \infty$$

3.213 a. The probability is $p(10) = \frac{\binom{40}{10}\binom{60}{10}}{\binom{100}{20}} = .1192$ (found by dhyper(10, 40, 60, 20) in R).

b. The binomial approximation is $\left(\frac{20!}{10!10!} \right) 4^{10} (.6)^{10} = .117$, a close value to the above (exact) answer.

3.214 Define: A = accident next year B = accident this year C = safe driver

Thus, $P(C) = .7$, $P(A|C) = .1 = P(B|C)$, and $P(A|\bar{C}) = P(B|\bar{C}) = .5$. From Bayes' rule,

$$P(C|B) = \frac{P(B|C)P(C)}{P(B|C)P(C) + P(B|\bar{C})P(\bar{C})} = \frac{.1(.7)}{.1(.7) + .5(.3)} = 7/22.$$

Now, we need $P(A|B)$. Note that since $C \cup \bar{C} = S$, this conditional probability is equal to

$$P(A \cap (C \cup \bar{C}) | B) = P(A \cap C | B) + P(A \cap \bar{C} | B) = P(A \cap C | B) + P(A \cap \bar{C} | B), \text{ or}$$

$$P(A|B) = P(C|B)P(A|C \cap B) + P(\bar{C}|B)P(A|\bar{C} \cap B) = 7/22(.1) + 15/22(.5) = .3727.$$

So, the premium should be $400(.3727) = \$149.09$.

3.215 a. Note that for (2), there are two possible values for N_2 , the number of tests performed: 1 and $k + 1$. If $N_2 = 1$, all of the k people are healthy and this probability is $(.95)^k$. Thus, $P(N_2 = k + 1) = 1 - (.95)^k$. Thus, $E(N_2) = 1(.95)^k + (k + 1)(1 - .95^k) = 1 + k(1 - .95^k)$. This expectation holds for each group, so that for n groups the expected number of tests is $n[1 + k(1 - .95^k)]$.

b. Writing the above as $g(k) = \frac{N}{k} [1 + k(1 - .95^k)]$, where $n = \frac{N}{k}$, we can minimize this with respect to k . Note that $g'(k) = \frac{1}{k^2} + (.95^k) \ln(.95)$, a strictly decreasing function. Since k must be an integer, it is found that $g(k)$ is minimized at $k = 5$ and $g(5) = .4262$.

c. The expected number of tests is $.4262N$, compared to the N tests is (1) is used. The savings is then $N - .4262N = .5738N$.

3.216 a.
$$P(Y = n) = \frac{\binom{r}{n} \binom{N-r}{n-n}}{\binom{N}{n}} = \frac{r!}{N!} \times \frac{(N-n)!}{(r-n)!} = \frac{r(r-1)(r-2) \cdots (r-n+1)}{N(N-1)(N-2) \cdots (N-n+1)}.$$

b. Since for integers $a > b$, $\frac{\binom{a}{b}}{\binom{a}{b+1}} = \frac{b+1}{a-b}$, apply this result to find that

$$\frac{p(y|r_1)}{p(y+1|r_1)} = \frac{y+1}{r_1-y} \cdot \frac{N-r_1+n+y+1}{n-y} \quad \text{and} \quad \frac{p(y|r_2)}{p(y+1|r_2)} = \frac{y+1}{r_2-y} \cdot \frac{N-r_2+n+y+1}{n-y}.$$

With $r_1 < r_2$, it follows that $\frac{p(y|r_1)}{p(y+1|r_1)} > \frac{p(y+1|r_2)}{p(y+1|r_2)}$.

c. Note that from the binomial theorem, $(1+a)^{N_1} = \sum_{k=0}^{N_1} \binom{N_1}{k} y^k$. So,

$$(1+a)^{N_1} (1+a)^{N_2} = \left[\binom{N_1}{0} + \binom{N_1}{1} y + \cdots + \binom{N_1}{N_1} y^{N_1} \right] \times \left[\binom{N_1}{0} + \binom{N_1}{1} y + \cdots + \binom{N_1}{N_1} y^{N_1} \right] =$$

$(1+a)^{N_1+N_2} = \left[\binom{N_1+N_2}{0} + \binom{N_1+N_2}{1}y + \dots + \binom{N_1+N_2}{N_1+N_2}y^{N_1+N_2} \right]$. Since these are equal, the coefficient of every y^n must be equal. In the second equality, the coefficient is

$$\binom{N_1+N_2}{n}.$$

In the first inequality, the coefficient is given by the sum

$$\binom{N_1}{0}\binom{N_2}{n} + \binom{N_1}{1}\binom{N_2}{n-1} + \dots + \binom{N_1}{n}\binom{N_2}{0} = \sum_{k=0}^n \binom{N_1}{k}\binom{N_2}{n-k}, \text{ thus the relation holds.}$$

d. The result follows from part **c** above.

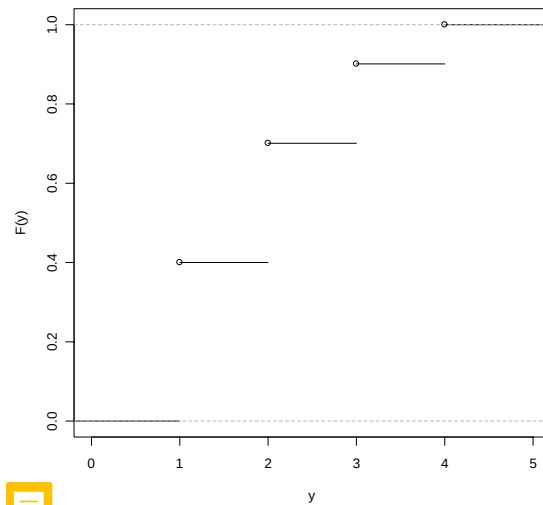
$$\begin{aligned} \mathbf{3.217} \quad E(Y) &= \sum_{y=0}^n \frac{y \binom{r}{y} \binom{N-r}{n-y}}{\binom{N}{n}} = r \sum_{y=1}^n \left[\frac{(r-1)!}{(y-1)!(r-y)!} \right] \left[\frac{\binom{N-r}{n-y}}{\binom{N}{n}} \right] = r \sum_{y=1}^n \left[\frac{\binom{r-1}{y-1} \binom{N-r}{n-y}}{\binom{N}{n}} \right]. \text{ In this sum, let } x = y-1: \\ & r \sum_{x=0}^{n-1} \left[\frac{\binom{r-1}{x} \binom{N-r}{n-x-1}}{\binom{N}{n}} \right] = r \sum_{x=0}^{n-1} \left[\frac{\binom{r-1}{x} \binom{N-r}{n-x-1}}{\binom{N}{n} \binom{N-1}{n-1}} \right] = \frac{nr}{N}. \end{aligned}$$

$$\mathbf{3.218} \quad E[Y(Y-1)] = \sum_{y=0}^n \frac{y(y-1) \binom{r}{y} \binom{N-r}{n-y}}{\binom{N}{n}} = \sum_{y=0}^n \left[\frac{r(r-1)(r-2)!}{y(y-1)(y-2)!(r-y)!} \right] \left[\frac{\binom{N-r}{n-y}}{\binom{N}{n}} \right] = r(r-1) \sum_{y=2}^n \left[\frac{\binom{r-2}{y-2} \binom{N-r}{n-y}}{\binom{N}{n}} \right]. \text{ In this}$$

sum, let $x = y-2$ to obtain the expectation $\frac{r(r-1)n(n-1)}{N(N-1)}$. From this result, the variance of the hypergeometric distribution can also be calculated.

Chapter 4: Continuous Variables and Their Probability Distributions

$$4.1 \quad \text{a. } F(y) = P(Y \leq y) = \begin{cases} 0 & y < 1 \\ .4 & 1 \leq y < 2 \\ .7 & 2 \leq y < 3 \\ .9 & 3 \leq y < 4 \\ 1 & y \geq 4 \end{cases}$$



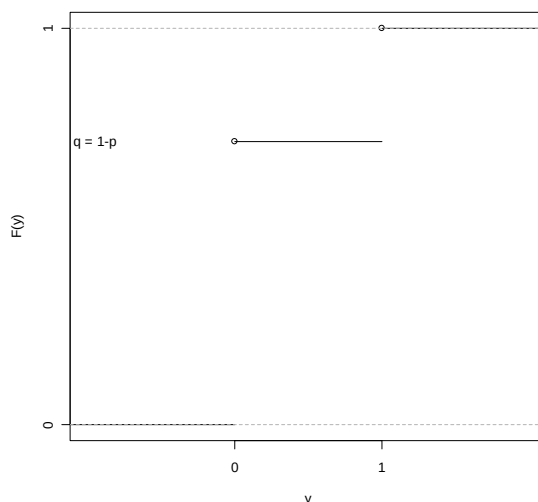
b. The graph is above. 

$$4.2 \quad \text{a. } p(1) = .2, p(2) = (1/4)4/5 = .2, p(3) = (1/3)(3/4)(4/5) = .2, p(4) = .2, p(5) = .2.$$

$$\text{b. } F(y) = P(Y \leq y) = \begin{cases} 0 & y < 1 \\ .2 & 1 \leq y < 2 \\ .4 & 2 \leq y < 3 \\ .6 & 3 \leq y < 4 \\ .8 & 4 \leq y < 5 \\ 1 & y \geq 5 \end{cases}$$

$$\text{c. } P(Y < 3) = F(2) = .4, P(Y \leq 3) = .6, P(Y = 3) = p(3) = .2$$

d. No, since Y is a discrete random variable.



4.3 a. The graph is above.

b. It is easily shown that all three properties hold.

4.4 A binomial variable with $n = 1$ has the Bernoulli distribution.

4.5 For $y = 2, 3, \dots$, $F(y) - F(y - 1) = P(Y \leq y) - P(Y \leq y - 1) = P(Y = y) = p(y)$. Also, $F(1) = P(Y \leq 1) = P(Y = 1) = p(1)$.

4.6 a. $F(i) = P(Y \leq i) = 1 - P(Y > i) = 1 - P(1^{\text{st}} i \text{ trials are failures}) = 1 - q^i$.

b. It is easily shown that all three properties hold.

4.7 a. $P(2 \leq Y < 5) = P(Y \leq 4) - P(Y \leq 1) = .967 - .376 = 0.591$
 $P(2 < Y < 5) = P(Y \leq 4) - P(Y \leq 2) = .967 - .678 = .289$.
 Y is a discrete variable, so they are not equal.

b. $P(2 \leq Y \leq 5) = P(Y \leq 5) - P(Y \leq 1) = .994 - .376 = 0.618$
 $P(2 < Y \leq 5) = P(Y \leq 5) - P(Y \leq 2) = .994 - .678 = 0.316$.
 Y is a discrete variable, so they are not equal.

c. Y is not a continuous random variable, so the earlier result do not hold.

4.8 a. The constant $k = 6$ is required so the density function integrates to 1.

b. $P(.4 \leq Y \leq 1) = .648$.

c. Same as part b. above.

d. $P(Y \leq .4 \mid Y \leq .8) = P(Y \leq .4)/P(Y \leq .8) = .352/.896 = 0.393$.

e. Same as part d. above.

4.9 a. Y is a discrete random variable because $F(y)$ is not a continuous function. Also, the set of possible values of Y represents a countable set.

b. These values are 2, 2.5, 4, 5.5, 6, and 7.

c. $p(2) = 1/8$, $p(2.5) = 3/16 - 1/8 = 1/16$, $p(4) = 1/2 - 3/16 = 5/16$, $p(5.5) = 5/8 - 1/2 = 1/8$, $p(6) = 11/16 - 5/8 = 1/16$, $p(7) = 1 - 11/16 = 5/16$.

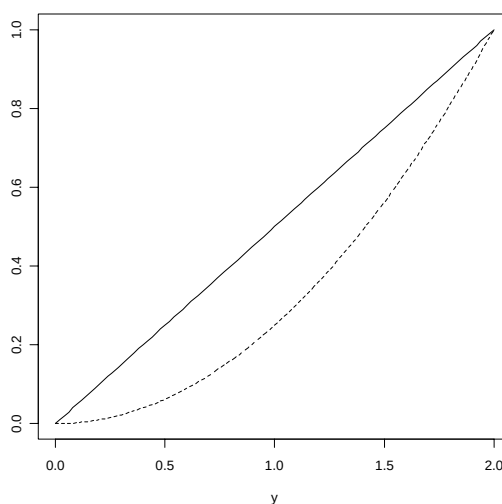
d. $P(Y \leq \phi_{.5}) = F(\phi_{.5}) = .5$, so $\phi_{.5} = 4$.

4.10 a. $F(\phi_{.95}) = \int_0^{\phi_{.95}} 6y(1-y)dy = .95$, so $\phi_{.95} = 0.865$.

b. Since Y is a continuous random variable, $y_0 = \phi_{.95} = 0.865$.

4.11 a. $\int_0^2 cydy = [cy^2/2]_0^2 = 2c = 1$, so $c = 1/2$.

b. $F(y) = \int_{-\infty}^y f(t)dt = \int_0^y \frac{t}{2} dt = \frac{y^2}{4}$, $0 \leq y \leq 2$. 



c. Solid line: $f(y)$; dashed line: $F(y)$

d. $P(1 \leq Y \leq 2) = F(2) - F(1) = 1 - .25 = .75$.

e. Note that $P(1 \leq Y \leq 2) = 1 - P(0 \leq Y < 1)$. The region $(0 \leq y < 1)$ forms a triangle (in the density graph above) with a base of 1 and a height of .5. So, $P(0 \leq Y < 1) = \frac{1}{2}(1)(.5) = .25$ and $P(1 \leq Y \leq 2) = 1 - .25 = .75$.

4.12 a. $F(-\infty) = 0$, $F(\infty) = 1$, and $F(y_1) - F(y_2) = e^{-y_2^2} - e^{-y_1^2} > 0$ provided $y_1 > y_2$.

b. $F(\phi_3) = 1 - e^{-\phi_3^2} = .3$, so $\phi_3 = \sqrt{-\ln(.7)} = 0.5972$.

c. $f(y) = F'(y) = 2ye^{-y^2}$ for $y \geq 0$ and 0 elsewhere.

d. $P(Y \geq 200) = 1 - P(Y < 200) = 1 - P(Y \leq 200) = 1 - F(2) = e^{-4}$.

e. $P(Y > 100 \mid Y \leq 200) = P(100 < Y \leq 200)/P(Y \leq 200) = [F(2) - F(1)]/F(2) = \frac{e^{-1} - e^{-4}}{1 - e^{-4}}$.

4.13 a. For $0 \leq y \leq 1$, $F(y) = \int_0^y t dt = y^2/2$. For $1 < y \leq 1.5$, $F(y) = \int_0^1 t dt + \int_1^y dt = 1/2 + y - 1 = y - 1/2$. Hence,

$$F(y) = \begin{cases} 0 & y < 0 \\ y^2/2 & 0 \leq y \leq 1 \\ y - 1/2 & 1 < y \leq 1.5 \\ 1 & y > 1.5 \end{cases}$$

b. $P(0 \leq Y \leq .5) = F(.5) = 1/8$.

c. $P(.5 \leq Y \leq 1.2) = F(1.2) - F(.5) = 1.2 - 1/2 - 1/8 = .575$.

4.14 a. A triangular distribution.

b. For $0 < y < 1$, $F(y) = \int_0^y t dt = y^2/2$. For $1 \leq y < 2$, $F(y) = \int_0^1 t dt + \int_1^y (2-t) dt = 2y - \frac{y^2}{2} - 1$.

c. $P(.8 \leq Y \leq 1.2) = F(1.2) - F(.8) = .36$.

d. $P(Y > 1.5 \mid Y > 1) = P(Y > 1.5)/P(Y > 1) = .125/.5 = .25$.

4.15 a. For $b \geq 0$, $f(y) \geq 0$. Also, $\int_{-\infty}^{\infty} f(y) dy = \int_b^{\infty} b/y^2 dy = -b/y \Big|_b^{\infty} = 1$.

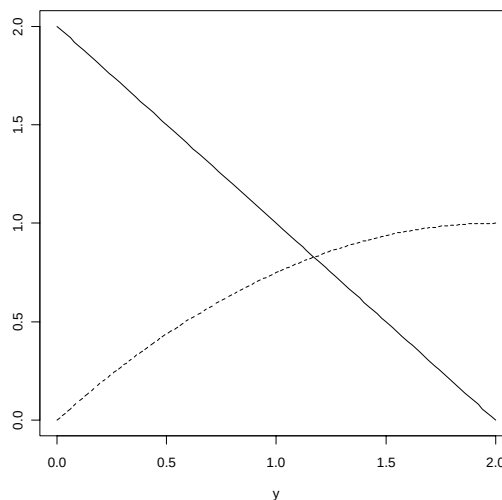
b. $F(y) = 1 - b/y$, for $y \geq b$, 0 elsewhere.

c. $P(Y > b + c) = 1 - F(b + c) = b/(b + c)$.

d. Applying part c., $P(Y > b + d \mid Y > b + c) = (b + c)/(b + d)$.

4.16 a. $\int_0^2 c(2-y)dy = c \left[2y - y^2/2 \right]_0^2 = 2c = 1$, so $c = 1/2$.

b. $F(y) = y - y^2/4$, for $0 \leq y \leq 2$.

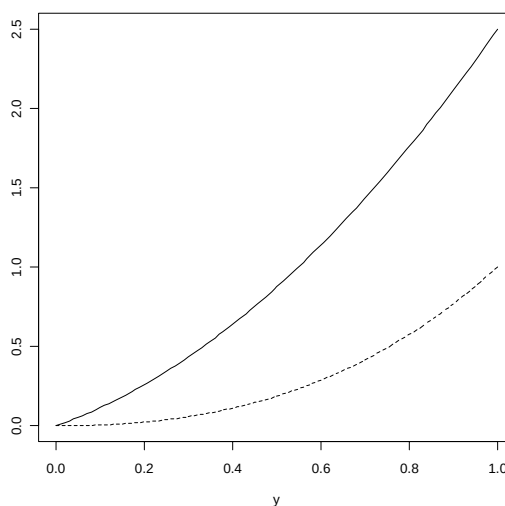


c. Solid line: $f(y)$; dashed line: $F(y)$

d. $P(1 \leq Y \leq 2) = F(2) - F(1) = 1/4$.

4.17 a. $\int_0^1 (cy^2 + y)dy = \left[cy^3/3 + y^2/2 \right]_0^1 = 1$, $c = 3/2$.

b. $F(y) = y^3/2 + y^2/2$ for $0 \leq y \leq 1$.



c. Solid line: $f(y)$; dashed line: $F(y)$

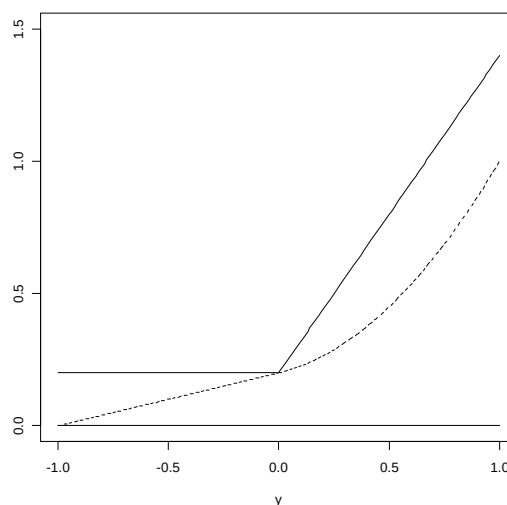
d. $F(-1) = 0, F(0) = 0, F(1) = 1.$

e. $P(Y < .5) = F(.5) = 3/16.$

f. $P(Y \geq .5 \mid Y \geq .25) = P(Y \geq .5)/P(Y \geq .25) = 104/123.$

4.18 a. $\int_{-1}^0 .2 dy + \int_0^1 (.2 + cy) dy = .4 + c/2 = 1, \text{ so } c = 1.2.$

b.
$$F(y) = \begin{cases} 0 & y \leq -1 \\ .2(1+y) & -1 < y \leq 0 \\ .2(1+y+3y^2) & 0 < y \leq 1 \\ 1 & y > 1 \end{cases}$$



c. Solid line: $f(y)$; dashed line: $F(y)$

d. $F(-1) = 0, F(0) = .2, F(1) = 1$

e. $P(Y > .5 \mid Y > .1) = P(Y > .5)/P(Y > .1) = .55/.774 = .71.$

4.19 a. Differentiating $F(y)$ with respect to y , we have

$$f(y) = \begin{cases} 0 & y \leq 0 \\ .125 & 0 < y < 2 \\ .125y & 2 \leq y < 4 \\ 0 & y \geq 4 \end{cases}$$

b. $F(3) - F(1) = 7/16$

c. $1 - F(1.5) = 13/16$

d. $7/16/(9/16) = 7/9.$

4.20 From Ex. 4.16:

$$E(Y) = \int_0^2 .5y(2-y)dy = \left[\frac{y^2}{2} - \frac{y^3}{6} \right]_0^2 = 2/3, \quad E(Y^2) = \int_0^2 .5y^2(2-y)dy = \left[\frac{y^3}{3} - \frac{y^4}{8} \right]_0^2 = 2/3.$$

$$\text{So, } V(Y) = 2/3 - (2/3)^2 = 2/9.$$

4.21 From Ex. 4.17:

$$E(Y) = \int_0^1 1.5y^3 + y^2 dy = \left[\frac{3y^4}{8} + \frac{y^3}{3} \right]_0^1 = 17/24 = .708.$$

$$E(Y^2) = \int_0^1 1.5y^4 + y^3 dy = \left[\frac{3y^5}{10} + \frac{y^4}{4} \right]_0^1 = 3/10 + 1/4 = .55.$$

$$\text{So, } V(Y) = .55 - (.708)^2 = .0487.$$

4.22 From Ex. 4.18:

$$E(Y) = \int_{-1}^0 .2ydy + \int_0^1 (.2y + 1.2y^2)dy = .4, \quad E(Y^2) = \int_{-1}^0 .2y^2dy + \int_0^1 (.2y^2 + 1.2y^3)dy = 1.3/3.$$

$$\text{So, } V(Y) = 1.3/3 - (.4)^2 = .2733.$$

4.23 1. $E(c) = \int_{-\infty}^{\infty} cf(y)dy = c \int_{-\infty}^{\infty} f(y)dy = c(1) = c.$

2. $E[cg(Y)] = \int_{-\infty}^{\infty} cg(y)f(y)dy = c \int_{-\infty}^{\infty} g(y)f(y)dy = cE[g(Y)].$

3. $E[g_1(Y) + g_2(Y) + \cdots + g_k(Y)] = \int_{-\infty}^{\infty} [g_1(y) + g_2(y) + \cdots + g_k(y)]f(y)dy$
 $= \int_{-\infty}^{\infty} g_1(y)f(y)dy + \int_{-\infty}^{\infty} g_2(y)f(y)dy + \cdots + \int_{-\infty}^{\infty} g_k(y)f(y)dy$
 $= E[g_1(Y)] + E[g_2(Y)] + \cdots + E[g_k(Y)].$

4.24 $V(Y) = E\{[Y - E(Y)]^2\} = E\{Y^2 - 2YE(Y) + [E(Y)]^2\} = E(Y^2) - 2[E(Y)]^2 + [E(Y)]^2$
 $= E(Y^2) - [E(Y)]^2 = \sigma^2.$

4.25 Ex. 4.19:

$$E(Y) = \int_0^2 .125ydy + \int_2^4 .125y^2dy = 31/12, \quad E(Y^2) = \int_0^2 .125y^2dy + \int_2^4 .125y^3dy = 47/6.$$

$$\text{So, } V(Y) = 47/6 - (31/12)^2 = 1.16.$$

4.26 a. $E(aY + b) = \int_{-\infty}^{\infty} (ay + b)f(y)dy = \int_{-\infty}^{\infty} ayf(y)dy + \int_{-\infty}^{\infty} bf(y)dy = aE(Y) + b = a\mu + b.$

b. $V(aY + b) = E\{[aY + b - E(aY + b)]^2\} = E\{[aY + b - a\mu - b]^2\} = E\{a^2[Y - \mu]^2\}$
 $= a^2V(Y) = a^2\sigma^2.$

4.27 First note that from Ex. 4.21, $E(Y) = .708$ and $V(Y) = .0487$. Then,
 $E(W) = E(5 - .5Y) = 5 - .5E(Y) = 5 - .5(.708) = \4.65 .
 $V(W) = V(5 - .5Y) = .25V(Y) = .25(.0487) = .012$.

4.28 a. By using the methods learned in this chapter, $c = 105$.

b. $E(Y) = 105 \int_0^1 y^3 (1-y)^4 dy = 3/8$.

4.29 $E(Y) = .5 \int_{59}^{61} y dy = .5 \frac{y^2}{2} \Big|_{59}^{61} = 60$, $E(Y^2) = .5 \int_{59}^{61} y^2 dy = .5 \frac{y^3}{3} \Big|_{59}^{61} = 3600 \frac{1}{3}$. Thus,
 $V(Y) = 3600 \frac{1}{3} - (60)^2 = \frac{1}{3}$.

4.30 a. $E(Y) = \int_0^1 2y^2 dy = 2/3$, $E(Y^2) = \int_0^1 2y^3 dy = 1/2$. Thus, $V(Y) = 1/2 - (2/3)^2 = 1/18$.

b. With $X = 200Y - 60$, $E(X) = 200(2/3) - 60 = 220/3$, $V(X) = 20000/9$.

c. Using Tchebysheff's theorem, a two standard deviation interval about the mean is given by $220/3 \pm 2\sqrt{20000/9}$ or $(-20.948, 167.614)$.

4.31 $E(Y) = \int_2^6 y \left(\frac{3}{32}\right)(y-2)(6-y) dy = 4$.

4.32 a. $E(Y) = \frac{3}{64} \int_0^4 y^3 (4-y) dy = \frac{3}{64} \left[y^4 - \frac{y^5}{5} \right]_0^4 = 2.4$. $V(Y) = .64$.

b. $E(200Y) = 200(2.4) = \$480$, $V(200Y) = 200^2(.64) = 25,600$.

c. $P(200Y > 600) = P(Y > 3) = \frac{3}{64} \int_3^4 y^2 (4-y) dy = .2616$, or about 26% of the time the cost will exceed \$600 (fairly common).

4.33 a. $E(Y) = \frac{3}{8} \int_5^7 y(7-y)^2 dy = \frac{3}{8} \left[\frac{49}{2} y^2 - \frac{14}{3} y^3 + \frac{y^4}{4} \right]_5^7 = 5.5$

$E(Y^2) = \frac{3}{8} \int_5^7 y^2 (7-y)^2 dy = \frac{3}{8} \left[\frac{49}{3} y^3 - \frac{14}{4} y^4 + \frac{y^5}{5} \right]_5^7 = 30.4$, so $V(Y) = .15$.

b. Using Tchebysheff's theorem, a two standard deviation interval about the mean is given by $5.5 \pm 2\sqrt{.15}$ or $(4.725, 6.275)$. Since $Y \geq 5$, the interval is $(5, 6.275)$.

c. $P(Y < 5.5) = \frac{3}{8} \int_5^{5.5} (7-y)^2 dy = .5781$, or about 58% of the time (quite common).

$$4.34 \quad E(Y) = \int_0^{\infty} yf(y)dy = \int_0^{\infty} \left(\int_0^y dt \right) f(y)dy = \int_0^{\infty} \left(\int_y^{\infty} f(y)dy \right) dt = \int_0^{\infty} P(Y > y)dy = \int_0^{\infty} [1 - F(y)]dy.$$

$$4.35 \quad \text{Let } \mu = E(Y). \text{ Then, } E[(Y - a)^2] = E[(Y - \mu + \mu - a)^2] \\ = E[(Y - \mu)^2] - 2E[(Y - \mu)(\mu - a)] + (\mu - a)^2 \\ = \sigma^2 + (\mu - a)^2.$$

The above quantity is minimized when $\mu = a$.

4.36 This is also valid for discrete random variables — the properties of expected values used in the proof hold for both continuous and discrete random variables.

$$4.37 \quad E(Y) = \int_{-\infty}^{\infty} yf(y)dy = \int_{-\infty}^0 yf(y)dy + \int_0^{\infty} yf(y)dy. \text{ In the first integral, let } w = -y. \text{ Then,}$$

$$E(Y) = -\int_0^{\infty} wf(-w)dy + \int_0^{\infty} yf(y)dy = -\int_0^{\infty} wf(w)dy + \int_0^{\infty} yf(y)dy = 0.$$

$$4.38 \quad \text{a. } F(y) = \begin{cases} 0 & y < 0 \\ \int_0^y 1dy = y & 0 \leq y \leq 1 \\ 1 & y > 1 \end{cases}$$

$$\text{b. } P(a \leq Y \leq a + b) = F(a + b) - F(a) = a + b - a = b.$$

4.39 The distance Y is uniformly distributed on the interval A to B . If she is closer to A , she has landed in the interval $(A, \frac{A+B}{2})$. This is one half the total interval length, so the probability is .5. If her distance to A is more than three times her distance to B , she has landed in the interval $(\frac{3B+A}{4}, B)$. This is one quarter the total interval length, so the probability is .25.

4.40 The probability of landing past the midpoint is 1/2 according to the uniform distribution. Let $X = \#$ parachutists that land past the midpoint of (A, B) . Therefore, X is binomial with $n = 3$ and $p = 1/2$. $P(X = 1) = 3(1/2)^3 = .375$.

$$4.41 \quad \text{First find } E(Y^2) = \frac{1}{\theta_2 - \theta_1} \int_{\theta_1}^{\theta_2} y^2 dy = \frac{1}{\theta_2 - \theta_1} \left[\frac{y^3}{3} \right]_{\theta_1}^{\theta_2} = \frac{\theta_2^3 - \theta_1^3}{3(\theta_2 - \theta_1)} = \frac{\theta_1^2 + \theta_1\theta_2 + \theta_2^2}{3}. \text{ Thus,}$$

$$V(Y) = \frac{\theta_1^2 + \theta_1\theta_2 + \theta_2^2}{3} - \left(\frac{\theta_2 + \theta_1}{2} \right)^2 = \frac{(\theta_2 - \theta_1)^2}{12}.$$

4.42 The distribution function is $F(y) = \frac{y - \theta_1}{\theta_2 - \theta_1}$, for $\theta_1 \leq y \leq \theta_2$. For $F(\phi_{.5}) = .5$, then $\phi_{.5} = \theta_1 + .5(\theta_2 - \theta_1) = .5(\theta_2 + \theta_1)$. This is also the mean of the distribution.

4.43 Let $A = \pi R^2$, where R has a uniform distribution on the interval $(0, 1)$. Then,

$$E(A) = \pi E(R^2) = \pi \int_0^1 r^2 dr = \frac{\pi}{3}$$

$$V(A) = \pi^2 V(R^2) = \pi^2 [E(R^4) - \left(\frac{1}{3}\right)^2] = \pi^2 \left[\int_0^1 r^4 dr - \left(\frac{1}{3}\right)^2 \right] = \pi^2 \left[\frac{1}{5} - \left(\frac{1}{3}\right)^2 \right] = \frac{4\pi^2}{45}.$$

4.44 a. Y has a uniform distribution (constant density function), so $k = 1/4$.

$$\text{b. } F(y) = \begin{cases} 0 & y < -2 \\ \int_{-2}^y \frac{1}{4} dy = \frac{y+2}{4} & -2 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$

4.45 Let Y = low bid (in thousands of dollars) on the next intrastate shipping contract. Then, Y is uniform on the interval $(20, 25)$.

a. $P(Y < 22) = 2/5 = .4$

b. $P(Y > 24) = 1/5 = .2$.

4.46 Mean of the uniform: $(25 + 20)/2 = 22.5$.

4.47 The density for Y = delivery time is $f(y) = \frac{1}{4}$, $1 \leq y \leq 5$. Also, $E(Y) = 3$, $V(Y) = 4/3$.

a. $P(Y > 2) = 3/4$.

b. $E(C) = E(c_0 + c_1 Y^2) = c_0 + c_1 E(Y^2) = c_0 + c_1 [V(Y) + (E(Y))^2] = c_0 + c_1 [4/3 + 9]$

4.48 Let Y = location of the selected point. Then, Y has a uniform distribution on the interval $(0, 500)$.

a. $P(475 \leq Y \leq 500) = 1/20$

b. $P(0 \leq Y \leq 25) = 1/20$

c. $P(0 < Y < 250) = 1/2$.

4.49 If Y has a uniform distribution on the interval $(0, 1)$, then $P(Y > 1/4) = 3/4$.

4.50 Let Y = time when the phone call comes in. Then, Y has a uniform distribution on the interval $(0, 5)$. The probability is $P(0 < Y < 1) + P(3 < Y < 4) = .4$.

4.51 Let Y = cycle time. Thus, Y has a uniform distribution on the interval $(50, 70)$. Then,

$$P(Y > 65 | Y > 55) = P(Y > 65) / P(Y > 55) = .25 / (.75) = 1/3.$$

4.52 Mean and variance of a uniform distribution: $\mu = 60$, $\sigma^2 = (70-50)^2/12 = 100/3$.

4.53 Let Y = time when the defective circuit board was produced. Then, Y has an approximate uniform distribution on the interval $(0, 8)$.

a. $P(0 < Y < 1) = 1/8$.

b. $P(7 < Y < 8) = 1/8$

c. $P(4 < Y < 5 | Y > 4) = P(4 < Y < 5)/P(Y > 4) = (1/8)/(1/2) = 1/4$.

4.54 Let Y = amount of measurement error. Then, Y is uniform on the interval $(-.05, .05)$.

a. $P(-.01 < Y < .01) = .2$

b. $E(Y) = 0$, $V(Y) = (.05 + .05)^2/12 = .00083$.

4.55 Let Y = amount of measurement error. Then, Y is uniform on the interval $(-.02, .05)$.

a. $P(-.01 < Y < .01) = 2/7$

b. $E(Y) = (-.02 + .05)/2 = .015$, $V(Y) = (.05 + .02)^2/12 = .00041$.

4.56 From Example 4.7, the arrival time Y has a uniform distribution on the interval $(0, 30)$. Then, $P(25 < Y < 30 | Y > 10) = 1/6/(2/3) = 1/4$.

4.57 The volume of a sphere is given by $(4/3)\pi r^3 = (1/6)\pi d^3$, where r is the radius and d is the diameter. Let D = diameter such that D is uniform distribution on the interval $(.01, .05)$.

Thus, $E(\frac{\pi}{6}D^3) = \frac{\pi}{6} \int_{.01}^{.05} d^3 \frac{1}{4} dd = .0000065\pi$. By similar logic used in Ex. 4.43, it can be found that $V(\frac{\pi}{6}D^3) = .0003525\pi^2$.

4.58 a. $P(0 \leq Z \leq 1.2) = .5 - .1151 = .3849$

b. $P(-.9 \leq Z \leq 0) = .5 - .1841 = .3159$.

c. $P(.3 \leq Z \leq 1.56) = .3821 - .0594 = .3227$.

d. $P(-.2 \leq Z \leq .2) = 1 - 2(.4207) = .1586$.

e. $P(-1.56 \leq Z \leq -.2) = .4207 - .0594 = .3613$

f. $P(0 \leq Z \leq 1.2) = .38493$. The desired probability is for a standard normal.

4.59 a. $z_0 = 0$.

b. $z_0 = 1.10$

c. $z_0 = 1.645$

d. $z_0 = 2.576$

4.60 The parameter σ must be positive, otherwise the density function could obtain a negative value (a violation).

4.61 Since the density function is symmetric about the parameter μ , $P(Y < \mu) = P(Y > \mu) = .5$. Thus, μ is the median of the distribution, regardless of the value of σ .

4.62 a. $P(Z^2 < 1) = P(-1 < Z < 1) = .6826$.

b. $P(Z^2 < 3.84146) = P(-1.96 < Z < 1.96) = .95$.

- 4.63** a. Note that the value 17 is $(17 - 16)/1 = 1$ standard deviation above the mean. So, $P(Z > 1) = .1587$.
b. The same answer is obtained.
- 4.64** a. Note that the value 450 is $(450 - 400)/20 = 2.5$ standard deviations above the mean. So, $P(Z > 2.5) = .0062$.
b. The probability is .00618.
c. The top scale is for the standard normal and the bottom scale is for a normal distribution with mean 400 and standard deviation 20.
- 4.65** For the standard normal, $P(Z > z_0) = .1$ if $z_0 = 1.28$. So, $y_0 = 400 + 1.28(20) = \$425.60$.
- 4.66** Let Y = bearing diameter, so Y is normal with $\mu = 3.0005$ and $\sigma = .0010$. Thus, Fraction of scrap = $P(Y > 3.002) + P(Y < 2.998) = P(Z > 1.5) + P(Z < -2.5) = .0730$.
- 4.67** In order to minimize the scrap fraction, we need the maximum amount in the specifications interval. Since the normal distribution is symmetric, the mean diameter should be set to be the midpoint of the interval, or $\mu = 3.000$ in.
- 4.68** The GPA 3.0 is $(3.0 - 2.4)/.8 = .75$ standard deviations above the mean. So, $P(Z > .75) = .2266$.
- 4.69** The z -score for 1.9 is $(1.9 - 2.4)/.8 = -.625$. Thus, $P(Z < -.625) = .2660$.
- 4.70** From Ex. 4.68, the proportion of students with a GPA greater than 3.0 is .2266. Let X = # in the sample with a GPA greater than 3.0. Thus, X is binomial with $n = 3$ and $p = .2266$. Then, $P(X = 3) = (.2266)^3 = .0116$.
- 4.71** Let Y = the measured resistance of a randomly selected wire.
a. $P(.12 \leq Y \leq .14) = P(\frac{.12-.13}{.005} \leq Z \leq \frac{.14-.13}{.005}) = P(-2 \leq Z \leq 2) = .9544$.
b. Let X = # of wires that ~~do not~~ meet specifications. Then, X is binomial with $n = 4$ and $p = .9544$. Thus, $P(X = 4) = (.9544)^4 = .8297$.
- 4.72** Let Y = interest rate forecast, so Y has a normal distribution with $\mu = .07$ and $\sigma = .026$.
a. $P(Y > .11) = P(Z > \frac{.11-.07}{.026}) = P(Z > 1.54) = .0618$.
b. $P(Y < .09) = P(Z > \frac{.09-.07}{.026}) = P(Z > .77) = .7794$.
- 4.73** Let Y = width of a bolt of fabric, so Y has a normal distribution with $\mu = 950$ mm and $\sigma = 10$ mm.
a. $P(947 \leq Y \leq 958) = P(\frac{947-950}{10} \leq Z \leq \frac{958-950}{10}) = P(-.3 \leq Z \leq .8) = .406$
b. It is necessary that $P(Y \leq c) = .8531$. Note that for the standard normal, we find that $P(Z \leq z_0) = .8531$ when $z_0 = 1.05$. So, $c = 950 + (1.05)(10) = 960.5$ mm.

- 4.74** Let Y = examination score, so Y has a normal distribution with $\mu = 78$ and $\sigma^2 = 36$.
- $P(Y > 72) = P(Z > -1) = .8413$.
 - We seek c such that $P(Y > c) = .1$. For the standard normal, $P(Z > z_0) = .1$ when $z_0 = 1.28$. So $c = 78 + (1.28)(6) = 85.68$.
 - We seek c such that $P(Y > c) = .281$. For the standard normal, $P(Z > z_0) = .281$ when $z_0 = .58$. So, $c = 78 + (.58)(6) = 81.48$.
 - For the standard normal, $P(Z < -.67) = .25$. So, the score that cuts off the lowest 25% is given by $(-.67)(6) + 78 = 73.98$.
 - Similar answers are obtained.
 - $P(Y > 84 | Y > 72) = P(Y > 84)/P(Y > 72) = P(Z > 1)/P(Z > -1) = .1587/.8413 = .1886$.
- 4.75** Let Y = volume filled, so that Y is normal with mean μ and $\sigma = .3$ oz. They require that $P(Y > 8) = .01$. For the standard normal, $P(Z > z_0) = .01$ when $z_0 = 2.33$. Therefore, it must hold that $2.33 = (8 - \mu)/.3$, so $\mu = 7.301$.
- 4.76** It follows that $.95 = P(|Y - \mu| < 1) = P(|Z| < 1/\sigma)$, so that $1/\sigma = 1.96$ or $\sigma = 1/1.96 = .5102$.
- 4.77** **a.** Let Y = SAT math score. Then, $P(Y < 550) = P(Z < .7) = 0.758$.
- b.** If we choose the same percentile, $18 + 6(.7) = 22.2$ would be comparable on the ACT math test.
- 4.78** Easiest way: maximize the function $\ln f(y) = -\ln(\sigma\sqrt{2\pi}) - \frac{(y-\mu)^2}{2\sigma^2}$ to obtain the maximum at $y = \mu$ and observe that $f(\mu) = 1/(\sigma\sqrt{2\pi})$.
- 4.79** The second derivative of $f(y)$ is found to be $f''(y) = \left(\frac{1}{\sigma^3\sqrt{2\pi}}\right)e^{-(y-\mu)^2/2\sigma^2} \left[1 - \frac{(\mu-y)^2}{\sigma^2}\right]$. Setting this equal to 0, we must have that $\left[1 - \frac{(\mu-y)^2}{\sigma^2}\right] = 0$ (the other quantities are strictly positive). The two solutions are $y = \mu + \sigma$ and $y = \mu - \sigma$.
- 4.80** Observe that $A = L*W = |Y| \times 3|Y| = 3Y^2$. Thus, $E(A) = 3E(Y^2) = 3(\sigma^2 + \mu^2)$.
- 4.81** **a.** $\Gamma(1) = \int_0^{\infty} e^{-y} dy = -e^{-y} \Big|_0^{\infty} = 1$.
- b.** $\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy = \left[-y^{\alpha-1} e^{-y}\right]_0^{\infty} + \int_0^{\infty} (\alpha-1)y^{\alpha-2} e^{-y} dy = (\alpha-1)\Gamma(\alpha-1)$.
- 4.82** From above we have $\Gamma(1) = 1$, so that $\Gamma(2) = 1\Gamma(1) = 1$, $\Gamma(3) = 2\Gamma(2) = 2(1)$, and generally $\Gamma(n) = (n-1)\Gamma(n-1) = (n-1)!$ $\Gamma(4) = 3! = 6$ and $\Gamma(7) = 6! = 720$.
- 4.83** Applet Exercise — the results should agree.

- 4.84** a. The larger the value of α , the more symmetric the density curve.
b. The location of the distribution centers are increasing with α .
c. The means of the distributions are increasing with α .
- 4.85** a. These are all exponential densities.
b. Yes, they are all skewed densities (decaying exponential).
c. The spread is increasing with β .
- 4.86** a. $P(Y < 3.5) = .37412$
b. $P(W < 1.75) = P(Y/2 < 1.75) = P(Y < 3.5) = .37412$.
c. They are identical.
- 4.87** a. For the gamma distribution, $\phi_{.05} = .70369$.
b. For the χ^2 distribution, $\phi_{.05} = .35185$.
c. The .05-quantile for the χ^2 distribution is exactly one-half that of the .05-quantile for the gamma distribution. It is due to the relationship stated in Ex. 4.86.
- 4.88** Let Y have an exponential distribution with $\beta = 2.4$.
a. $P(Y > 3) = \int_3^{\infty} \frac{1}{2.4} e^{-y/2.4} dy = e^{-3/2.4} = .2865$.
b. $P(2 \leq Y \leq 3) = \int_2^3 \frac{1}{2.4} e^{-y/2.4} dy = .1481$.
- 4.89** a. Note that $\int_2^{\infty} \frac{1}{\beta} e^{-y/\beta} dy = e^{-2/\beta} = .0821$, so $\beta = .8$
b. $P(Y \leq 1.7) = 1 - e^{-1.7/.8} = .5$ 
- 4.90** Let Y = magnitude of the earthquake which is exponential with $\beta = 2.4$. Let X = # of earthquakes that exceed 5.0 on the Richter scale. Therefore, X is binomial with $n = 10$ and $p = P(Y > 5) = \int_5^{\infty} \frac{1}{2.4} e^{-y/2.4} dy = e^{-5/2.4} = .1245$. Finally, the probability of interest is
$$P(X \geq 1) = 1 - P(X = 0) = 1 - (.8755)^{10} = 1 - .2646 = .7354$$
.
- 4.91** Let Y = water demand in the early afternoon. Then, Y is exponential with $\beta = 100$ cfs.
a. $P(Y > 200) = \int_{200}^{\infty} \frac{1}{100} e^{-y/100} dy = e^{-2} = .1353$.
b. We require the 99th percentile of the distribution of Y :
$$P(Y > \phi_{.99}) = \int_{\phi_{.99}}^{\infty} \frac{1}{100} e^{-y/100} dy = e^{-\phi_{.99}/100} = .01$$
. So, $\phi_{.99} = -100 \ln(.01) = 460.52$ cfs.

- 4.92** The random variable Y has an exponential distribution with $\beta = 10$. The cost C is related to Y by the formula $C = 100 + 40Y + 3Y^2$. Thus,

$$E(C) = E(100 + 40Y + 3Y^2) = 100 + 40(10) + 3E(Y^2) = 100 + 400 + 3(100 + 10^2) = 1100.$$

To find $V(C)$, note that $V(C) = E(C^2) - [E(C)]^2$. Therefore,

$$E(C^2) = E[(100 + 40Y + 3Y^2)^2] = 10,000 + 2200E(Y^2) + 9E(Y^4) + 8000E(Y) + 240E(Y^3).$$

$$E(Y) = 10 \qquad E(Y^2) = 200$$

$$E(Y^3) = \int_0^{\infty} y^3 \frac{1}{100} e^{-y/100} dy = \Gamma(4)100^3 = 6000.$$

$$E(Y^4) = \int_0^{\infty} y^4 \frac{1}{100} e^{-y/100} dy = \Gamma(5)100^4 = 240,000.$$

$$\text{Thus, } E(C^2) = 10,000 + 2200(200) + 9(240,000) + 8000(10) + 240(6000) = 4,130,000.$$

$$\text{So, } V(C) = 4,130,000 - (1100)^2 = 2,920,000.$$

- 4.93** Let Y = time between fatal airplane accidents. So, Y is exponential with $\beta = 44$ days.

$$\text{a. } P(Y \leq 31) = \int_0^{31} \frac{1}{44} e^{-y/44} dy = 1 - e^{-31/44} = .5057.$$

$$\text{b. } V(Y) = 44^2 = 1936.$$

- 4.94** Let Y = CO concentration in air samples. So, Y is exponential with $\beta = 3.6$ ppm.

$$\text{a. } P(Y > 9) = \int_9^{\infty} \frac{1}{3.6} e^{-y/3.6} dy = e^{-9/3.6} = .0821$$

$$\text{b. } P(Y > 9) = \int_9^{\infty} \frac{1}{2.5} e^{-y/3.6} dy = e^{-9/2.5} = .0273$$

- 4.95** a. For any $k = 1, 2, 3, \dots$

$$P(X = k) = P(k-1 \leq Y < k) = P(Y < k) - P(Y \leq k-1) = 1 - e^{-k/\beta} - (1 - e^{-(k-1)/\beta}) = e^{-(k-1)/\beta} - e^{-k/\beta}.$$

$$\text{b. } P(X = k) = e^{-(k-1)/\beta} - e^{-k/\beta} = e^{-(k-1)/\beta} - e^{-(k-1)/\beta} (e^{1/\beta}) = e^{-(k-1)/\beta} (1 - e^{1/\beta}) = [e^{-1/\beta}]^{k-1} (1 - e^{1/\beta}).$$

Thus, X has a geometric distribution with $p = 1 - e^{1/\beta}$.

- 4.96** a. The density function $f(y)$ is in the form of a gamma density with $\alpha = 4$ and $\beta = 2$.

Thus, $k = \frac{1}{\Gamma(4)2^4} = \frac{1}{96}$.

- b. Y has a χ^2 distribution with $v = 2(4) = 8$ degrees of freedom.

c. $E(Y) = 4(2) = 8$, $V(Y) = 4(2^2) = 16$. 

- d. Note that $\sigma = \sqrt{16} = 4$. Thus, $P(|Y - 8| < 2(4)) = P(0 < Y < 16) = .95762$.

4.97 $P(Y > 4) = \int_4^{\infty} \frac{1}{4} e^{-y/4} dy = e^{-1} = .3679$.

- 4.98** We require the 95th percentile of the distribution of Y :

$$P(Y > \phi_{.95}) = \int_{\phi_{.95}}^{\infty} \frac{1}{4} e^{-y/4} dy = e^{-\phi_{.95}/4} = .05. \text{ So, } \phi_{.95} = -4\ln(.05) = 11.98.$$

4.99 a. $P(Y > 1) = \sum_{y=0}^{\infty} \frac{e^{-1}}{y!} = e^{-1} + e^{-1} = .7358$.

- b. The same answer is found.

4.100 a. $P(X_1 = 0) = e^{-\lambda_1}$ and $P(X_2 = 0) = e^{-\lambda_2}$. Since $\lambda_2 > \lambda_1$, $e^{-\lambda_2} < e^{-\lambda_1}$.

- b. The result follows from Ex. 4.100.

- c. Since distribution function is a nondecreasing function, it follows from part b that

$$P(X_1 \leq k) = P(Y > \lambda_1) > P(Y > \lambda_2) = P(X_2 \leq k)$$

- d. We say that X_2 is “stochastically greater” than X_1 .

- 4.101** Let Y have a gamma distribution with $\alpha = .8$, $\beta = 2.4$.

a. $E(Y) = (.8)(2.4) = 1.92$

b. $P(Y > 3) = .21036$

- c. The probability found in Ex. 4.88 (a) is larger. There is greater variability with the exponential distribution.

d. $P(2 \leq Y \leq 3) = P(Y > 2) - P(Y > 3) = .33979 - .21036 = .12943$.

- 4.102** Let Y have a gamma distribution with $\alpha = 1.5$, $\beta = 3$.

a. $P(Y > 4) = .44592$.

- b. We require the 95th percentile: $\phi_{.95} = 11.72209$.

- 4.103** Let R denote the radius of a crater. Therefore, R is exponential w/ $\beta = 10$ and the area is $A = \pi R^2$. Thus,

$$E(A) = E(\pi R^2) = \pi E(R^2) = \pi(100 + 100) = 200\pi.$$

$$V(A) = E(A^2) - [E(A)]^2 = \pi^2[E(R^4) - 200^2] = \pi^2[240,000 - 200^2] = 200,000\pi^2,$$

$$\text{where } E(R^4) = \int_0^{\infty} \frac{1}{10} r^4 e^{-r/10} dr = 10^4 \Gamma(5) = 240,000.$$

- 4.104** Y has an exponential distribution with $\beta = 100$. Then, $P(Y > 200) = e^{-200/100} = e^{-2}$. Let the random variable $X = \#$ of componential that operate in the equipment for more than 200 hours. Then, X has a binomial distribution and

$$P(\text{equipment operates}) = P(X \geq 2) = P(X = 2) + P(X = 3) = 3(e^{-2})^2(1 - e^{-2}) + (e^{-2})^3 = .05.$$

- 4.105** Let the random variable $Y =$ four-week summer rainfall totals

a. $E(Y) = 1.6(2) = 3.2$, $V(Y) = 1.6(2^2) = 6.4$

b. $P(Y > 4) = .28955$.

- 4.106** Let $Y =$ response time. If $\mu = 4$ and $\sigma^2 = 8$, then it is clear that $\alpha = 2$ and $\beta = 2$.

a. $f(y) = \frac{1}{4} y e^{-y/2}$, $y > 0$.

b. $P(Y < 5) = 1 - .2873 = .7127$.

- 4.107** a. Using Tchebysheff's theorem, two standard deviations about the mean is given by $4 \pm 2\sqrt{8} = 4 \pm 5.657$ or $(-1.657, 9.657)$, or simply $(0, 9.657)$ since Y must be positive.

b. $P(Y < 9.657) = 1 - .04662 = 0.95338$.

- 4.108** Let $Y =$ annual income. Then, Y has a gamma distribution with $\alpha = 20$ and $\beta = 1000$.

a. $E(Y) = 20(1000) = 20,000$, $V(Y) = 20(1000)^2 = 20,000,000$.

b. The standard deviation $\sigma = \sqrt{20,000,000} = 4472.14$. The value 30,000 is $\frac{30,000 - 20,000}{4472.14} = 2.236$ standard deviations above the mean. This represents a fairly extreme value.

c. $P(Y > 30,000) = .02187$

- 4.109** Let Y have a gamma distribution with $\alpha = 3$ and $\beta = 2$. Then, the loss $L = 30Y + 2Y^2$. Then,

$$E(L) = E(30Y + 2Y^2) = 30E(Y) + 2E(Y^2) = 30(6) + 2(12 + 6^2) = 276,$$

$$V(L) = E(L^2) - [E(L)]^2 = E(900Y^2 + 120Y^3 + 4Y^4) - 276^2.$$

$$E(Y^3) = \int_0^{\infty} \frac{y^5}{16} e^{-y/2} dy = 480$$

$$E(Y^4) = \int_0^{\infty} \frac{y^6}{16} e^{-y/2} dy = 5760$$

$$\text{Thus, } V(L) = 900(480) + 120(480) + 4(5760) - 276^2 = 47,664.$$

4.110 Y has a gamma distribution with $\alpha = 3$ and $\beta = .5$. Thus, $E(Y) = 1.5$ and $V(Y) = .75$.

4.111 a. $E(Y^a) = \int_0^{\infty} y^a \frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} e^{-y/\beta} dy = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^{\infty} y^{a+\alpha-1} e^{-y/\beta} dy = \frac{1}{\Gamma(\alpha)\beta^\alpha} \frac{\Gamma(a+\alpha)\beta^{a+\alpha}}{1} = \frac{\beta^a \Gamma(a+\alpha)}{\Gamma(\alpha)}.$

b. For the gamma function $\Gamma(t)$, we require $t > 0$.

c. $E(Y^1) = \frac{\beta^1 \Gamma(1+\alpha)}{\Gamma(\alpha)} = \frac{\beta \alpha \Gamma(\alpha)}{\Gamma(\alpha)} = \alpha \beta.$

d. $E(\sqrt{Y}) = E(Y^{.5}) = \frac{\beta^{.5} \Gamma(.5+\alpha)}{\Gamma(\alpha)}, \alpha > 0.$

e. $E(1/Y) = E(Y^{-1}) = \frac{\beta^{-1} \Gamma(-1+\alpha)}{\Gamma(\alpha)} = \frac{1}{\beta(\alpha-1)}, \alpha > 1.$

$E(1/\sqrt{Y}) = E(Y^{-.5}) = \frac{\beta^{-.5} \Gamma(-.5+\alpha)}{\Gamma(\alpha)} = \frac{\Gamma(\alpha-.5)}{\sqrt{\beta} \Gamma(\alpha)}, \alpha > .5.$

$E(1/Y^2) = E(Y^{-2}) = \frac{\beta^{-2} \Gamma(-2+\alpha)}{\Gamma(\alpha)} = \frac{\beta^{-2} \Gamma(\alpha-2)}{(\alpha-1)(\alpha-2)\Gamma(\alpha-2)} \frac{1}{\beta^2(\alpha-1)(\alpha-2)}, \alpha > 2.$

4.112 The chi-square distribution with v degrees of freedom is the same as a gamma distribution with $\alpha = v/2$ and $\beta = 2$.

a. From Ex. 4.111, $E(Y^a) = \frac{2^a \Gamma(a+\frac{v}{2})}{\Gamma(\frac{v}{2})}.$

b. As in Ex. 4.111 with $\alpha + a > 0$ and $\alpha = v/2$, it must hold that $v > -2a$

c. $E(\sqrt{Y}) = E(Y^{.5}) = \frac{\sqrt{2} \Gamma(\frac{v+1}{2})}{\Gamma(\frac{v}{2})}, v > 0.$

d. $E(1/Y) = E(Y^{-1}) = \frac{2^{-1} \Gamma(-1+\frac{v}{2})}{\Gamma(\frac{v}{2})} = \frac{1}{v-2}, v > 2.$

$E(1/\sqrt{Y}) = E(Y^{-.5}) = \frac{\Gamma(\frac{v-1}{2})}{\sqrt{2} \Gamma(\frac{v}{2})}, v > 1.$

$E(1/Y^2) = E(Y^{-2}) = \frac{1}{2^2 (\frac{v}{2}-1)(\frac{v}{2}-2)} = \frac{1}{(v-2)(v-4)}, \alpha > 4.$

4.113 Applet exercise.

4.114 a. This is the (standard) uniform distribution.

b. The beta density with $\alpha = 1, \beta = 1$ is symmetric.

c. The beta density with $\alpha = 1, \beta = 2$ is skewed right.

d. The beta density with $\alpha = 2, \beta = 1$ is skewed left.

e. Yes.

4.115 a. The means of all three distributions are .5.

b. They are all symmetric.

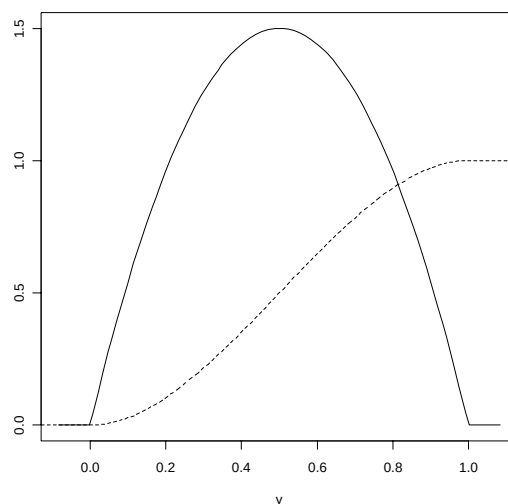
c. The spread decreases with larger (and equal) values of α and β .

d. The standard deviations are .2236, .1900, and .1147 respectively. The standard deviations are decreasing which agrees with the density plots.

e. They are always symmetric when $\alpha = \beta$.

- 4.116** a. All of the densities are skewed right.
 b. The density obtains a more symmetric appearance.
 c. They are always skewed right when $\alpha < \beta$ and $\alpha > 1$ and $\beta > 1$.
- 4.117** a. All of the densities are skewed left.
 b. The density obtains a more symmetric appearance.
 c. They are always skewed right when $\alpha > \beta$ and $\alpha > 1$ and $\beta > 1$.
- 4.118** a. All of the densities are skewed right (similar to an exponential shape).
 b. The spread decreases as the value of β gets closer to 12.
 c. The distribution with $\alpha = .3$ and $\beta = 4$ has the highest probability.
 d. The shapes are all similar.
- 4.119** a. All of the densities are skewed left (a mirror image of those from Ex. 4.118).
 b. The spread decreases as the value of α gets closer to 12.
 c. The distribution with $\alpha = 4$ and $\beta = .3$ has the highest probability.
 d. The shapes are all similar.
- 4.120** Yes, the mapping explains the mirror image.
- 4.121** a. These distributions exhibit a “U” shape.
 b. The area beneath the curve is greater closer to “1” than “0”.
- 4.122** a. $P(Y > .1) = .13418$
 b. $P(Y < .1) = 1 - .13418 = .86582$.
 c. Values smaller than .1 have greater probability.
 d. $P(Y < .1) = 1 - .45176 = .54824$
 e. $P(Y > .9) = .21951$.
 f. $P(0.1 < Y < 0.9) = 1 - .54824 - .21951 = .23225$.
 g. Values of $Y < .1$ have the greatest probability.
- 4.123** a. The random variable Y follows the beta distribution with $\alpha = 4$ and $\beta = 3$, so the constant $k = \frac{\Gamma(4+3)}{\Gamma(4)\Gamma(3)} = \frac{6!}{3!2!} = 60$.
 b. We require the 95th percentile of this distribution, so it is found that $\phi_{.95} = 0.84684$.
- 4.124** a. $P(Y > .4) = \int_{.4}^1 (12y^2 - 12y^3)dy = [4y^3 - 3y^4]_{.4}^1 = .8208$.
 b. $P(Y > .4) = .82080$.
- 4.125** From Ex. 4.124 and using the formulas for the mean and variance of beta random variables, $E(Y) = 3/5$ and $V(Y) = 1/25$.

4.126 a. $F(y) = \int_0^y (6t - 6t^2) dt = 3y^2 - 2y^3$, $0 \leq y \leq 1$. $F(y) = 0$ for $y < 0$ and $F(y) = 1$ for $y > 1$.



b. Solid line: $f(y)$; dashed line: $F(y)$

c. $P(.5 \leq Y \leq .8) = F(.8) - F(.5) = 1.92 - 1.092 - .75 + .25 = .396$.

4.127 For $\alpha = \beta = 1$, $f(y) = \frac{\Gamma(2)}{\Gamma(1)\Gamma(1)} y^{1-1} (1-y)^{1-1} = 1$, $0 \leq y \leq 1$, which is the uniform distribution.

4.128 The random variable Y = weekly repair cost (in hundreds of dollars) has a beta distribution with $\alpha = 1$ and $\beta = 3$. We require the 90th percentile of this distribution:

$$P(Y > \phi_{.9}) = \int_{\phi_{.9}}^1 3(1-y)^2 dy = (1 - \phi_{.9})^3 = .1.$$

Therefore, $\phi_{.9} = 1 - (.1)^{1/3} = .5358$. So, the budgeted cost should be \$53.58.

4.129 $E(C) = 10 + 20E(Y) + 4E(Y^2) = 10 + 20\left(\frac{1}{3}\right) + 4\left(\frac{2}{9 \cdot 4} + \frac{1}{9}\right) = \frac{52}{3}$

$$V(C) = E(C^2) - [E(C)]^2 = E[(10 + 20Y + 4Y^2)^2] - \left(\frac{52}{3}\right)^2$$

$$E[(10 + 20Y + 4Y^2)^2] = 100 + 400E(Y) + 480E(Y^2) + 160E(Y^3) + 16E(Y^4)$$

Using mathematical expectation, $E(Y^3) = \frac{1}{10}$ and $E(Y^4) = \frac{1}{15}$. So,

$$V(C) = E(C^2) - [E(C)]^2 = (100 + 400/3 + 480/6 + 160/10 + 16/15) - (52/3)^2 = 29.96.$$

4.130 To find the variance $\sigma^2 = E(Y^2) - \mu^2$:

$$E(Y^2) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{\alpha+1} (1-y)^{\beta-1} dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+2+\beta)} = \frac{(\alpha+1)\alpha}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$\sigma^2 = \frac{(\alpha+1)\alpha}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

4.131 This is the same beta distribution used in Ex. 4.129.

a. $P(Y < .5) = \int_0^{.5} 2(1-y)dy = 2y - y^2 \Big|_0^{.5} = .75$

b. $E(Y) = 1/3$, $V(Y) = 1/18$, so $\sigma = 1/\sqrt{18} = .2357$.

4.132 Let Y = proportion of weight contributed by the fine powders

a. $E(Y) = .5$, $V(Y) = 9/(36*7) = 1/28$

b. $E(Y) = .5$, $V(Y) = 4/(16*5) = 1/20$

c. $E(Y) = .5$, $V(Y) = 1/(4*3) = 1/12$

d. Case (a) will yield the most homogenous blend since the variance is the smallest.

4.133 The random variable Y has a beta distribution with $\alpha = 3$, $\beta = 5$.

a. The constant $c = \frac{\Gamma(3+5)}{\Gamma(3)\Gamma(5)} = \frac{7!}{2!4!} = 105$.

b. $E(Y) = 3/8$.

c. $V(Y) = 15/(64*9) = 5/192$, so $\sigma = .1614$.

d. $P(Y > .375 + 2(.1614)) = P(Y > .6978) = .02972$.

4.134 a. If $\alpha = 4$ and $\beta = 7$, then we must find

$$P(Y \leq .7) = F(.7) = \sum_{i=4}^{10} \binom{10}{i} (.7)^i (.3)^{10-i} = P(4 \leq X \leq 10), \text{ for the random variable } X$$

distributed as binomial with $n = 10$ and $p = .7$. Using Table I in Appendix III, this is .989.

b. Similarly, $F(.6) = P(12 \leq X \leq 25)$, for the random variable X distributed as binomial with $n = 25$ and $p = .6$. Using Table I in Appendix III, this is .922.

c. Similar answers are found.

4.135 a. $P(Y_1 = 0) = (1 - p_1)^n > P(Y_2 = 0) = (1 - p_2)^n$, since $p_1 < p_2$.

b. $P(Y_1 \leq k) = 1 - P(Y_1 \geq k+1) = 1 - \sum_{i=k+1}^n \binom{n}{i} p_1^i (1-p_1)^{n-i} = 1 - \int_0^{p_1} \frac{t^k (1-t)^{n-k-1}}{B(k+1, n-k)} dt$
 $= 1 - P(X \leq p_1) = P(X > p_1)$, where X is beta with parameters $k+1$, $n-k$.

c. From part b, we see the integrands for $P(Y_1 \leq k)$ and $P(Y_2 \leq k)$ are identical but since $p_1 < p_2$, the regions of integration are different. So, Y_2 is “stochastically greater” than Y_1 .

- 4.136 a.** Observing that the exponential distribution is a special case of the gamma distribution, we can use the gamma moment-generating function with $\alpha = 1$ and $\beta = \theta$:

$$m(t) = \frac{1}{1 - \theta t}, \quad t < 1/\theta.$$

- b.** The first two moments are found by $m'(t) = \frac{\theta}{(1 - \theta t)^2}$, $E(Y) = m'(0) = \theta$.

$$m''(t) = \frac{2\theta}{(1 - \theta t)^3}, \quad E(Y^2) = m''(0) = 2\theta^2. \quad \text{So, } V(Y) = 2\theta^2 - \theta^2 = \theta^2.$$

- 4.137** The mgf for U is $m_U(t) = E(e^{tU}) = E(e^{t(aY+b)}) = E(e^{bt} e^{(at)Y}) = e^{bt} m(at)$. Thus,

$$m'_U(t) = be^{bt} m(at) + ae^{bt} m'(at). \quad \text{So, } m'_U(0) = b + am'(0) = b + a\mu = E(U).$$

$$m''_U(t) = b^2 e^{bt} m(at) + abe^{bt} m'(at) + abe^{bt} m'(at) + a^2 e^{bt} m''(at), \quad \text{so}$$

$$m''_U(0) = b^2 + 2ab\mu + a^2 E(Y^2) = E(U^2).$$

$$\text{Therefore, } V(U) = b^2 + 2ab\mu + a^2 E(Y^2) - (b + a\mu)^2 = a^2 [E(Y^2) - \mu^2] = a^2 \sigma^2.$$

- 4.138 a.** For $U = Y - \mu$, the mgf $m_U(t)$ is given in Example 4.16. To find the mgf for $Y = U + \mu$, use the result in Ex. 4.137 with $a = 1$, $b = -\mu$:

$$m_Y(t) = e^{-\mu t} m_U(t) = e^{\mu t + \sigma^2 t^2 / 2}$$

- b.** $m'_Y(t) = (\mu + t\sigma^2) e^{\mu t + \sigma^2 t^2 / 2}$, so $m'_Y(0) = \mu$

$$m''_Y(t) = (\mu + t\sigma^2)^2 e^{\mu t + \sigma^2 t^2 / 2} + \sigma^2 e^{\mu t + \sigma^2 t^2 / 2}, \quad \text{so } m''_Y(0) = \mu^2 + \sigma^2. \quad \text{Finally, } V(Y) = \sigma^2.$$

- 4.139** Using Ex. 4.137 with $a = -3$ and $b = 4$, it is trivial to see that the mgf for X is

$$m_X(t) = e^{4t} m(-3t) = e^{(4-3\mu)t + 9\sigma^2 t^2 / 2}.$$

By the uniqueness of mgfs, X is normal with mean $4 - 3\mu$ and variance $9\sigma^2$.

- 4.140 a.** Gamma with $\alpha = 2$, $\beta = 4$
b. Exponential with $\beta = 3.2$
c. Normal with $\mu = -5$, $\sigma^2 = 12$

4.141 $m(t) = E(e^{tY}) = \int_{\theta_1}^{\theta_2} \frac{e^{ty}}{\theta_2 - \theta_1} dy = \frac{e^{\theta_2 t} - e^{\theta_1 t}}{t(\theta_2 - \theta_1)}.$

4.142 a. $m_Y(t) = \frac{e^t - 1}{t}$

b. From the cited exercises, $m_W(t) = \frac{e^{at} - 1}{at}$. From the uniqueness property of mgfs, W is uniform on the interval $(0, a)$.

c. The mgf for X is $m_X(t) = \frac{e^{-at} - 1}{-at}$, which implies that X is uniform on the interval $(-a, 0)$.

d. The mgf for V is $m_V(t) = e^{bt} \frac{e^{at} - 1}{at} = \frac{e^{(b+a)t} - e^{bt}}{at}$, which implies that V is uniform on the interval $(b, b + a)$.

4.143 The mgf for the gamma distribution is $m(t) = (1 - \beta t)^{-\alpha}$. Thus,

$$m'(t) = \alpha\beta(1 - \beta t)^{-\alpha-1}, \text{ so } m'(0) = \alpha\beta = E(Y)$$

$$m''(t) = (\alpha + 1)\alpha\beta^2(1 - \beta t)^{-\alpha-2}, \text{ so } m''(0) = (\alpha + 1)\alpha\beta^2 = E(Y^2). \text{ So,}$$

$$V(Y) = (\alpha + 1)\alpha\beta^2 - (\alpha\beta)^2 = \alpha\beta^2.$$

4.144 a. The density shown is a normal distribution with $\mu = 0$ and $\sigma^2 = 1$. Thus, $k = 1/\sqrt{2\pi}$.

b. From Ex. 4.138, the mgf is $m(t) = e^{t^2/2}$.

c. $E(Y) = 0$ and $V(Y) = 1$.

4.145 a. $E(e^{3Y/2}) = \int_{-\infty}^0 e^{3y/2} e^y dy = \frac{2}{5} e^{5y/2} \Big|_{-\infty}^0 = \frac{2}{5}.$

b. $m(t) = E(e^{tY}) = \int_{-\infty}^0 e^{ty} e^y dy = \frac{1}{t+1}, t > -1.$

c. By using the methods with mgfs, $E(Y) = -1$, $E(Y^2) = 2$, so $V(Y) = 2 - (-1)^2 = 1$.

4.146 We require $P(|Y - \mu| \leq k\sigma) \geq .90 = 1 - 1/k^2$. Solving for k , we see that $k = 3.1622$. Thus, the necessary interval is $|Y - 25,000| \leq (3.1622)(4000)$ or $12,351 \leq 37,649$. 

4.147 We require $P(|Y - \mu| \leq .1) \geq .75 = 1 - 1/k^2$. Thus, $k = 2$. Using Tchebysheff's inequality, $1 = k\sigma$ and so $\sigma = 1/2$.

4.148 In Exercise 4.16, $\mu = 2/3$ and $\sigma = \sqrt{2/9} = .4714$. Thus,

$$P(|Y - \mu| \leq 2\sigma) = P(|Y - 2/3| \leq .9428) = P(-.2761 \leq Y \leq 1.609) = F(1.609) = .962.$$

Note that the negative portion of the interval in the probability statement is irrelevant since Y is non-negative. According to Tchebysheff's inequality, the probability is at least 75%. The empirical rule states that the probability is approximately 95%. The above probability is closest to the empirical rule, even though the density function is far from mound shaped.

4.149 For the uniform distribution on (θ_1, θ_2) , $\mu = \frac{\theta_1 + \theta_2}{2}$ and $\sigma^2 = \frac{(\theta_2 - \theta_1)^2}{12}$. Thus,

$$2\sigma = \frac{(\theta_2 - \theta_1)}{\sqrt{3}}.$$

The probability of interest is

$$P(|Y - \mu| \leq 2\sigma) = P(\mu - 2\sigma \leq Y \leq \mu + 2\sigma) = P\left(\frac{\theta_1 + \theta_2}{2} - \frac{(\theta_2 - \theta_1)}{\sqrt{3}} \leq Y \leq \frac{\theta_1 + \theta_2}{2} + \frac{(\theta_2 - \theta_1)}{\sqrt{3}}\right)$$

It is not difficult to show that the range in the last probability statement is greater than the actual interval that Y is restricted to, so

$$P\left(\frac{\theta_1 + \theta_2}{2} - \frac{(\theta_2 - \theta_1)}{\sqrt{3}} \leq Y \leq \frac{\theta_1 + \theta_2}{2} + \frac{(\theta_2 - \theta_1)}{\sqrt{3}}\right) = P(\theta_1 \leq Y \leq \theta_2) = 1.$$

Note that Tchebysheff's theorem is satisfied, but the probability is greater than what is given by the empirical rule. The uniform is not a mound-shaped distribution.

4.150 For the exponential distribution, $\mu = \beta$ and $\sigma^2 = \beta^2$. Thus, $2\sigma = 2\beta$. The probability of interest is

$$P(|Y - \mu| \leq 2\sigma) = P(\mu - 2\sigma \leq Y \leq \mu + 2\sigma) = P(-\beta \leq Y \leq 3\beta) = P(0 \leq Y \leq 3\beta)$$

This is simply $F(3\beta) = 1 - e^{-3\beta} = .9502$. The empirical rule and Tchebysheff's theorem are both valid.

4.151 From Exercise 4.92, $E(C) = 1000$ and $V(C) = 2,920,000$ so that the standard deviation is $\sqrt{2,920,000} = 1708.80$. The value 2000 is only $(2000 - 1100)/1708.8 = .53$ standard deviations above the mean. Thus, we would expect C to exceed 2000 fair often.

4.152 We require $P(|L - \mu| \leq k\sigma) \geq .89 = 1 - 1/k^2$. Solving for k , we have $k = 3.015$. From Ex. 4.109, $\mu = 276$ and $\sigma = 218.32$. The interval is

$$|L - 276| \leq 3.015(218.32) \text{ or } (-382.23, 934.23)$$

Since L must be positive, the interval is $(0, 934.23)$

4.153 From Ex. 4.129, it is shown that $E(C) = \frac{52}{3}$ and $V(C) = 29.96$, so, the standard deviation is $\sqrt{29.96} = 5.474$. Thus, using Tchebysheff's theorem with $k = 2$, the interval is

$$|Y - \frac{52}{3}| \leq 2(5.474) \text{ or } (6.38, 28.28)$$

4.154 a. $\mu = 7$, $\sigma^2 = 2(7) = 14$.

b. Note that $\sigma = \sqrt{14} = 3.742$. The value 23 is $(23 - 7)/3.742 = 4.276$ standard deviations above the mean, which is unlikely.

c. With $\alpha = 3.5$ and $\beta = 2$, $P(Y > 23) = .00170$.

- 4.155** The random variable Y is uniform over the interval $(1, 4)$. Thus, $f(y) = \frac{1}{3}$ for $1 \leq y \leq 4$ and $f(y) = 0$ elsewhere. The random variable $C = \text{cost of the delay}$ is given as

$$C = g(Y) = \begin{cases} 100 & 1 \leq y \leq 2 \\ 100 + 20(Y - 2) & 2 < y \leq 4 \end{cases}$$

$$\text{Thus, } E(C) = E[g(Y)] = \int_{-\infty}^{\infty} g(y) f(y) dy = \int_1^2 \frac{100}{3} dy + \int_2^4 [100 + 20(y - 2)] \frac{1}{3} dy = \$113.33.$$

- 4.156** Note that Y is a discrete random variable with probability $.2 + .1 = .3$ and it is continuous with probability $1 - .3 = .7$. Hence, by using Definition 4.15, we can write Y as a mixture of two random variables X_1 and X_2 . The random variable X_1 is discrete and can assume two values with probabilities $P(X_1 = 3) = .2/.3 = 2/3$ and $P(X_1 = 6) = .1/.3 = 1/3$. Thus, $E(X_1) = 3(2/3) + 6(1/3) = 4$. The random variable X_2 is continuous and follows a gamma distribution (as given in the problem) so that $E(X_2) = 2(2) = 4$. Therefore,

$$E(Y) = .3(4) + .7(4) = 4.$$

- 4.157 a.** The distribution function for X is $F(x) = \begin{cases} 0 & x < 0 \\ \int_0^x \frac{1}{100} e^{-t/100} dt = 1 - e^{-x/100} & 0 \leq x < 200 \\ 1 & x \geq 200 \end{cases}$.

b. $E(X) = \int_0^{200} x \frac{1}{100} e^{-x/100} dx + .1353(200) = 86.47$, where $.1353 = P(Y > 200)$.

- 4.158** The distribution for V is gamma with $\alpha = 4$ and $\beta = 500$. Since there is one discrete point at 0 with probability .02, using Definition 4.15 we have that $c_1 = .02$ and $c_2 = .98$. Denoting the kinetic energy as $K = \frac{m}{2} V^2$ we can solve for the expected value:

$$E(K) = (.98) \frac{m}{2} E(V^2) = (.98) \frac{m}{2} \{V(V) + [E(V)]^2\} = (.98) \frac{m}{2} \{4(500)^2 + 2000^2\} = 2,450,000m.$$

- 4.159 a.** The distribution function has jumps at two points: $y = 0$ (of size .1) and $y = .5$ (of size .15). So, the discrete component of $F(y)$ is given by

$$F_1(y) = \begin{cases} 0 & y < 0 \\ \frac{.1}{.1+.15} = .4 & 0 \leq y < .5 \\ 1 & y \geq .5 \end{cases}$$

The continuous component of $F(y)$ can then be determined:

$$F_2(y) = \begin{cases} 0 & y < 0 \\ 4y^2/3 & 0 \leq y < .5 \\ (4y-1)/3 & .5 \leq y < 1 \\ 1 & y \geq 1 \end{cases}$$

b. Note that $c_1 = .1 + .15 = .25$. So, $F(y) = 0.25F_1(y) + 0.75F_2(y)$.

c. First, observe that $f_2(y) = F_2'(y) = \begin{cases} 8y/3 & 0 \leq y < .5 \\ 4/3 & y \geq .5 \end{cases}$. Thus,

$$E(Y) = .25(.6)(.5) + \int_0^{.5} 8y^2/3 dy + \int_{.5}^1 4y/3 dy = .533. \text{ Similarly, } E(Y^2) = .3604 \text{ so}$$

that $V(Y) = .076$.

4.160 a. $F(y) = \int_{-1}^y \frac{2}{\pi(1+y^2)} dy = \frac{2}{\pi} \tan^{-1}(y) + \frac{1}{2}, -1 \leq y \leq 1, F(y) = 0 \text{ if } y < 0, F(y) = 1 \text{ if } y > 1.$

b. Find $E(Y)$ directly using mathematical expectation, or observe that $f(y)$ is symmetric about 0 so using the result from Ex. 4.27, $E(Y) = 0$.

4.161 Here, $\mu = 70$ and $\sigma = 12$ with the normal distribution. We require $\phi_{.9}$, the 90th percentile of the distribution of test times. Since for the standard normal distribution, $P(Z < z_0) = .9$ for $z_0 = 1.28$, thus

$$\phi_{.9} = 70 + 12(1.28) = 85.36.$$

4.162 Here, $\mu = 500$ and $\sigma = 50$ with the normal distribution. We require $\phi_{.01}$, the 1st percentile of the distribution of light bulb lives. For the standard normal distribution, $P(Z < z_0) = .01$ for $z_0 = -2.33$, thus

$$\phi_{.01} = 500 + 50(-2.33) = 383.5$$

4.163 Referring to Ex. 4.66, let $X = \#$ of defective bearings. Thus, X is binomial with $n = 5$ and $p = P(\text{defective}) = .073$. Thus,

$$P(X > 1) = 1 - P(X = 0) = 1 - (.927)^5 = .3155.$$

4.164 Let $Y =$ lifetime of a drill bit. Then, Y has a normal distribution with $\mu = 75$ hours and $\sigma = 12$ hours.

a. $P(Y < 60) = P(Z < -1.25) = .1056$

b. $P(Y \geq 60) = 1 - P(Y < 60) = 1 - .1056 = .8944$.

c. $P(Y > 90) = P(Z > 1.25) = .1056$

4.165 The density function for Y is in the form of a gamma density with $\alpha = 2$ and $\beta = .5$.

a. $c = \frac{1}{\Gamma(2)(.5)^2} = 4$.

b. $E(Y) = 2(.5) = 1, V(Y) = 2(.5)^2 = .5$.

c. $m(t) = \frac{1}{(1-.5t)^2}, t < 2$.

4.166 In Example 4.16, the mgf is $m(t) = e^{t^2\sigma^2/2}$. The infinite series expansion of this is

$$m(t) = 1 + \left(\frac{t^2\sigma^2}{2}\right) + \left(\frac{t^2\sigma^2}{2}\right)^2 \frac{1}{2!} + \left(\frac{t^2\sigma^2}{2}\right)^3 \frac{1}{3!} + \cdots = 1 + \frac{t^2\sigma^2}{2} + \frac{t^4\sigma^4}{8} + \frac{t^6\sigma^6}{48} + \cdots$$

Then, $\mu_1 = \text{coefficient of } t, \text{ so } \mu_1 = 0$

$\mu_2 = \text{coefficient of } t^2/2!, \text{ so } \mu_2 = \sigma^2$

$\mu_3 = \text{coefficient of } t^3/3!, \text{ so } \mu_3 = 0$

$\mu_4 = \text{coefficient of } t^4/4!, \text{ so } \mu_4 = 3\sigma^4$

4.167 For the beta distribution,

$$E(Y^k) = \int_0^1 y^k \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1} (1-y)^{\beta-1} dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{k+\alpha-1} (1-y)^{\beta-1} dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(k+\alpha)\Gamma(\beta)}{\Gamma(k+\alpha+\beta)}.$$

$$\text{Thus, } E(Y^k) = \frac{\Gamma(\alpha+\beta)\Gamma(k+\alpha)}{\Gamma(\alpha)\Gamma(k+\alpha+\beta)}.$$

4.168 Let T = length of time until the first arrival. Thus, the distribution function for T is given by

$$F(t) = P(T \leq t) = 1 - P(T > t) = 1 - P[\text{no arrivals in } (0, t)] = 1 - P[N = 0 \text{ in } (0, t)]$$

The probability $P[N = 0 \text{ in } (0, t)]$ is given by $\frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-\lambda t}$. Thus, $F(t) = 1 - e^{-\lambda t}$ and

$$f(t) = \lambda e^{-\lambda t}, t > 0.$$

This is the exponential distribution with $\beta = 1/\lambda$.

4.169 Let Y = time between the arrival of two call, measured in hours. To find $P(Y > .25)$, note that $\lambda t = 10$ and $t = 1$. So, the density function for Y is given by $f(y) = 10e^{-10y}$, $y > 0$. Thus,

$$P(Y > .25) = e^{-10(.25)} = e^{-2.5} = .082.$$

4.170 a. Similar to Ex. 4.168, the second arrival will occur after time t if either one arrival has occurred in $(0, t)$ or no arrivals have occurred in $(0, t)$. Thus:

$$P(U > t) = P[\text{one arrival in } (0, t)] + P[\text{no arrivals in } (0, t)] = \frac{(\lambda t)^0 e^{-\lambda t}}{0!} + \frac{(\lambda t)^1 e^{-\lambda t}}{1!}. \text{ So,}$$

$$F(t) = 1 - P(U > t) = 1 - \frac{(\lambda t)^0 e^{-\lambda t}}{0!} - \frac{(\lambda t)^1 e^{-\lambda t}}{1!} = 1 - (\lambda t + 1)e^{-\lambda t}.$$

The density function is given by $f(t) = F'(t) = \lambda^2 t e^{-\lambda t}$, $t > 0$. This is a gamma density with $\alpha = 2$ and $\beta = 1/\lambda$.

b. Similar to part a, but let X = time until the k^{th} arrival. Thus, $P(X > t) = \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!}$. So,

$$F(t) = 1 - \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

The density function is given by

$$f(t) = F'(t) = -\left[-\lambda e^{-\lambda t} \sum_{n=0}^{k-1} \frac{(\lambda t)^n}{n!} + e^{-\lambda t} \sum_{n=1}^{k-1} \frac{\lambda^n t^{n-1}}{(n-1)!} \right] = \lambda e^{-\lambda t} \left[\sum_{n=0}^{k-1} \frac{(\lambda t)^n}{n!} - \sum_{n=1}^{k-1} \frac{(\lambda t)^{n-1}}{(n-1)!} \right]. \text{ Or,}$$

$$f(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!}, t > 0. \text{ This is a gamma density with } \alpha = k \text{ and } \beta = 1/\lambda.$$

4.171 From Ex. 4.169, W = waiting time follow an exponential distribution with $\beta = 1/2$.

a. $E(W) = 1/2, V(W) = 1/4.$

b. $P(\text{at least one more customer waiting}) = 1 - P(\text{no customers waiting in three minutes})$
 $= 1 - e^{-6}.$

4.172 Twenty seconds is $1/5$ a minute. The random variable Y = time between calls follows an exponential distribution with $\beta = .25$. Thus:

$$P(Y > 1/5) = \int_{1/5}^{\infty} 4e^{-4y} dy = e^{-4/5}.$$

4.173 Let R = distance to the nearest neighbor. Then,

$$P(R > r) = P(\text{no plants in a circle of radius } r)$$

Since the number of plants in a area of one unit has a Poisson distribution with mean λ , the number of plants in a area of πr^2 units has a Poisson distribution with mean $\lambda \pi r^2$. Thus,

$$F(r) = P(R \leq r) = 1 - P(R > r) = 1 - e^{-\lambda \pi r^2}.$$

So, $f(r) = F'(r) = 2\lambda \pi r e^{-\lambda \pi r^2}, r > 0.$

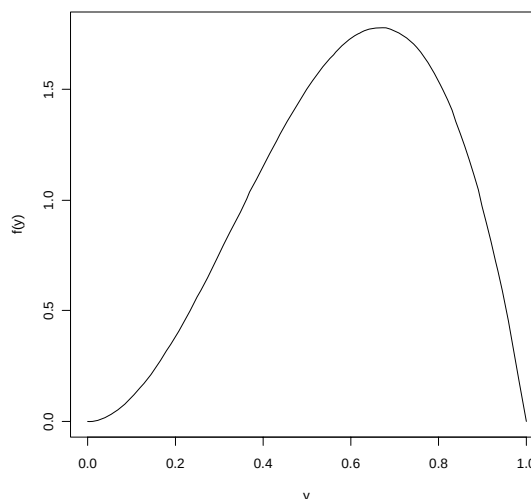
4.174 Let Y = interview time (in hours). The second applicant will have to wait only if the time to interview the first applicant exceeds 15 minutes, or .25 hour. So,

$$P(Y > .25) = \int_{.25}^{\infty} 2e^{-2y} dy = e^{-.5} = .61.$$

4.175 From Ex. 4.11, the median value will satisfy $F(y) = y^2/2 = .5$. Thus, the median is given by $\sqrt{2} = 1.414$.

4.176 The distribution function for the exponential distribution with mean β is $F(y) = 1 - e^{-y/\beta}$. Thus, we require the value y such that $F(y) = 1 - e^{-y/\beta} = .5$. Solving for y , this is $\beta \ln(2)$.

- 4.177** a. $2.07944 = 3\ln(2)$
 b. $3.35669 < 4$, the mean of the gamma distribution.
 c. $46.70909 < 50$, the mean of the gamma distribution.
 d. In all cases the median is *less* than the mean, indicating right skewed distributions.



- 4.178** The graph of this beta density is above.

- a. Using the relationship with binomial probabilities,
 $P(.1 \leq Y \leq .2) = 4(.2)^3(.8) + (.2)^4 - 4(.1)^3(.9) - (.1)^4 = .0235$.
 b. $P(.1 \leq Y \leq .2) = .9963 - .9728 = .0235$, which is the same answer as above.
 c. $\phi_{.05} = .24860$, $\phi_{.95} = .90239$.
 d. $P(\phi_{.05} \leq Y \leq \phi_{.95}) = .9$.
- 4.179** Let X represent the grocer's profit. In general, her profit (in cents) on a order of $100k$ pounds of food will be $X = 1000Y - 600k$ as long as $Y < k$. But, when $Y \geq k$ the grocer's profit will be $X = 1000k - 600k = 400k$. Define the random variable Y' as

$$Y' = \begin{cases} Y & 0 \leq Y < k \\ k & Y \geq k \end{cases}$$

Then, we can write $g(Y') = X = 1000Y' - 600k$. The random variable Y' has a mixed distribution with one discrete point at k . Therefore,

$$c_1 = P(Y' = k) = P(Y \geq k) = \int_k^1 3y^2 dy = 1 - k^3, \text{ so that } c_2 = k^3.$$

$$\text{Thus, } F_2(y) = \begin{cases} 0 & 0 \leq y < k \\ 1 & y \geq k \end{cases} \text{ and } F_1(y) = P(Y_2 \leq y | 0 \leq Y' < k) = \frac{\int_0^y 3t^2 dt}{k^3} = \frac{y^3}{k^3}, 0 \leq y < k.$$

Thus, from Definition 4.15,

$$E(X) = E[g(Y')] = c_1 E[g(Y_1)] + c_2 E[g(Y_2)] = (1 - k^3)400k + k^3 \int_0^k (1000y - 600k) \frac{3y^2}{k^3} dy,$$

or $E(X) = 400k - 250k^2$. This is maximized at $k = (.4)^{1/3} = .7368$. (2nd derivative is -.)

4.180 a. Using the result of Ex. 4.99, $P(Y \leq 4) = 1 - \sum_{y=0}^2 \frac{4^y e^{-4}}{y!} = .7619$.

b. A similar result is found.

4.181 The mgf for Z is $m_Z(t) = E(e^{Zt}) = E(e^{(\frac{y-\mu}{\sigma})t}) = e^{-\frac{\mu}{\sigma}t} m_Y(t/\sigma) = e^{t^2/2}$, which is a mgf for a normal distribution with $\mu = 0$ and $\sigma = 1$.

4.182 a. $P(Y \leq 4) = P(X \leq \ln 4) = P[Z \leq (\ln 4 - 4)/1] = P(Z \leq -2.61) = .0045$.

b. $P(Y > 8) = P(X > \ln 8) = P[Z > (\ln 8 - 4)/1] = P(Z > -1.92) = .9726$.

4.183 a. $E(Y) = e^{3+16/2} = e^{11}$ (598.74 g), $V(Y) = e^{22}(e^{16} - 1)10^{-4}$.

b. With $k = 2$, the interval is given by $E(Y) \pm 2\sqrt{V(Y)}$ or $598.74 \pm 3,569,038.7$. Since the weights must be positive, the final interval is $(0, 3,570,236.1)$

c. $P(Y < 598.74) = P(\ln Y < 6.3948) = P[Z < (6.3948 - 3)/4] = P(Z < .8487) = .8020$

4.184 The mgf for Y is $m_Y(t) = E(e^{tY}) = \frac{1}{2} \int_{-\infty}^0 e^{ty} e^y dy + \frac{1}{2} \int_0^{\infty} e^{ty} e^{-y} dy = \frac{1}{2} \int_{-\infty}^0 e^{(t+1)y} dy + \frac{1}{2} \int_0^{\infty} e^{-y(1-t)} dy$.

This simplifies to $m_Y(t) = \frac{1}{1-t^2}$. Using this, $E(Y) = m'(t)|_{t=0} = \frac{2t}{(1-t^2)}|_{t=0} = 0$.

4.185 a. $\int_{-\infty}^{\infty} f(y) dy = a \int_{-\infty}^{\infty} f_1(y) dy + (1-a) \int_{-\infty}^{\infty} f_2(y) dy = a + (1-a) = 1$.

b. i. $E(Y) = \int_{-\infty}^{\infty} yf(y) dy = a \int_{-\infty}^{\infty} yf_1(y) dy + (1-a) \int_{-\infty}^{\infty} yf_2(y) dy = a\mu_1 + (1-a)\mu_2$

ii. $E(Y_2) = a \int_{-\infty}^{\infty} y^2 f_1(y) dy + (1-a) \int_{-\infty}^{\infty} y^2 f_2(y) dy = a(\mu_1^2 + \sigma_1^2) + (1-a)(\mu_2^2 + \sigma_2^2)$. So,

$V(Y) = E(Y^2) - [E(Y)]^2 = a(\mu_1^2 + \sigma_1^2) + (1-a)(\mu_2^2 + \sigma_2^2) - [a\mu_1 + (1-a)\mu_2]^2$, which simplifies to $a\sigma_1^2 + (1-a)\sigma_2^2 + a(1-a)[\mu_1 - \mu_2]^2$

4.186 For $m = 2$, $E(Y) = \int_0^{\infty} y \frac{2y}{\alpha} e^{-y^2/\alpha} dy$. Let $u = y^2/\alpha$. Then, $dy = \frac{\sqrt{\alpha}}{2\sqrt{u}} du$. Then,

$E(Y) = \int_0^{\infty} \frac{2y^2}{\alpha} e^{-y^2/\alpha} dy = \sqrt{\alpha} \int_0^{\infty} u^{1/2} e^{-u} du = \sqrt{\alpha} \Gamma(3/2) = \frac{\sqrt{\alpha} \Gamma(1/2)}{2}$. Using similar methods,

it can be shown that $E(Y^2) = \alpha$ so that $V(Y) = \alpha - \left[\frac{\sqrt{\alpha} \Gamma(1/2)}{2} \right]^2 = \alpha \left[1 - \frac{\pi}{4} \right]$, since it will

be shown in Ex. 4.196 that $\Gamma(1/2) = \sqrt{\pi}$.

4.187 The density for Y = the life length of a resistor (in thousands of hours) is

$$f(y) = \frac{2ye^{-y^2/10}}{10}, \quad y > 0.$$

a. $P(Y > 5) = \int_5^{\infty} \frac{2ye^{-y^2/10}}{10} dy = -e^{-y^2/10} \Big|_5^{\infty} = e^{-2.5} = .082.$

b. Let X = # of resistors that burn out prior to 5000 hours. Thus, X is a binomial random variable with $n = 3$ and $p = .082$. Then, $P(X = 1) = 3(1 - .082)(.082)^2 = .0186$.

4.188 a. This is the exponential distribution with $\beta = \alpha$.

b. Using the substitution $u = y^m/\alpha$ in the integrals below, we find:

$$E(Y) = \int_0^{\infty} \frac{m}{\alpha} y^m e^{-y^m/\alpha} dy = \alpha^{1/m} \int_0^{\infty} u^{1/m} e^{-u} du = \alpha^{1/m} \Gamma(1 + 1/m)$$

$$E(Y^2) = \int_0^{\infty} \frac{m}{\alpha} y^{m+1} e^{-y^m/\alpha} dy = \alpha^{2/m} \int_0^{\infty} u^{2/m} e^{-u} du = \alpha^{2/m} \Gamma(1 + 2/m).$$

Thus,

$$V(Y) = \alpha^{2/m} [\Gamma(1 + 2/m) + \Gamma^2(1 + 1/m)].$$

4.189 Since this density is symmetric about 0, so using the result from Ex. 4.27, $E(Y) = 0$.

Also, it is clear that $V(Y) = E(Y^2)$. Thus,

$$E(Y^2) = \int_{-1}^1 \frac{1}{B(1/2, (n-2)/2)} y^2 (1-y^2)^{(n-4)/2} dy = \frac{B(3/2, (n-2)/2)}{B(1/2, (n-2)/2)} = \frac{1}{n-1} = V(Y). \text{ This}$$

equality follows after making the substitution $u = y^2$.

4.190 a. For the exponential distribution, $f(t) = \frac{1}{\beta} e^{-t/\beta}$ and $F(t) = 1 - e^{-t/\beta}$. Thus, $r(t) = 1/\beta$.

b. For the Weibull, $f(y) = \frac{m}{\alpha} y^{m-1} e^{-y^m/\alpha}$ and $F(y) = 1 - e^{-y^m/\alpha}$. Thus, $r(t) = \frac{my^{m-1}}{\alpha}$, which is an increasing function of t when $m > 1$.

4.191 a. $G(y) = P(Y \leq y | Y \geq c) = \frac{P(c \leq Y \leq y)}{P(Y \geq c)} = \frac{F(y) - F(c)}{1 - F(c)}.$

b. Refer to the properties of distribution functions; namely, show $G(-\infty) = 0$, $G(\infty) = 1$, and for constants a and b such that $a \leq b$, $G(a) \leq G(b)$.

c. It is given that $F(y) = 1 - e^{-y^2/3}$. Thus, by using the result in part b above,

$$P(Y \leq 4 | Y \geq 2) = \frac{1 - e^{-4^2/3} - (1 - e^{-2^2/3})}{e^{-2^2/3}} = 1 - e^{-4}.$$

4.192 a. $E(V) = 4\pi \left(\frac{m}{2\pi KT}\right)^{3/2} \int_0^{\infty} v^3 e^{-v^2(m/2KT)} dv$. To evaluate this integral, let $u = v^2 \left(\frac{m}{2KT}\right)$ so that

$$dv = \sqrt{\frac{2KT}{m}} \frac{1}{2\sqrt{u}} du \text{ to obtain } E(V) = 2\sqrt{\frac{2KT}{m\pi}} \int_0^{\infty} u e^{-u} du = 2\sqrt{\frac{2KT}{m\pi}} \Gamma(2) = 2\sqrt{\frac{2KT}{m\pi}}.$$

b. $E(\frac{1}{2}mV^2) = \frac{1}{2}mE(V^2) = 2\pi m \left(\frac{m}{2\pi KT}\right)^{3/2} \int_0^\infty v^4 e^{-v^2(m/2KT)} dv$. Here, let $u = v^2 \left(\frac{m}{2KT}\right)$ so that $dv = \sqrt{\frac{2KT}{m}} \frac{1}{2\sqrt{u}} du$ to obtain $E(\frac{1}{2}mV^2) = \frac{2KT}{\sqrt{\pi}} \Gamma(\frac{5}{2}) = \frac{3}{2}KT$ (here, we again used the result from Ex. 4.196 where it is shown that $\Gamma(1/2) = \sqrt{\pi}$).

4.193 For $f(y) = \frac{1}{100}e^{-y/100}$, we have that $F(y) = 1 - e^{-y/100}$. Thus,

$$E(Y | Y \geq 50) = \frac{1}{e^{-1/2}} \int_{50}^{\infty} \frac{ye^{-y/100}}{100} dy = 150.$$

Note that this value is $50 + 100$, where 100 is the (unconditional) mean of Y . This illustrates the memoryless property of the exponential distribution.

4.194 $\left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1/2)uy^2} dy \right] \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1/2)ux^2} dx \right] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(1/2)u(x^2+y^2)} dx dy$. By changing to polar coordinates, $x^2 + y^2 = r^2$ and $dx dy = r dr d\theta$. Thus, the desired integral becomes

$$\frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} e^{-(1/2)ur^2} r dr d\theta = \frac{1}{u}.$$

Note that the result proves that the standard normal density integrates to 1 with $u = 1$.

4.195 a. First note that $W = (Z^2 + 3Z)^2 = Z^4 + 6Z^3 + 9Z^2$. The odd moments of the standard normal are equal to 0, and $E(Z^2) = V(Z) + [E(Z)]^2 = 1 + 0 = 1$. Also, using the result in Ex. 4.199, $E(Z^4) = 3$ so that $E(W) = 3 + 9(1) = 12$.

b. Applying Ex. 4.198 and the result in part a:

$$P(W \leq w) \geq 1 - \frac{E(W)}{w} = .9,$$

so that $w = 120$.

4.196 $\Gamma(1/2) = \int_0^\infty y^{-1/2} e^{-y} dy = \int_0^\infty \sqrt{2} e^{-(1/2)x^2} dx = \sqrt{2} \sqrt{2\pi} \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-(1/2)x^2} dx = 2\sqrt{\pi} \left[\frac{1}{2}\right] = \sqrt{\pi}$ (relating the last integral to that $P(Z > 0)$, where Z is a standard normal random variable).

4.197 a. Let $y = \sin^2 \theta$, so that $dy = 2\sin \theta \cos \theta d\theta$. Thus,

$$B(\alpha, \beta) = \int_0^1 y^{\alpha-1} (1-y)^{\beta-1} dy = 2 \int_0^{\pi/2} \sin^{2\alpha-2} \theta (1 - \sin^2 \theta)^{\beta-1} d\theta = 2 \int_0^{\pi/2} \sin^{2\alpha-1} \theta \cos^{2\beta-1} \theta d\theta, \text{ using}$$

the trig identity $1 - \sin^2 \theta = \cos^2 \theta$.

b. Following the text, $\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty y^{\alpha-1} e^{-y} dy \int_0^\infty z^{\beta-1} e^{-z} dz = \int_0^\infty \int_0^\infty y^{\alpha-1} z^{\beta-1} e^{-y-z} dy dz$. Now,

use the transformation $y = r^2 \cos^2 \theta$, $x = r^2 \sin^2 \theta$ so that $dydz = 4r^3 \cos\theta \sin\theta$.

Following this and using the result in part a, we find

$$\Gamma(\alpha)\Gamma(\beta) = B(\alpha, \beta) \int_0^\infty r^{2(\alpha+\beta-1)} e^{-r^2} 2r dr.$$

A final transformation with $x = r^2$ gives $\Gamma(\alpha)\Gamma(\beta) = B(\alpha, \beta)\Gamma(\alpha + \beta)$, proving the result.

4.198 Note that

$$E[|g(Y)|] = \int_{-\infty}^{\infty} |g(y)| f(y) dy \geq \int_{|g(y)| > k} |g(y)| f(y) dy > \int_{|g(y)| > k} k f(y) dy = kP(|g(Y)| > k),$$

Since $|g(y)| > k$ for this integral. Therefore,

$$P(|g(Y)| \leq k) \geq 1 - E(|g(Y)|)/k.$$

4.199 a. Define $g(y) = y^{2i-1} e^{-y^2/2}$ for positive integer values of i . Observe that $g(-y) = -g(y)$ so that $g(y)$ is an odd function. The expected value $E(Z^{2i-1})$ can be written

as $E(Z^{2i-1}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} g(y) dy$ which is thus equal to 0.

b. Now, define $h(y) = y^{2i} e^{-y^2/2}$ for positive integer values of i . Observe that $h(-y) = h(y)$ so that $h(y)$ is an even function. The expected value $E(Z^{2i})$ can be written

as $E(Z^{2i}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} h(y) dy = \int_0^\infty \frac{2}{\sqrt{2\pi}} h(y) dy$. Therefore, the integral becomes

$$E(Z^{2i}) = \int_0^\infty \frac{2}{\sqrt{2\pi}} y^{2i} e^{-y^2/2} dy = \frac{1}{\sqrt{\pi}} \int_0^\infty 2^i w^{i-1/2} e^{-w} dw = \frac{1}{\sqrt{\pi}} 2^i \Gamma(i + 1/2).$$

In the last integral, we applied the transformation $w = z^2/2$.

$$\begin{aligned} \text{c. } E(Z^2) &= \frac{1}{\sqrt{\pi}} 2^1 \Gamma(1 + 1/2) = \frac{1}{\sqrt{\pi}} 2^1 (1/2) \sqrt{\pi} = 1 \\ E(Z^4) &= \frac{1}{\sqrt{\pi}} 2^2 \Gamma(2 + 1/2) = \frac{1}{\sqrt{\pi}} 2^2 (3/2)(1/2) \sqrt{\pi} = 3 \\ E(Z^6) &= \frac{1}{\sqrt{\pi}} 2^3 \Gamma(3 + 1/2) = \frac{1}{\sqrt{\pi}} 2^3 (5/2)(3/2)(1/2) \sqrt{\pi} = 15 \\ E(Z^8) &= \frac{1}{\sqrt{\pi}} 2^4 \Gamma(4 + 1/2) = \frac{1}{\sqrt{\pi}} 2^4 (7/2)(5/2)(3/2)(1/2) \sqrt{\pi} = 105. \end{aligned}$$

d. The result follows from:

$$\prod_{j=i}^i (2j-1) = \prod_{j=i}^i 2(j-1/2) = 2^i \prod_{j=i}^i (j-1/2) = 2^i \Gamma(i+1/2) \left(\frac{1}{\sqrt{\pi}}\right) = E(Z^{2i}).$$

$$4.200 \quad \mathbf{a.} \quad E(Y^a) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{a+\alpha-1} (1-y)^{\beta-1} dy = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(a+\alpha)\Gamma(\beta)}{\Gamma(a+\alpha+\beta)} = \frac{\Gamma(\alpha+\beta)\Gamma(a+\alpha)}{\Gamma(\alpha)\Gamma(a+\alpha+\beta)}.$$

b. The value $\alpha + a$ must be positive in the beta density.

c. With $a = 1$, $E(Y^1) = \frac{\Gamma(\alpha+\beta)\Gamma(1+\alpha)}{\Gamma(\alpha)\Gamma(1+\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}.$

d. With $a = 1/2$, $E(Y^{1/2}) = \frac{\Gamma(\alpha+\beta)\Gamma(1/2+\alpha)}{\Gamma(\alpha)\Gamma(1/2+\alpha+\beta)}.$

e. With $a = -1$, $E(Y^{-1}) = \frac{\Gamma(\alpha+\beta)\Gamma(\alpha-1)}{\Gamma(\alpha)\Gamma(\alpha+\beta-1)} = \frac{\alpha+\beta-1}{\alpha-1}, \alpha > 1$

With $a = -1/2$, $E(Y^{-1/2}) = \frac{\Gamma(\alpha+\beta)\Gamma(\alpha-1/2)}{\Gamma(\alpha)\Gamma(\alpha+\beta-1/2)}, \alpha > 1/2$

With $a = -2$, $E(Y^{-2}) = \frac{\Gamma(\alpha+\beta)\Gamma(\alpha-2)}{\Gamma(\alpha)\Gamma(\alpha+\beta-2)} = \frac{(\alpha+\beta-1)(\alpha+\beta-2)}{(\alpha-1)(\alpha-2)}, \alpha > 2.$

Chapter 5: Multivariate Probability Distributions

5.1 a. The sample space S gives the possible values for Y_1 and Y_2 :

S	AA	AB	AC	BA	BB	BC	CA	CB	CC
(y_1, y_2)	(2, 0)	(1, 1)	(1, 0)	(1, 1)	(0, 2)	(1, 0)	(1, 0)	(0, 1)	(0, 0)

Since each sample point is equally likely with probability $1/9$, the joint distribution for Y_1 and Y_2 is given by

		y_1			
		0	1	2	
y_2	0	1/9	2/9	1/9	
	1	2/9	2/9	0	
	2	1/9	0	0	

b. $F(1, 0) = p(0, 0) + p(1, 0) = 1/9 + 2/9 = 3/9 = 1/3$.

5.2 a. The sample space for the toss of three balanced coins w/ probabilities are below:

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
(y_1, y_2)	(3, 1)	(3, 1)	(2, 1)	(1, 1)	(2, 2)	(1, 2)	(1, 3)	(0, -1)
probability	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8

		y_1				
		0	1	2	3	
y_2	-1	1/8	0	0	0	
	1	0	1/8	2/8	1/8	
	2	0	1/8	1/8	0	
	3	0	1/8	0	0	

b. $F(2, 1) = p(0, -1) + p(1, 1) + p(2, 1) = 1/2$.

5.3 Note that using material from Chapter 3, the joint probability function is given by

$$p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2) = \frac{\binom{4}{y_1} \binom{3}{y_2} \binom{2}{3-y_1-y_2}}{\binom{9}{3}}, \text{ where } 0 \leq y_1, 0 \leq y_2, \text{ and } y_1 + y_2 \leq 3.$$

In table format, this is

		y_1			
		0	1	2	3
y_2	0	0	3/84	6/84	1/84
	1	4/84	24/84	12/84	0
	2	12/84	18/84	0	0
	3	4/84	0	0	0

- 5.4** **a.** All of the probabilities are at least 0 and sum to 1.
b. $F(1, 2) = P(Y_1 \leq 1, Y_2 \leq 2) = 1$. Every child in the experiment either survived or didn't and used either 0, 1, or 2 seatbelts.

5.5 **a.** $P(Y_1 \leq 1/2, Y_2 \leq 1/3) = \int_0^{1/2} \int_0^{1/3} 3y_1 dy_1 dy_2 = .1065$. 

b. $P(Y_2 \leq Y_1/2) = \int_0^1 \int_0^{y_1/2} 3y_1 dy_1 dy_2 = .5$. 

5.6 **a.** $P(Y_1 - Y_2 > .5) = P(Y_1 > .5 + Y_2) = \int_0^{.5} \int_{y_2+.5}^1 1 dy_1 dy_2 = \int_0^{.5} [y_1]_{y_2+.5}^1 dy_2 = \int_0^{.5} (.5 - y_2) dy_2 = .125$.

b. $P(Y_1 Y_2 < .5) = 1 - P(Y_1 Y_2 > .5) = 1 - P(Y_1 > .5/Y_2) = 1 - \int_{.5}^1 \int_{.5/y_2}^1 1 dy_1 dy_2 = 1 - \int_{.5}^1 (1 - .5/y_2) dy_2$
 $= 1 - [.5 + .5 \ln(.5)] = .8466$.

5.7 **a.** $P(Y_1 < 1, Y_2 > 5) = \int_0^1 \int_5^\infty e^{-(y_1+y_2)} dy_1 dy_2 = \left[\int_0^1 e^{-y_1} dy_1 \right] \left[\int_5^\infty e^{-y_2} dy_2 \right] = [1 - e^{-1}] e^{-5} = .00426$. 

b. $P(Y_1 + Y_2 < 3) = P(Y_1 < 3 - Y_2) = \int_0^3 \int_0^{3-y_2} e^{-(y_1+y_2)} dy_1 dy_2 = 1 - 4e^{-3} = .8009$.

5.8 **a.** Since the density must integrate to 1, evaluate $\int_0^1 \int_0^1 ky_1 y_2 dy_1 dy_2 = k/4 = 1$, so $k = 4$.

 **b.** $F(y_1, y_2) = P(Y_1 \leq y_1, Y_2 \leq y_2) = 4 \int_0^{y_2} \int_0^{y_1} t_1 t_2 dt_1 dt_2 = y_1^2 y_2^2, 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1$.

c. $P(Y_1 \leq 1/2, Y_2 \leq 3/4) = (1/2)^2 (3/4)^2 = 9/64$.

5.9 **a.** Since the density must integrate to 1, evaluate $\int_0^1 \int_0^{y_2} k(1 - y_2) dy_1 dy_2 = k/6 = 1$, so $k = 6$.

b. Note that since $Y_1 \leq Y_2$, the probability must be found in two parts (drawing a picture is useful):

$P(Y_1 \leq 3/4, Y_2 \geq 1/2) = \int_{1/2}^1 \int_{1/2}^{y_2} (1 - y_2) dy_1 dy_2 + \int_{1/2}^{3/4} \int_{y_1}^1 6(1 - y_2) dy_2 dy_1 = 24/64 + 7/64 = 31/64$.  

- 5.10** **a.** Geometrically, since Y_1 and Y_2 are distributed uniformly over the triangular region, using the area formula for a triangle $k = 1$.

b. This probability can also be calculated using geometric considerations. The area of the triangle specified by $Y_1 \geq 3Y_2$ is $2/3$, so this is the probability.

5.11 The area of the triangular region is 1, so with a uniform distribution this is the value of the density function. Again, using geometry (drawing a picture is again useful):

a. $P(Y_1 \leq 3/4, Y_2 \leq 3/4) = 1 - P(Y_1 > 3/4) - P(Y_2 > 3/4) = 1 - \frac{1}{2}\left(\frac{1}{2}\right)\left(\frac{1}{4}\right) - \frac{1}{2}\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{29}{32}.$

b. $P(Y_1 - Y_2 \geq 0) = P(Y_1 \geq Y_2)$. The region specified in this probability statement represents 1/4 of the total region of support, so $P(Y_1 \geq Y_2) = 1/4$.

5.12 Similar to Ex. 5.11:

a. $P(Y_1 \leq 3/4, Y_2 \leq 3/4) = 1 - P(Y_1 > 3/4) - P(Y_2 > 3/4) = \frac{1}{2}\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) - \frac{1}{2}\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{7}{8}.$

b. $P(Y_1 \leq 1/2, Y_2 \leq 1/2) = \int_0^{1/2} \int_0^{1/2} 2 dy_1 dy_2 = 1/2.$

5.13 a. $F(1/2, 1/2) = \int_0^{1/2} \int_{y_1-1}^{1/2} 30 y_1 y_2^2 dy_2 dy_1 = \frac{9}{16}.$

b. Note that:

$$F(1/2, 2) = F(1/2, 1) = P(Y_1 \leq 1/2, Y_2 \leq 1) = P(Y_1 \leq 1/2, Y_2 \leq 1/2) + P(Y_1 \leq 1/2, Y_2 > 1/2)$$

So, the first probability statement is simply $F(1/2, 1/2)$ from part a. The second probability statement is found by

$$P(Y_1 \leq 1/2, Y_2 > 1/2) = \int_{1/2}^1 \int_0^{1-y_2} 30 y_1 y_2^2 dy_2 dy_1 = \frac{4}{16}.$$

Thus, $F(1/2, 2) = \frac{9}{16} + \frac{4}{16} = \frac{13}{16}.$

c. $P(Y_1 > Y_2) = 1 - P(Y_1 \leq Y_2) = 1 - \int_0^{1/2} \int_{y_1}^{1-y_1} 30 y_1 y_2^2 dy_2 dy_1 = 1 - \frac{11}{32} = \frac{21}{32} = .65625.$

5.14 a. Since $f(y_1, y_2) \geq 0$, simply show $\int_0^1 \int_{y_1}^{2-y_1} 6 y_1^2 y_2 dy_2 dy_1 = 1.$

b. $P(Y_1 + Y_2 < 1) = P(Y_2 < 1 - Y_1) = \int_0^{.5} \int_{y_1}^{1-y_1} 6 y_1^2 y_2 dy_2 dy_1 = 1/16.$

5.15 a. $P(Y_1 < 2, Y_2 > 1) = \int_1^2 \int_1^{y_1} e^{-y_1} dy_2 dy_1 = \int_1^2 \int_{y_2}^2 e^{-y_1} dy_1 dy_2 = e^{-1} - 2e^{-2}.$

b. $P(Y_1 \geq 2Y_2) = \int_0^\infty \int_{2y_2}^\infty e^{-y_1} dy_1 dy_2 = 1/2.$

c. $P(Y_1 - Y_2 \geq 1) = P(Y_1 \geq Y_2 + 1) = \int_0^\infty \int_{y_2+1}^\infty e^{-y_1} dy_1 dy_2 = e^{-1}.$

5.16 a. $P(Y_1 < 1/2, Y_2 > 1/4) = \int_{1/4}^1 \int_0^{1/2} (y_1 + y_2) dy_1 dy_2 = 21/64 = .328125.$

b. $P(Y_1 + Y_2 \leq 1) = P(Y_1 \leq 1 - Y_2) = \int_0^1 \int_0^{1-y_2} (y_1 + y_2) dy_1 dy_2 = 1/3.$

5.17 This can be found using integration (polar coordinates are helpful). But, note that this is a bivariate uniform distribution over a circle of radius 1, and the probability of interest represents 50% of the support. Thus, the probability is .50.

5.18 $P(Y_1 > 1, Y_2 > 1) = \int_1^\infty \int_1^\infty \frac{1}{8} y_1 e^{-(y_1+y_2)/2} dy_1 dy_2 = \left[\int_1^\infty \frac{1}{4} y_1 e^{-y_1/2} dy_1 \right] \left[\int_1^\infty \frac{1}{2} e^{-y_2/2} dy_2 \right] = \frac{3}{2} e^{-1/2} \left(e^{-1/2} \right) = \frac{3}{2} e^{-1}$

5.19 a. The marginal probability function is given in the table below.

y_1	0	1	2
$p_1(y_1)$	4/9	4/9	1/9

b. No, evaluating binomial probabilities with $n = 3$, $p = 1/3$ yields the same result.

5.20 a. The marginal probability function is given in the table below.

y_2	-1	1	2	3
$p_2(y_2)$	1/8	4/8	2/8	1/8

b. $P(Y_1 = 3 | Y_2 = 1) = \frac{P(Y_1=3, Y_2=1)}{P(Y_2=1)} = \frac{1/8}{4/8} = 1/4.$

5.21 a. The marginal distribution of Y_1 is hypergeometric with $N = 9$, $n = 3$, and $r = 4$.

b. Similar to part a, the marginal distribution of Y_2 is hypergeometric with $N = 9$, $n = 3$, and $r = 3$. Thus,

$$P(Y_1 = 1 | Y_2 = 2) = \frac{P(Y_1=1, Y_2=2)}{P(Y_2=2)} = \frac{\binom{4}{1} \binom{3}{2} \binom{2}{0}}{\binom{9}{3}} \bigg/ \frac{\binom{3}{2} \binom{6}{1}}{\binom{9}{3}} = 2/3.$$

c. Similar to part b,

$$P(Y_3 = 1 | Y_2 = 1) = P(Y_1 = 1 | Y_2 = 1) = \frac{P(Y_1=1, Y_2=1)}{P(Y_2=1)} = \frac{\binom{3}{1} \binom{2}{1} \binom{4}{1}}{\binom{9}{3}} \bigg/ \frac{\binom{3}{1} \binom{6}{2}}{\binom{9}{3}} = 8/15.$$

5.22 a. The marginal distributions for Y_1 and Y_2 are given in the margins of the table.

b. $P(Y_2 = 0 | Y_1 = 0) = .38/.76 = .5$ $P(Y_2 = 1 | Y_1 = 0) = .14/.76 = .18$

$P(Y_2 = 2 | Y_1 = 0) = .24/.76 = .32$

c. The desired probability is $P(Y_1 = 0 | Y_2 = 0) = .38/.55 = .69.$

5.23 a. $f_2(y_2) = \int_{y_2}^1 3y_1 dy_1 = \frac{3}{2} - \frac{3}{2}y_2^2, 0 \leq y_2 \leq 1.$

b. Defined over $y_2 \leq y_1 \leq 1$, with the constant $y_2 \geq 0$.

c. First, we have $f_1(y_1) = \int_0^{y_1} 3y_1 dy_2 = 3y_1^2, 0 \leq y_1 \leq 1.$ Thus,

$f(y_2 | y_1) = 1/y_1, 0 \leq y_2 \leq y_1.$ So, conditioned on $Y_1 = y_1$, we see Y_2 has a uniform distribution on the interval $(0, y_1)$. Therefore, the probability is simple:

$$P(Y_2 > 1/2 | Y_1 = 3/4) = (3/4 - 1/2)/(3/4) = 1/3.$$

5.24 a. $f_1(y_1) = 1, 0 \leq y_1 \leq 1, f_2(y_2) = 1, 0 \leq y_2 \leq 1.$

b. Since both Y_1 and Y_2 are uniformly distributed over the interval $(0, 1)$, the probabilities are the same: .2

c. $0 \leq y_2 \leq 1.$

d. $f(y_1 | y_2) = f(y_1) = 1, 0 \leq y_1 \leq 1$

e. $P(.3 < Y_1 < .5 | Y_2 = .3) = .2$

f. $P(.3 < Y_2 < .5 | Y_2 = .5) = .2$

g. The answers are the same.

5.25 a. $f_1(y_1) = e^{-y_1}, y_1 > 0, f_2(y_2) = e^{-y_2}, y_2 > 0.$ These are both exponential density functions with $\beta = 1.$

b. $P(1 < Y_1 < 2.5) = P(1 < Y_2 < 2.5) = e^{-1} - e^{-2.5} = .2858.$

c. $y_2 > 0.$

d. $f(y_1 | y_2) = f_1(y_1) = e^{-y_1}, y_1 > 0.$

e. $f(y_2 | y_1) = f_2(y_2) = e^{-y_2}, y_2 > 0.$

f. The answers are the same.

g. The probabilities are the same.

5.26 a. $f_1(y_1) = \int_0^1 4y_1y_2 dy_2 = 2y_1, 0 \leq y_1 \leq 1; f(y_2) = 2y_2, 0 \leq y_2 \leq 1.$

b. $P(Y_1 \leq 1/2 | Y_2 \geq 3/4) = \frac{\int_0^{1/2} \int_{3/4}^1 4y_1y_2 dy_1 dy_2}{\int_{3/4}^1 2y_2 dy_2} = \int_0^{1/2} 2y_1 dy_1 = 1/4.$

c. $f(y_1 | y_2) = f_1(y_1) = 2y_1, 0 \leq y_1 \leq 1.$

d. $f(y_2 | y_1) = f_2(y_2) = 2y_2, 0 \leq y_2 \leq 1.$

e. $P(Y_1 \leq 3/4 | Y_2 = 1/2) = P(Y_1 \leq 3/4) = \int_0^{3/4} 2y_1 dy_1 = 9/16.$

5.27 a. $f_1(y_1) = \int_{y_1}^1 6(1-y_2)dy_2 = 3(1-y_1)^2, 0 \leq y_1 \leq 1;$

$$f_2(y_2) = \int_0^{y_2} 6(1-y_2)dy_1 = 6y_2(1-y_2), 0 \leq y_2 \leq 1.$$

b. $P(Y_2 \leq 1/2 | Y_1 \leq 3/4) = \frac{\int_0^{1/2} \int_0^{y_2} 6(1-y_2)dy_1dy_2}{\int_0^{3/4} 3(1-y_1)^2 dy_1} = 32/63.$

c. $f(y_1 | y_2) = 1/y_2, 0 \leq y_1 \leq y_2 \leq 1.$

d. $f(y_2 | y_1) = 2(1-y_2)/(1-y_1)^2, 0 \leq y_1 \leq y_2 \leq 1.$

e. From part **d**, $f(y_2 | 1/2) = 8(1-y_2), 1/2 \leq y_2 \leq 1.$ Thus, $P(Y_2 \geq 3/4 | Y_1 = 1/2) = 1/4.$

5.28 Referring to Ex. 5.10:

a. First, find $f_2(y_2) = \int_{2y_2}^2 1dy_1 = 2(1-y_2), 0 \leq y_2 \leq 1.$ Then, $P(Y_2 \geq .5) = .25.$

b. First find $f(y_1 | y_2) = \frac{1}{2(1-y_2)}, 2y_2 \leq y_1 \leq 2.$ Thus, $f(y_1 | .5) = 1, 1 \leq y_1 \leq 2$ — the conditional distribution is uniform on $(1, 2).$ Therefore, $P(Y_1 \geq 1.5 | Y_2 = .5) = .5$

5.29 Referring to Ex. 5.11:

a. $f_2(y_2) = \int_{y_2-1}^{1-y_2} 1dy_1 = 2(1-y_2), 0 \leq y_2 \leq 1.$ In order to find $f_1(y_1)$, notice that the limits of integration are different for $0 \leq y_1 \leq 1$ and $-1 \leq y_1 \leq 0.$ For the first case:

$f_1(y_1) = \int_0^{1-y_1} 1dy_2 = 1-y_1, \text{ for } 0 \leq y_1 \leq 1.$ For the second case, $f_1(y_1) = \int_0^{1+y_1} 1dy_2 = 1+y_1, \text{ for } -1 \leq y_1 \leq 0.$ This can be written as $f_1(y_1) = 1 - |y_1|, \text{ for } -1 \leq y_1 \leq 1.$

b. The conditional distribution is $f(y_2 | y_1) = \frac{1}{1-|y_1|}, \text{ for } 0 \leq y_1 \leq 1 - |y_1|.$ Thus,

$f(y_2 | 1/4) = 4/3.$ Then, $P(Y_2 > 1/2 | Y_1 = 1/4) = \int_{1/2}^{3/4} 4/3 dy_2 = 1/3.$

5.30 a. $P(Y_1 \geq 1/2, Y_2 \leq 1/4) = \int_0^{1/4} \int_{1/2}^{1-y_2} 2dy_1dy_2 = \frac{3}{16}.$ And, $P(Y_2 \leq 1/4) = \int_0^{1/4} 2(1-y_2)dy_2 = \frac{7}{16}.$

Thus, $P(Y_1 \geq 1/2 | Y_2 \leq 1/4) = \frac{3}{7}.$

b. Note that $f(y_1 | y_2) = \frac{1}{1-y_2}, 0 \leq y_1 \leq 1-y_2.$ Thus, $f(y_1 | 1/4) = 4/3, 0 \leq y_1 \leq 3/4.$

Thus, $P(Y_2 > 1/2 | Y_1 = 1/4) = \int_{1/2}^{3/4} 4/3 dy_2 = 1/3.$

5.31 a. $f_1(y_1) = \int_{y_1-1}^{1-y_1} 30y_1y_2^2 dy_2 = 20y_1(1-y_1)^2, 0 \leq y_1 \leq 1.$

b. This marginal density must be constructed in two parts:

$$f_2(y_2) = \begin{cases} \int_{y_2}^{1+y_2} 30y_1y_2^2 dy_1 = 15y_2^2(1+y_2) & -1 \leq y_2 \leq 0 \\ \int_0^{1-y_2} 30y_1y_2^2 dy_1 = 15y_2^2(1-y_2) & 0 \leq y_2 \leq 1 \end{cases}.$$

c. $f(y_2 | y_1) = \frac{3}{2}y_2^2(1-y_1)^{-3}, \text{ for } y_1 - 1 \leq y_2 \leq 1 - y_1.$

d. $f(y_2 | .75) = \frac{3}{2}y_2^2(.25)^{-3}, \text{ for } -.25 \leq y_2 \leq .25, \text{ so } P(Y_2 > 0 | Y_1 = .75) = .5.$

5.32 a. $f_1(y_1) = \int_{y_1}^{2-y_1} 6y_1^2y_2 dy_2 = 12y_1^2(1-y_1), 0 \leq y_1 \leq 1.$

b. This marginal density must be constructed in two parts:

$$f_2(y_2) = \begin{cases} \int_0^{y_2} 6y_1^2y_2 dy_1 = 2y_2^4 & 0 \leq y_2 \leq 1 \\ \int_{2-y_2}^{2-y_2} 6y_1^2y_2 dy_1 = 2y_2(2-y_2)^3 & 1 \leq y_2 \leq 2 \end{cases}.$$

c. $f(y_2 | y_1) = \frac{1}{2}y_2 / (1-y_1), y_1 \leq y_2 \leq 2-y_1.$

d. Using

the density found in part c, $P(Y_2 < 1.1 | Y_1 = .6) = \frac{1}{2} \int_{.6}^{1.1} y_2 / .4 dy_2 = .53$

5.33 Refer to Ex. 5.15:

a. $f_1(y_1) = \int_0^{y_1} e^{-y_1} dy_2 = y_1 e^{-y_1}, y_1 \geq 0. \quad f_2(y_2) = \int_{y_2}^{\infty} e^{-y_1} dy_1 = e^{-y_2}, y_2 \geq 0.$

b. $f(y_1 | y_2) = e^{-(y_1-y_2)}, y_1 \geq y_2.$

c. $f(y_2 | y_1) = 1/y_1, 0 \leq y_2 \leq y_1.$

d. The density functions are different.

e. The marginal and conditional probabilities can be different.

5.34 a. Given $Y_1 = y_1$, Y_2 has a uniform distribution on the interval $(0, y_1)$.

b. Since $f_1(y_1) = 1, 0 \leq y_1 \leq 1, f(y_1, y_2) = f(y_2 | y_1)f_1(y_1) = 1/y_1, 0 \leq y_2 \leq y_1 \leq 1.$

c. $f_2(y_2) = \int_{y_2}^1 1/y_1 dy_1 = -\ln(y_2), 0 \leq y_2 \leq 1.$

5.35 With $Y_1 = 2$, the conditional distribution of Y_2 is uniform on the interval $(0, 2)$. Thus, $P(Y_2 < 1 | Y_1 = 2) = .5.$

5.36 a. $f_1(y_1) = \int_0^1 (y_1 + y_2) dy_2 = y_1 + \frac{1}{2}$, $0 \leq y_1 \leq 1$. Similarly $f_2(y_2) = y_2 + \frac{1}{2}$, $0 \leq y_2 \leq 1$.

b. First, $P(Y_2 \geq \frac{1}{2}) = \int_{1/2}^1 (y_2 + \frac{1}{2}) dy_2 = \frac{5}{8}$, and $P(Y_1 \geq \frac{1}{2}, Y_2 \geq \frac{1}{2}) = \int_{1/2}^1 \int_{1/2}^1 (y_1 + y_2) dy_1 dy_2 = \frac{3}{8}$.

Thus, $P(Y_1 \geq \frac{1}{2} | Y_2 \geq \frac{1}{2}) = \frac{3}{5}$.

c. $P(Y_1 > .75 | Y_2 = .5) = \frac{\int_{.75}^1 (y_1 + \frac{1}{2}) dy_1}{\frac{1}{2} + \frac{1}{2}} = .34375$.

5.37 Calculate $f_2(y_2) = \int_0^\infty \frac{y_1}{8} e^{-(y_1+y_2)/2} dy_1 = \frac{1}{2} e^{-y_2/2}$, $y_2 > 0$. Thus, Y_2 has an exponential distribution with $\beta = 2$ and $P(Y_2 > 2) = 1 - F(2) = e^{-1}$.

5.38 This is the identical setup as in Ex. 5.34.

a. $f(y_1, y_2) = f(y_2 | y_1) f_1(y_1) = 1/y_1$, $0 \leq y_2 \leq y_1 \leq 1$.

b. Note that $f(y_2 | 1/2) = 1/2$, $0 \leq y_2 \leq 1/2$. Thus, $P(Y_2 < 1/4 | Y_1 = 1/2) = 1/2$.

c. The probability of interest is $P(Y_1 > 1/2 | Y_2 = 1/4)$. So, the necessary conditional density is $f(y_1 | y_2) = f(y_1, y_2)/f_2(y_2) = \frac{1}{y_1(-\ln y_2)}$, $0 \leq y_2 \leq y_1 \leq 1$. Thus,

$$P(Y_1 > 1/2 | Y_2 = 1/4) = \int_{1/2}^1 \frac{1}{y_1 \ln 4} dy_1 = 1/2.$$

5.39 The result follows from:

$$P(Y_1 = y_1 | W = w) = \frac{P(Y_1 = y_1, W = w)}{P(W = w)} = \frac{P(Y_1 = y_1, Y_1 + Y_2 = w)}{P(W = w)} = \frac{P(Y_1 = y_1, Y_2 = w - y_1)}{P(W = w)}.$$

Since Y_1 and Y_2 are independent, this is

$$P(Y_1 = y_1 | W = w) = \frac{P(Y_1 = y_1)P(Y_2 = w - y_1)}{P(W = w)} = \frac{\frac{\lambda_1^{y_1} e^{-\lambda_1}}{y_1!} \left(\frac{\lambda_2^{w-y_1} e^{-\lambda_2}}{(w-y_1)!} \right)}{\frac{(\lambda_1 + \lambda_2)^w e^{-(\lambda_1 + \lambda_2)}}{w!}}$$

$$= \binom{w}{y_1} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^{y_1} \left(1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^{w-y_1}.$$

This is the binomial distribution with $n = w$ and $p = \frac{\lambda_1}{\lambda_1 + \lambda_2}$.

5.40 As the Ex. 5.39 above, the result follows from:

$$P(Y_1 = y_1 | W = w) = \frac{P(Y_1 = y_1, W = w)}{P(W = w)} = \frac{P(Y_1 = y_1, Y_1 + Y_2 = w)}{P(W = w)} = \frac{P(Y_1 = y_1, Y_2 = w - y_1)}{P(W = w)}.$$

Since Y_1 and Y_2 are independent, this is (all terms involving p_1 and p_2 drop out)

$$P(Y_1 = y_1 | W = w) = \frac{P(Y_1 = y_1)P(Y_2 = w - y_1)}{P(W = w)} = \frac{\binom{n_1}{y_1} \binom{n_2}{w - y_1}}{\binom{n_1 + n_2}{w}}, \quad \begin{array}{l} 0 \leq y_1 \leq n_1 \\ 0 \leq w - y_1 \leq n_2 \end{array}.$$

5.41 Let $Y = \#$ of defectives in a random selection of three items. Conditioned on p , we have

$$P(Y = y | p) = \binom{3}{y} p^y (1 - p)^{3-y}, \quad y = 0, 1, 2, 3.$$

We are given that the proportion of defectives follows a uniform distribution on $(0, 1)$, so the unconditional probability that $Y = 2$ can be found by

$$\begin{aligned} P(Y = 2) &= \int_0^1 P(Y = 2, p) dp = \int_0^1 P(Y = 2 | p) f(p) dp = \int_0^1 3p^2(1 - p)^{3-1} dp = 3 \int_0^1 (p^2 - p^3) dp \\ &= 1/4. \end{aligned}$$

5.42 (Similar to Ex. 5.41) Let $Y = \#$ of defects per yard. Then,

$$p(y) = \int_0^\infty P(Y = y, \lambda) d\lambda = \int_0^\infty P(Y = y | \lambda) f(\lambda) d\lambda = \int_0^\infty \frac{\lambda^y e^{-\lambda}}{y!} e^{-\lambda} d\lambda = \left(\frac{1}{2}\right)^{y+1}, \quad y = 0, 1, 2, \dots$$

Note that this is essentially a geometric distribution (see Ex. 3.88).

5.43 Assume $f(y_1 | y_2) = f_1(y_1)$. Then, $f(y_1, y_2) = f(y_1 | y_2) f_2(y_2) = f_1(y_1) f_2(y_2)$ so that Y_1 and Y_2 are independent. Now assume that Y_1 and Y_2 are independent. Then, there exists functions g and h such that $f(y_1, y_2) = g(y_1)h(y_2)$ so that

$$1 = \iint f(y_1, y_2) dy_1 dy_2 = \int g(y_1) dy_1 \times \int h(y_2) dy_2.$$

Then, the marginals for Y_1 and Y_2 can be defined by

$$f_1(y_1) = \int \frac{g(y_1)h(y_2)}{\int g(y_1) dy_1 \times \int h(y_2) dy_2} dy_2 = \frac{g(y_1)}{\int g(y_1) dy_1}, \text{ so } f_2(y_2) = \frac{h(y_2)}{\int h(y_2) dy_2}.$$

Thus, $f(y_1, y_2) = f_1(y_1) f_2(y_2)$. Now it is clear that

$$f(y_1 | y_2) = f(y_1, y_2) / f_2(y_2) = f_1(y_1) f_2(y_2) / f_2(y_2) = f_1(y_1),$$

provided that $f_2(y_2) > 0$ as was to be shown.

5.44 The argument follows exactly as Ex. 5.43 with integrals replaced by sums and densities replaced by probability mass functions.

5.45 No. Counterexample: $P(Y_1 = 2, Y_2 = 2) = 0 \neq P(Y_1 = 2)P(Y_2 = 2) = (1/9)(1/9)$.

5.46 No. Counterexample: $P(Y_1 = 3, Y_2 = 1) = 1/8 \neq P(Y_1 = 3)P(Y_2 = 1) = (1/8)(4/8)$.

5.47 Dependent. For example: $P(Y_1 = 1, Y_2 = 2) \neq P(Y_1 = 1)P(Y_2 = 2)$.

5.48 Dependent. For example: $P(Y_1 = 0, Y_2 = 0) \neq P(Y_1 = 0)P(Y_2 = 0)$.

5.49 Note that $f_1(y_1) = \int_0^{y_1} 3y_1 dy_2 = 3y_1^2$, $0 \leq y_1 \leq 1$, $f_2(y_2) = \int_{y_1}^1 3y_1 dy_1 = \frac{3}{2}[1 - y_2^2]$, $0 \leq y_2 \leq 1$.

Thus, $f(y_1, y_2) \neq f_1(y_1)f_2(y_2)$ so that Y_1 and Y_2 are dependent.

5.50 a. Note that $f_1(y_1) = \int_0^1 1 dy_2 = 1$, $0 \leq y_1 \leq 1$ and $f_2(y_2) = \int_0^1 1 dy_1 = 1$, $0 \leq y_2 \leq 1$. Thus,

$f(y_1, y_2) = f_1(y_1)f_2(y_2)$ so that Y_1 and Y_2 are independent.

b. Yes, the conditional probabilities are the same as the marginal probabilities.

5.51 a. Note that $f_1(y_1) = \int_0^\infty e^{-(y_1+y_2)} dy_2 = e^{-y_1}$, $y_1 > 0$ and $f_2(y_2) = \int_0^\infty e^{-(y_1+y_2)} dy_1 = e^{-y_2}$, $y_2 > 0$.

Thus, $f(y_1, y_2) = f_1(y_1)f_2(y_2)$ so that Y_1 and Y_2 are independent.

b. Yes, the conditional probabilities are the same as the marginal probabilities.

5.52 Note that $f(y_1, y_2)$ can be factored and the ranges of y_1 and y_2 do not depend on each other so by Theorem 5.5 Y_1 and Y_2 are independent.

5.53 The ranges of y_1 and y_2 depend on each other so Y_1 and Y_2 cannot be independent.

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5.58 Following Ex. 5.32, it is seen that $f(y_1, y_2) \neq f_1(y_1)f_2(y_2)$ so that Y_1 and Y_2 are dependent.

5.59 The ranges of y_1 and y_2 depend on each other so Y_1 and Y_2 cannot be independent.

5.60 From Ex. 5.36, $f_1(y_1) = y_1 + \frac{1}{2}$, $0 \leq y_1 \leq 1$, and $f_2(y_2) = y_2 + \frac{1}{2}$, $0 \leq y_2 \leq 1$. But, $f(y_1, y_2) \neq f_1(y_1)f_2(y_2)$ so Y_1 and Y_2 are dependent.

5.61 Note that $f(y_1, y_2)$ can be factored and the ranges of y_1 and y_2 do not depend on each other so by Theorem 5.5, Y_1 and Y_2 are independent.

- 5.62** Let X, Y denote the number on which person A, B flips a head on the coin, respectively. Then, X and Y are geometric random variables and the probability that the stop on the same number toss is:

$$P(X=1, Y=1) + P(X=2, Y=2) + \cdots = P(X=1)P(Y=1) + P(X=2)P(Y=2) + \cdots \\ = \sum_{i=1}^{\infty} P(X=i)P(Y=i) = \sum_{i=1}^{\infty} p(1-p)^{i-1} p(1-p)^{i-1} = p^2 \sum_{k=0}^{\infty} [(1-p)^2]^k = \frac{p^2}{1-(1-p)^2}.$$

- 5.63** $P(Y_1 > Y_2, Y_1 < 2Y_2) = \int_0^{\infty} \int_{y_1/2}^{y_1} e^{-(y_1+y_2)} dy_2 dy_1 = \frac{1}{6}$ and $P(Y_1 < 2Y_2) = \int_0^{\infty} \int_0^{y_1/2} e^{-(y_1+y_2)} dy_2 dy_1 = \frac{2}{3}$. So, $P(Y_1 > Y_2 | Y_1 < 2Y_2) = 1/4$.

- 5.64** $P(Y_1 > Y_2, Y_1 < 2Y_2) = \int_0^1 \int_{y_1/2}^{y_1} 1 dy_2 dy_1 = \frac{1}{4}$, $P(Y_1 < 2Y_2) = 1 - P(Y_1 \geq 2Y_2) = 1 - \int_0^1 \int_0^{y_1/2} 1 dy_2 dy_1 = \frac{3}{4}$. So, $P(Y_1 > Y_2 | Y_1 < 2Y_2) = 1/3$.

- 5.65 a.** The marginal density for Y_1 is $f_1(y_1) = \int_0^{\infty} [(1 - \alpha(1 - 2e^{-y_1}))(1 - 2e^{-y_2})] e^{-y_1 - y_2} dy_2$
- $$= e^{-y_1} \left[\int_0^{\infty} e^{-y_2} dy_2 - \alpha(1 - 2e^{-y_1}) \int_0^{\infty} (e^{-y_2} - 2e^{-2y_2}) dy_2 \right]$$
- $$= e^{-y_1} \left[\int_0^{\infty} e^{-y_2} dy_2 - \alpha(1 - 2e^{-y_1})(1 - 1) \right] = e^{-y_1},$$

which is the exponential density with a mean of 1.

b. By symmetry, the marginal density for Y_2 is also exponential with $\beta = 1$.

c. When $\alpha = 0$, then $f(y_1, y_2) = e^{-y_1 - y_2} = f_1(y_1)f_2(y_2)$ and so Y_1 and Y_2 are independent. Now, suppose Y_1 and Y_2 are independent. Then, $E(Y_1 Y_2) = E(Y_1)E(Y_2) = 1$. So,

$$E(Y_1 Y_2) = \int_0^{\infty} \int_0^{\infty} y_1 y_2 [(1 - \alpha(1 - 2e^{-y_1}))(1 - 2e^{-y_2})] e^{-y_1 - y_2} dy_1 dy_2 \\ = \int_0^{\infty} \int_0^{\infty} y_1 y_2 e^{-y_1 - y_2} dy_1 dy_2 - \alpha \left[\int_0^{\infty} y_1 (1 - 2e^{-y_1}) e^{-y_1} dy_1 \right] \times \left[\int_0^{\infty} y_2 (1 - 2e^{-y_2}) e^{-y_2} dy_2 \right] \\ = 1 - \alpha(1 - \frac{1}{2})(1 - \frac{1}{2}) = 1 - \alpha/4. \text{ This equals 1 only if } \alpha = 0.$$

- 5.66 a.** Since $F_2(\infty) = 1$, $F(y_1, \infty) = F_1(y_1) \cdot 1 \cdot [1 - \alpha\{1 - F_1(y_1)\}\{1 - 1\}] = F_1(y_1)$.
b. Similarly, it is $F_2(y_2)$ from $F(y_1, y_2)$.
c. If $\alpha = 0$, $F(y_1, y_2) = F_1(y_1)F_2(y_2)$, so by Definition 5.8 they are independent.
d. If $\alpha \neq 0$, $F(y_1, y_2) \neq F_1(y_1)F_2(y_2)$, so by Definition 5.8 they are not independent.

$$\begin{aligned}
 5.67 \quad P(a < Y_1 \leq b, c < Y_2 \leq d) &= F(b, d) - F(b, c) - F(a, d) + F(a, c) \\
 &= F_1(b)F_2(d) - F_1(b)F_2(c) - F_1(a)F_2(d) + F_1(a)F_2(c) \\
 &= F_1(b)[F_2(d) - F_2(c)] - F_1(a)[F_2(d) - F_2(c)] \\
 &= [F_1(b) - F_1(a)] \times [F_2(d) - F_2(c)] \\
 &= P(a < Y_1 \leq b) \times P(c < Y_2 \leq d).
 \end{aligned}$$

$$5.68 \quad \text{Given that } p_1(y_1) = \binom{2}{y_1} (.2)^{y_1} (.8)^{2-y_1}, y_1 = 0, 1, 2, \text{ and } p_2(y_2) = (.3)^{y_2} (.7)^{1-y_2}, y_2 = 0, 1:$$

$$\text{a. } p(y_1, y_2) = p_1(y_1)p_2(y_2) = \binom{2}{y_1} (.2)^{y_1} (.8)^{2-y_1} (.3)^{y_2} (.7)^{1-y_2}, y_1 = 0, 1, 2 \text{ and } y_2 = 0, 1.$$

$$\text{b. The probability of interest is } P(Y_1 + Y_2 \leq 1) = p(0, 0) + p(1, 0) + p(0, 1) = .864.$$

$$5.69 \quad \text{a. } f(y_1, y_2) = f_1(y_1)f_2(y_2) = (1/9)e^{-(y_1+y_2)/3}, y_1 > 0, y_2 > 0.$$

$$\text{b. } P(Y_1 + Y_2 \leq 1) = \int_0^1 \int_0^{1-y_2} (1/9)e^{-(y_1+y_2)/3} dy_1 dy_2 = 1 - \frac{4}{3}e^{-1/3} = .0446.$$

$$5.70 \quad \text{With } f(y_1, y_2) = f_1(y_1)f_2(y_2) = 1, 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1,$$

$$P(Y_2 \leq Y_1 \leq Y_2 + 1/4) = \int_0^{1/4} \int_0^{y_1} 1 dy_2 dy_1 + \int_{1/4}^1 \int_{y_1-1/4}^{y_1} 1 dy_2 dy_1 = 7/32.$$

$$5.71 \quad \text{Assume uniform distributions for the call times over the 1-hour period. Then,}$$

$$\text{a. } P(Y_1 \leq 1/2, Y_2 \leq 1/2) = P(Y_1 \leq 1/2)P(Y_2 \leq 1/2) = (1/2)(1/2) = 1/4.$$

$$\text{b. Note that 5 minutes} = 1/12 \text{ hour. To find } P(|Y_1 - Y_2| \leq 1/12), \text{ we must break the region into three parts in the integration:}$$

$$P(|Y_1 - Y_2| \leq 1/12) = \int_0^{1/12} \int_0^{y_1+1/12} 1 dy_2 dy_1 + \int_{1/12}^{11/12} \int_{y_1-1/12}^{y_1+1/12} 1 dy_2 dy_1 + \int_{11/12}^1 \int_{y_1-1/12}^1 1 dy_2 dy_1 = 23/144.$$

$$5.72 \quad \text{a. } E(Y_1) = 2(1/3) = 2/3.$$

$$\text{b. } V(Y_1) = 2(1/3)(2/3) = 4/9$$

$$\text{c. } E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0.$$

$$5.73 \quad \text{Use the mean of the hypergeometric: } E(Y_1) = 3(4)/9 = 4/3.$$

$$5.74 \quad \text{The marginal distributions for } Y_1 \text{ and } Y_2 \text{ are uniform on the interval } (0, 1). \text{ And it was found in Ex. 5.50 that } Y_1 \text{ and } Y_2 \text{ are independent. So:}$$

$$\text{a. } E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0.$$

$$\text{b. } E(Y_1 Y_2) = E(Y_1)E(Y_2) = (1/2)(1/2) = 1/4.$$

$$\text{c. } E(Y_1^2 + Y_2^2) = E(Y_1^2) + E(Y_2^2) = (1/12 + 1/4) + (1/12 + 1/4) = 2/3$$

$$\text{d. } V(Y_1 Y_2) = V(Y_1)V(Y_2) = (1/12)(1/12) = 1/144.$$

5.75 The marginal distributions for Y_1 and Y_2 are exponential with $\beta = 1$. And it was found in Ex. 5.51 that Y_1 and Y_2 are independent. So:

a. $E(Y_1 + Y_2) = E(Y_1) + E(Y_2) = 2$, $V(Y_1 + Y_2) = V(Y_1) + V(Y_2) = 2$.

b. $P(Y_1 - Y_2 > 3) = P(Y_1 > 3 + Y_2) = \int_0^\infty \int_{3+y_2}^\infty e^{-y_1-y_2} dy_1 dy_2 = (1/2)e^{-3} = .0249$.

c. $P(Y_1 - Y_2 < -3) = P(Y_1 > Y_2 - 3) = \int_0^\infty \int_{3+y_1}^\infty e^{-y_1-y_2} dy_2 dy_1 = (1/2)e^{-3} = .0249$.

d. $E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0$, $V(Y_1 - Y_2) = V(Y_1) + V(Y_2) = 2$.

e. They are equal.

5.76 From Ex. 5.52, we found that Y_1 and Y_2 are independent. So,

a. $E(Y_1) = \int_0^1 2y_1^2 dy_1 = 2/3$.

b. $E(Y_1^2) = \int_0^1 2y_1^3 dy_1 = 2/4$, so $V(Y_1) = 2/4 - 4/9 = 1/18$.

c. $E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0$.

5.77 Following Ex. 5.27, the marginal densities can be used:

a. $E(Y_1) = \int_0^1 3y_1(1-y_1)^2 dy_1 = 1/4$, $E(Y_2) = \int_0^1 6y_2(1-y_2) dy_2 = 1/2$.

b. $E(Y_1^2) = \int_0^1 3y_1^2(1-y_1)^2 dy_1 = 1/10$, $V(Y_1) = 1/10 - (1/4)^2 = 3/80$,

$E(Y_2^2) = \int_0^1 6y_2^2(1-y_2) dy_2 = 3/10$, $V(Y_2) = 3/10 - (1/2)^2 = 1/20$.

c. $E(Y_1 - 3Y_2) = E(Y_1) - 3 \cdot E(Y_2) = 1/4 - 3/2 = -5/4$.

5.78 a. The marginal distribution for Y_1 is $f_1(y_1) = y_1/2$, $0 \leq y_1 \leq 2$. $E(Y_1) = 4/3$, $V(Y_1) = 2/9$.

b. Similarly, $f_2(y_2) = 2(1-y_2)$, $0 \leq y_2 \leq 1$. So, $E(Y_2) = 1/3$, $V(Y_1) = 1/18$.

c. $E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 4/3 - 1/3 = 1$.

d. $V(Y_1 - Y_2) = E[(Y_1 - Y_2)^2] - [E(Y_1 - Y_2)]^2 = E(Y_1^2) - 2E(Y_1 Y_2) + E(Y_2^2) - 1$.

Since $E(Y_1 Y_2) = \int_0^1 \int_{2y_2}^2 y_1 y_2 dy_1 dy_2 = 1/2$, we have that

$$V(Y_1 - Y_2) = [2/9 + (4/3)^2] - 1 + [1/18 + (1/3)^2] - 1 = 1/6.$$

Using Tchebysheff's theorem, two standard deviations about the mean is (.19, 1.81).

5.79 Referring to Ex. 5.16, integrating the joint density over the two regions of integration:

$$E(Y_1 Y_2) = \int_{-1}^0 \int_0^{1+y_1} y_1 y_2 dy_2 dy_1 + \int_0^1 \int_0^{1-y_1} y_1 y_2 dy_2 dy_1 = 0$$

5.80 From Ex. 5.36, $f_1(y_1) = y_1 + \frac{1}{2}$, $0 \leq y_1 \leq 1$, and $f_2(y_2) = y_2 + \frac{1}{2}$, $0 \leq y_2 \leq 1$. Thus, $E(Y_1) = 7/12$ and $E(Y_2) = 7/12$. So, $E(30Y_1 + 25Y_2) = 30(7/12) + 25(7/12) = 32.08$.

5.81 Since Y_1 and Y_2 are independent, $E(Y_2/Y_1) = E(Y_2)E(1/Y_1)$. Thus, using the marginal densities found in Ex. 5.61,

$$E(Y_2/Y_1) = E(Y_2)E(1/Y_1) = \frac{1}{2} \int_0^{\infty} y_2 e^{-y_2/2} dy_2 \left[\frac{1}{4} \int_0^{\infty} e^{-y_1/2} dy_1 \right] = 2(\frac{1}{2}) = 1.$$

5.82 The marginal densities were found in Ex. 5.34. So,

$$E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 1/2 - \int_0^1 -y_2 \ln(y_2) dy_2 = 1/2 - 1/4 = 1/4.$$

5.83 From Ex. 3.88 and 5.42, $E(Y) = 2 - 1 = 1$.

5.84 All answers use results proven for the geometric distribution and independence:

- a. $E(Y_1) = E(Y_2) = 1/p$, $E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0$.
- b. $E(Y_1^2) = E(Y_2^2) = (1-p)/p^2 + (1/p)^2 = (2-p)/p^2$. $E(Y_1 Y_2) = E(Y_1)E(Y_2) = 1/p^2$.
- c. $E[(Y_1 - Y_2)^2] = E(Y_1^2) - 2E(Y_1 Y_2) + E(Y_2^2) = 2(1-p)/p^2$.
 $V(Y_1 - Y_2) = V(Y_1) + V(Y_2) = 2(1-p)/p^2$.
- d. Use Tchebysheff's theorem with $k = 3$.

5.85

- a. $E(Y_1) = E(Y_2) = 1$ (both marginal distributions are exponential with mean 1)
- b. $V(Y_1) = V(Y_2) = 1$
- c. $E(Y_1 - Y_2) = E(Y_1) - E(Y_2) = 0$.
- d. $E(Y_1 Y_2) = 1 - \alpha/4$, so $\text{Cov}(Y_1, Y_2) = -\alpha/4$.
- e. $V(Y_1 - Y_2) = V(Y_1) + V(Y_2) - 2\text{Cov}(Y_1, Y_2) = 1 + \alpha/2$. Using Tchebysheff's theorem with $k = 2$, the interval is $(-2\sqrt{2 + \alpha/2}, 2\sqrt{2 + \alpha/2})$.

5.86 Using the hint and Theorem 5.9:

- a. $E(W) = E(Z)E(Y_1^{-1/2}) = 0E(Y_1^{-1/2}) = 0$. Also, $V(W) = E(W^2) - [E(W)]^2 = E(W^2)$.
Now, $E(W^2) = E(Z^2)E(Y_1^{-1}) = 1 \cdot E(Y_1^{-1}) = E(Y_1^{-1}) = \frac{1}{v_1 - 2}$, $v_1 > 2$ (using Ex. 4.82).
- b. $E(U) = E(Y_1)E(Y_2^{-1}) = \frac{v_1}{v_2 - 2}$, $v_2 > 2$, $V(U) = E(U^2) - [E(U)]^2 = E(Y_1^2)E(Y_2^{-2}) - \left(\frac{v_1}{v_2 - 2}\right)^2$
 $= v_1(v_1 + 2) \frac{1}{(v_2 - 2)(v_2 - 4)} - \left(\frac{v_1}{v_2 - 2}\right)^2 = \frac{2v_1(v_1 + v_2 - 2)}{(v_2 - 2)^2(v_2 - 4)}$, $v_2 > 4$.

- 5.87** a. $E(Y_1 + Y_2) = E(Y_1) + E(Y_2) = v_1 + v_2$.
 b. By independence, $V(Y_1 + Y_2) = V(Y_1) + V(Y_2) = 2v_1 + 2v_2$.

- 5.88** It is clear that $E(Y) = E(Y_1) + E(Y_2) + \dots + E(Y_6)$. Using the result that Y_i follows a geometric distribution with success probability $(7 - i)/6$, we have

$$E(Y) = \sum_{i=1}^6 \frac{6}{7-i} = 1 + 6/5 + 6/4 + 6/3 + 6/2 + 6 = 14.7.$$

- 5.89** $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = \sum_{y_1} \sum_{y_2} y_1 y_2 p(y_1, y_2) - [2(1/3)]^2 = 2/9 - 4/9 = -2/9$.

As the value of Y_1 increases, the value of Y_2 tends to decrease.

- 5.90** From Ex. 5.3 and 5.21, $E(Y_1) = 4/3$ and $E(Y_2) = 1$. Thus,

$$E(Y_1 Y_2) = 1(1)\frac{24}{84} + 2(1)\frac{12}{84} + 1(2)\frac{18}{84} = 1$$

 So, $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = 1 - (4/3)(1) = -1/3$.

- 5.91** From Ex. 5.76, $E(Y_1) = E(Y_2) = 2/3$. $E(Y_1 Y_2) = \int_0^1 \int_0^1 4y_1^2 y_2^2 dy_1 dy_2 = 4/9$. So,
 $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = 4/9 - 4/9 = 0$ as expected since Y_1 and Y_2 are independent.

- 5.92** From Ex. 5.77, $E(Y_1) = 1/4$ and $E(Y_2) = 1/2$. $E(Y_1 Y_2) = \int_0^1 \int_0^{y_2} 6y_1 y_2 (1 - y_2) dy_1 dy_2 = 3/20$.
 So, $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = 3/20 - 1/8 = 1/40$ as expected since Y_1 and Y_2 are dependent.

- 5.93** a. From Ex. 5.55 and 5.79, $E(Y_1 Y_2) = 0$ and $E(Y_1) = 0$. So,

$$\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = 0 - 0E(Y_2) = 0.$$

 b. Y_1 and Y_2 are dependent.
 c. Since $\text{Cov}(Y_1, Y_2) = 0$, $\rho = 0$.
 d. If $\text{Cov}(Y_1, Y_2) = 0$, Y_1 and Y_2 are not necessarily independent.

- 5.94** a. $\text{Cov}(U_1, U_2) = E[(Y_1 + Y_2)(Y_1 - Y_2)] - E(Y_1 + Y_2)E(Y_1 - Y_2)$

$$= E(Y_1^2) - E(Y_2^2) - [E(Y_1)]^2 - [E(Y_2)]^2$$

$$= (\sigma_1^2 + \mu_1^2) - (\sigma_2^2 + \mu_2^2) - (\mu_1^2 - \mu_2^2) = \sigma_1^2 - \sigma_2^2.$$

- b. Since $V(U_1) = V(U_2) = \sigma_1^2 + \sigma_2^2$ (Y_1 and Y_2 are uncorrelated), $\rho = \frac{\sigma_1^2 - \sigma_2^2}{\sigma_1^2 + \sigma_2^2}$.

- c. If $\sigma_1^2 = \sigma_2^2$, U_1 and U_2 are uncorrelated.

5.95 Note that the marginal distributions for Y_1 and Y_2 are

y_1	-1	0	1
$p_1(y_1)$	1/3	1/3	1/3

y_2	0	1
$p_2(y_2)$	2/3	1/3

So, Y_1 and Y_2 not independent since $p(-1, 0) \neq p_1(-1)p_2(0)$. However, $E(Y_1) = 0$ and $E(Y_1 Y_2) = (-1)(0)1/3 + (0)(1)(1/3) + (1)(0)(1/3) = 0$, so $\text{Cov}(Y_1, Y_2) = 0$.

5.96 a. $\text{Cov}(Y_1, Y_2) = E[(Y_1 - \mu_1)(Y_2 - \mu_2)] = E[(Y_2 - \mu_2)(Y_1 - \mu_1)] = \text{Cov}(Y_2, Y_1)$.

b. $\text{Cov}(Y_1, Y_1) = E[(Y_1 - \mu_1)(Y_1 - \mu_1)] = E[(Y_1 - \mu_1)^2] = V(Y_1)$.

5.97 a. From Ex. 5.96, $\text{Cov}(Y_1, Y_1) = V(Y_1) = 2$.

b. If $\text{Cov}(Y_1, Y_2) = 7$, $\rho = 7/4 > 1$, impossible.

c. With $\rho = 1$, $\text{Cov}(Y_1, Y_2) = 1(4) = 4$ (a perfect positive linear association).

d. With $\rho = -1$, $\text{Cov}(Y_1, Y_2) = -1(4) = -4$ (a perfect negative linear association).

5.98 Since $\rho^2 \leq 1$, we have that $-1 \leq \rho \leq 1$ or $-1 \leq \frac{\text{Cov}(Y_1, Y_2)}{\sqrt{V(Y_1)}\sqrt{V(Y_2)}} \leq 1$.

5.99 Since $E(c) = c$, $\text{Cov}(c, Y) = E[(c - c)(Y - \mu)] = 0$.

5.100 a. $E(Y_1) = E(Z) = 0$, $E(Y_2) = E(Z^2) = 1$.

b. $E(Y_1 Y_2) = E(Z^3) = 0$ (odd moments are 0).

c. $\text{Cov}(Y_1, Y_1) = E(Z^3) - E(Z)E(Z^2) = 0$.

d. $P(Y_2 > 1 \mid Y_1 > 1) = P(Z^2 > 1 \mid Z > 1) = 1 \neq P(Z^2 > 1)$. Thus, Y_1 and Y_2 are dependent.

5.101 a. $\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2) = 1 - \alpha/4 - (1)(1) = -\frac{\alpha}{4}$.

b. This is clear from part a.

c. We showed previously that Y_1 and Y_2 are independent only if $\alpha = 0$. If $\rho = 0$, it must be true that $\alpha = 0$.

5.102 The quantity $3Y_1 + 5Y_2$ = dollar amount spend per week. Thus:

$$E(3Y_1 + 5Y_2) = 3(40) + 5(65) = 445.$$

$$\text{Var}(3Y_1 + 5Y_2) = 9V(Y_1) + 25V(Y_2) = 9(4) + 25(8) = 236.$$

5.103 $E(3Y_1 + 4Y_2 - 6Y_3) = 3E(Y_1) + 4E(Y_2) - 6E(Y_3) = 3(2) + 4(-1) - 6(-4) = -22$,

$$V(3Y_1 + 4Y_2 - 6Y_3) = 9V(Y_1) + 16V(Y_2) + 36V(Y_3) + 24\text{Cov}(Y_1, Y_2) - 36\text{Cov}(Y_1, Y_3) - 48\text{Cov}(Y_2, Y_3) = 9(4) + 16(6) + 36(8) + 24(1) - 36(-1) - 48(0) = 480.$$

5.104 a. Let $X = Y_1 + Y_2$. Then, the probability distribution for X is

x	1	2	3
$p(x)$	7/84	42/84	35/84

Thus, $E(X) = 7/3$ and $V(X) = .3889$.

b. $E(Y_1 + Y_2) = E(Y_1) + E(Y_2) = 4/3 + 1 = 7/3$. We have that $V(Y_1) = 10/18$, $V(Y_2) = 42/84$, and $\text{Cov}(Y_1, Y_2) = -1/3$, so

$$V(Y_1 + Y_2) = V(Y_1) + V(Y_2) + 2\text{Cov}(Y_1, Y_2) = 10/18 + 42/84 - 2/3 = 7/18 = .3889.$$

5.105 Since Y_1 and Y_2 are independent, $V(Y_1 + Y_2) = V(Y_1) + V(Y_2) = 1/18 + 1/18 = 1/9$.

5.106 $V(Y_1 - 3Y_2) = V(Y_1) + 9V(Y_2) - 6\text{Cov}(Y_1, Y_2) = 3/80 + 9(1/20) - 6(1/40) = 27/80 = .3375$.

5.107 Since $E(Y_1) = E(Y_2) = 1/3$, $V(Y_1) = V(Y_2) = 1/18$ and $E(Y_1 Y_2) = \int_0^1 \int_0^{1-y_2} 2y_1 y_2 dy_1 dy_2 = 1/12$, we have that $\text{Cov}(Y_1, Y_2) = 1/12 - 1/9 = -1/36$. Therefore,

$$E(Y_1 + Y_2) = 1/3 + 1/3 = 2/3 \text{ and } V(Y_1 + Y_2) = 1/18 + 1/18 + 2(-1/36) = 1/18.$$

5.108 From Ex. 5.33, Y_1 has a gamma distribution with $\alpha = 2$ and $\beta = 1$, and Y_2 has an exponential distribution with $\beta = 1$. Thus, $E(Y_1 + Y_2) = 2(1) + 1 = 3$. Also, since

$$E(Y_1 Y_2) = \int_0^\infty \int_0^{y_1} y_1 y_2 e^{-y_1} dy_2 dy_1 = 3, \text{ Cov}(Y_1, Y_2) = 3 - 2(1) = 1,$$

$$V(Y_1 - Y_2) = 2(1)^2 + 1^2 - 2(1) = 1.$$

Since a value of 4 minutes is four three standard deviations above the mean of 1 minute, this is not likely.

5.109 We have $E(Y_1) = E(Y_2) = 7/12$. Intermediate calculations give $V(Y_1) = V(Y_2) = 11/144$.

$$\text{Thus, } E(Y_1 Y_2) = \int_0^1 \int_0^1 y_1 y_2 (y_1 + y_2) dy_1 dy_2 = 1/3, \text{ Cov}(Y_1, Y_2) = 1/3 - (7/12)^2 = -1/144.$$

From Ex. 5.80, $E(30Y_1 + 25Y_2) = 32.08$, so

$$V(30Y_1 + 25Y_2) = 900V(Y_1) + 625V(Y_2) + 2(30)(25) \text{Cov}(Y_1, Y_2) = 106.08.$$

The standard deviation of $30Y_1 + 25Y_2$ is $\sqrt{106.08} = 10.30$. Using Tchebysheff's theorem with $k = 2$, the interval is $(11.48, 52.68)$.

5.110 a. $V(1 + 2Y_1) = 4V(Y_1)$, $V(3 + 4Y_2) = 16V(Y_2)$, and $\text{Cov}(1 + 2Y_1, 3 + 4Y_2) = 8\text{Cov}(Y_1, Y_2)$.

$$\text{So, } \frac{8\text{Cov}(Y_1, Y_2)}{\sqrt{4V(Y_1)}\sqrt{16V(Y_2)}} = \rho = .2.$$

b. $V(1 + 2Y_1) = 4V(Y_1)$, $V(3 - 4Y_2) = 16V(Y_2)$, and $\text{Cov}(1 + 2Y_1, 3 - 4Y_2) = -8\text{Cov}(Y_1, Y_2)$.

$$\text{So, } \frac{-8\text{Cov}(Y_1, Y_2)}{\sqrt{4V(Y_1)}\sqrt{16V(Y_2)}} = -\rho = -.2.$$

c. $V(1 - 2Y_1) = 4V(Y_1)$, $V(3 - 4Y_2) = 16V(Y_2)$, and $\text{Cov}(1 - 2Y_1, 3 - 4Y_2) = 8\text{Cov}(Y_1, Y_2)$.

$$\text{So, } \frac{8\text{Cov}(Y_1, Y_2)}{\sqrt{4V(Y_1)}\sqrt{16V(Y_2)}} = \rho = .2.$$

- 5.111 a.** $V(a + bY_1) = b^2V(Y_1)$, $V(c + dY_2) = d^2V(Y_2)$, and $\text{Cov}(a + bY_1, c + dY_2) = bd\text{Cov}(Y_1, Y_2)$.

So, $\rho_{w_1, w_2} = \frac{bd\text{Cov}(Y_1, Y_2)}{\sqrt{b^2V(Y_1)}\sqrt{d^2V(Y_2)}} = \frac{bd}{|bd|} \rho_{Y_1, Y_2}$. Provided that the constants b and d are nonzero, $\frac{bd}{|bd|}$ is either 1 or -1 . Thus, $|\rho_{w_1, w_2}| = |\rho_{Y_1, Y_2}|$.

b. Yes, the answers agree.

- 5.112** In Ex. 5.61, it was showed that Y_1 and Y_2 are independent. In addition, Y_1 has a gamma distribution with $\alpha = 2$ and $\beta = 2$, and Y_2 has an exponential distribution with $\beta = 2$. So, with $C = 50 + 2Y_1 + 4Y_2$, it is clear that

$$E(C) = 50 + 2E(Y_1) + 4E(Y_2) = 50 + (2)(4) + (4)(2) = 66$$

$$V(C) = 4V(Y_1) + 16V(Y_2) = 4(2)(4) + 16(4) = 96.$$

- 5.113** The net daily gain is given by the random variable $G = X - Y$. Thus, given the distributions for X and Y in the problem,

$$E(G) = E(X) - E(Y) = 50 - (4)(2) = 42$$

$$V(G) = V(X) + V(Y) = 3^2 + 4(2^2) = 25.$$

The value \$70 is $(70 - 42)/5 = 5.6$ standard deviations above the mean, an unlikely value.

- 5.114** Observe that Y_1 has a gamma distribution with $\alpha = 4$ and $\beta = 1$ and Y_2 has an exponential distribution with $\beta = 2$. Thus, with $U = Y_1 - Y_2$,

a. $E(U) = 4(1) - 2 = 2$

b. $V(U) = 4(1^2) + 2^2 = 8$

- c.** The value 0 has a z-score of $(0 - 2)/\sqrt{8} = -.707$, or it is $-.707$ standard deviations below the mean. This is not extreme so it is likely the profit drops below 0.

- 5.115** Following Ex. 5.88:

- a.** Note that for non-negative integers a and b and $i \neq j$,

$$P(Y_i = a, Y_j = b) = P(Y_j = b | Y_i = a)P(Y_i = a)$$

But, $P(Y_j = b | Y_i = a) = P(Y_j = b)$ since the trials (i.e. die tosses) are independent — the experiments that generate Y_i and Y_j represent independent experiments via the memoryless property. So, Y_i and Y_j are independent and thus $\text{Cov}(Y_i, Y_j) = 0$.

b. $V(Y) = V(Y_1) + \dots + V(Y_6) = 0 + \frac{1/6}{(5/6)^2} + \frac{2/6}{(4/6)^2} + \frac{3/6}{(3/6)^2} + \frac{4/6}{(2/6)^2} + \frac{5/6}{(1/6)^2} = 38.99$.

- c.** From Ex. 5.88, $E(Y) = 14.7$. Using Tchebysheff's theorem with $k = 2$, the interval is $14.7 \pm 2\sqrt{38.99}$ or $(0, 27.188)$

5.116 $V(Y_1 + Y_2) = V(Y_1) + V(Y_2) + 2\text{Cov}(Y_1, Y_2)$, $V(Y_1 - Y_2) = V(Y_1) + V(Y_2) - 2\text{Cov}(Y_1, Y_2)$.
When Y_1 and Y_2 are independent, $\text{Cov}(Y_1, Y_2) = 0$ so the quantities are the same.

5.117 Refer to Example 5.29 in the text. The situation here is analogous to drawing n balls from an urn containing N balls, r_1 of which are red, r_2 of which are black, and $N - r_1 - r_2$ are neither red nor black. Using the argument given there, we can deduce that:

$$E(Y_1) = np_1 \quad V(Y_1) = np_1(1 - p_1)\left(\frac{N-n}{N-1}\right) \quad \text{where } p_1 = r_1/N$$

$$E(Y_2) = np_2 \quad V(Y_2) = np_2(1 - p_2)\left(\frac{N-n}{N-1}\right) \quad \text{where } p_2 = r_2/N$$

Now, define new random variables for $i = 1, 2, \dots, n$:

$$U_i = \begin{cases} 1 & \text{if alligator } i \text{ is a mature female} \\ 0 & \text{otherwise} \end{cases} \quad V_i = \begin{cases} 1 & \text{if alligator } i \text{ is a mature male} \\ 0 & \text{otherwise} \end{cases}$$

Then, $Y_1 = \sum_{i=1}^n U_i$ and $Y_2 = \sum_{i=1}^n V_i$. Now, we must find $\text{Cov}(Y_1, Y_2)$. Note that:

$$E(Y_1 Y_2) = E\left(\sum_{i=1}^n U_i, \sum_{i=1}^n V_i\right) = \sum_{i=1}^n E(U_i V_i) + \sum_{i \neq j} E(U_i V_j).$$

Now, since for all i , $E(U_i, V_i) = P(U_i = 1, V_i = 1) = 0$ (an alligator can't be both female and male), we have that $E(U_i, V_i) = 0$ for all i . Now, for $i \neq j$,

$$E(U_i, V_j) = P(U_i = 1, V_j = 1) = P(U_i = 1)P(V_j = 1|U_i = 1) = \frac{n}{N} \left(\frac{r_2}{N-1}\right) = \frac{N}{N-1} p_1 p_2.$$

Since there are $n(n-1)$ terms in $\sum_{i \neq j} E(U_i V_j)$, we have that $E(Y_1 Y_2) = n(n-1) \frac{N}{N-1} p_1 p_2$.

Thus, $\text{Cov}(Y_1, Y_2) = n(n-1) \frac{N}{N-1} p_1 p_2 - (np_1)(np_2) = -\frac{n(N-n)}{N-1} p_1 p_2$.

So, $E\left[\frac{Y_1}{n} - \frac{Y_2}{n}\right] = \frac{1}{n}(np_1 - np_2) = p_1 - p_2$,

$$V\left[\frac{Y_1}{n} - \frac{Y_2}{n}\right] = \frac{1}{n^2} [V(Y_1) + V(Y_2) - 2\text{Cov}(Y_1, Y_2)] = \frac{N-n}{n(N-1)} (p_1 + p_2 - (p_1 - p_2)^2)$$

5.118 Let $Y = X_1 + X_2$, the total sustained load on the footing.

a. Since X_1 and X_2 have gamma distributions and are independent, we have that

$$E(Y) = 50(2) + 20(2) = 140$$

$$V(Y) = 50(2^2) + 20(2^2) = 280.$$

b. Consider Tchebysheff's theorem with $k = 4$: the corresponding interval is

$$140 + 4\sqrt{280} \text{ or } (73.07, 206.93).$$

So, we can say that the sustained load will exceed 206.93 kips with probability less than $1/16$.

- 5.119 a.** Using the multinomial distribution with $p_1 = p_2 = p_3 = 1/3$,

$$P(Y_1 = 3, Y_2 = 1, Y_3 = 2) = \frac{6!}{3!1!2!} \left(\frac{1}{3}\right)^6 = .0823.$$

b. $E(Y_1) = n/3$, $V(Y_1) = n(1/3)(2/3) = 2n/9$.

c. $\text{Cov}(Y_2, Y_3) = -n(1/3)(1/3) = -n/9$.

d. $E(Y_2 - Y_3) = n/3 - n/3 = 0$, $V(Y_2 - Y_3) = V(Y_2) + V(Y_3) - 2\text{Cov}(Y_2, Y_3) = 2n/3$.

- 5.120** $E(C) = E(Y_1) + 3E(Y_2) = np_1 + 3np_2$.

$$V(C) = V(Y_1) + 9V(Y_2) + 6\text{Cov}(Y_1, Y_2) = np_1q_1 + 9np_2q_2 - 6np_1p_2.$$

- 5.121** If N is large, the multinomial distribution is appropriate:

a. $P(Y_1 = 2, Y_2 = 1) = \frac{5!}{2!1!2!} (.3)^2 (.1)^1 (.6)^2 = .0972$.

b. $E\left[\frac{Y_1}{n} - \frac{Y_2}{n}\right] = p_1 - p_2 = .3 - .1 = .2$

$$V\left[\frac{Y_1}{n} - \frac{Y_2}{n}\right] = \frac{1}{n^2} [V(Y_1) + V(Y_2) - 2\text{Cov}(Y_1, Y_2)] = \frac{p_1q_1}{n} + \frac{p_2q_2}{n} + 2\frac{p_1p_2}{n} = .072.$$

- 5.122** Let $Y_1 = \#$ of mice weighing between 80 and 100 grams, and let $Y_2 = \#$ weighing over 100 grams. Thus, with X having a normal distribution with $\mu = 100$ g. and $\sigma = 20$ g.,

$$p_1 = P(80 \leq X \leq 100) = P(-1 \leq Z \leq 0) = .3413$$

$$p_2 = P(X > 100) = P(Z > 0) = .5$$

a. $P(Y_1 = 2, Y_2 = 1) = \frac{4!}{2!1!1!} (.3413)^2 (.5)^1 (.1587)^1 = .1109$.

b. $P(Y_2 = 4) = \frac{4!}{0!4!0!} (.5)^4 = .0625$.

- 5.123** Let $Y_1 = \#$ of family home fires, $Y_2 = \#$ of apartment fires, and $Y_3 = \#$ of fires in other types. Thus, (Y_1, Y_2, Y_3) is multinomial with $n = 4$, $p_1 = .73$, $p_2 = .2$ and $p_3 = .07$. Thus,

$$P(Y_1 = 2, Y_2 = 1, Y_3 = 1) = 6(.73)^2(.2)(.07) = .08953.$$

- 5.124** Define $C = \text{total cost} = 20,000Y_1 + 10,000Y_2 + 2000Y_3$

a. $E(C) = 20,000E(Y_1) + 10,000E(Y_2) + 2000E(Y_3)$
 $= 20,000(2.92) + 10,000(.8) + 2000(.28) = 66,960.$

b. $V(C) = (20,000)^2V(Y_1) + (10,000)^2V(Y_2) + (2000)^2V(Y_3) + \text{covariance terms}$
 $= (20,000)^2(4)(.73)(.27) + (10,000)^2(4)(.8)(.2) + (2000)^2(4)(.07)(.93)$
 $+ 2[20,000(10,000)(-4)(.73)(.2) + 20,000(2000)(-4)(.73)(.07) +$
 $10,000(2000)(-4)(.2)(.07)] = 380,401,600 - 252,192,000 = 128,209,600.$

- 5.125** Let $Y_1 = \#$ of planes with no wing cracks, $Y_2 = \#$ of planes with detectable wing cracks, and $Y_3 = \#$ of planes with critical wing cracks. Therefore, (Y_1, Y_2, Y_3) is multinomial with $n = 5$, $p_1 = .7$, $p_2 = .25$ and $p_3 = .05$.

a. $P(Y_1 = 2, Y_2 = 2, Y_3 = 1) = 30(.7)^2(.25)^2(.05) = .046.$

- b.** The distribution of Y_3 is binomial with $n = 5$, $p_3 = .05$, so

$$P(Y_3 \geq 1) = 1 - P(Y_3 = 0) = 1 - (.95)^5 = .2262.$$

5.126 Using formulas for means, variances, and covariances for the multinomial:

$$E(Y_1) = 10(.1) = 1$$

$$V(Y_1) = 10(.1)(.9) = .9$$

$$E(Y_2) = 10(.05) = .5$$

$$V(Y_2) = 10(.05)(.95) = .475$$

$$\text{Cov}(Y_1, Y_2) = -10(.1)(.05) = -.05$$

So,

$$E(Y_1 + 3Y_2) = 1 + 3(.5) = 2.5$$

$$V(Y_1 + 3Y_2) = .9 + 9(.475) + 6(-.05) = 4.875.$$

5.127 Y is binomial with $n = 10$, $p = .10 + .05 = .15$.

a. $P(Y = 2) = \binom{10}{2} (.15)^2 (.85)^8 = .2759.$

b. $P(Y \geq 1) = 1 - P(Y = 0) = 1 - (.85)^{10} = .8031.$

5.128 The marginal distribution for Y_1 is found by

$$f_1(y_1) = \int_{-\infty}^{\infty} f(y_1, y_2) dy_2.$$

Making the change of variables $u = (y_1 - \mu_1)/\sigma_1$ and $v = (y_2 - \mu_2)/\sigma_2$ yields

$$f_1(y_1) = \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}(u^2 + v^2 - 2\rho uv)\right] dv.$$

To evaluate this, note that $u^2 + v^2 - 2\rho uv = (v - \rho u)^2 + u^2(1 - \rho^2)$ so that

$$f_1(y_1) = \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} e^{-u^2/2} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}(v - \rho u)^2\right] dv,$$

So, the integral is that of a normal density with mean ρu and variance $1 - \rho^2$. Therefore,

$$f_1(y_1) = \frac{1}{2\pi\sigma_1} e^{-(y_1 - \mu_1)^2 / 2\sigma_1^2}, \quad -\infty < y_1 < \infty,$$

which is a normal density with mean μ_1 and standard deviation σ_1 . A similar procedure will show that the marginal distribution of Y_2 is normal with mean μ_2 and standard deviation σ_2 .

5.129 The result follows from Ex. 5.128 and defining $f(y_1 | y_2) = f(y_1, y_2) / f_2(y_2)$, which yields a density function of a normal distribution with mean $\mu_1 + \rho(\sigma_1 / \sigma_2)(y_2 - \mu_2)$ and variance $\sigma_1^2(1 - \rho^2)$.

5.130 a. $\text{Cov}(U_1, U_2) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \text{Cov}(Y_i, Y_j) = \sum_{i=1}^n a_i b_j V(Y_i) = \sigma^2 \sum_{i=1}^n a_i b_j$, since the Y_i 's are

independent. If $\text{Cov}(U_1, U_2) = 0$, it must be true that $\sum_{i=1}^n a_i b_j = 0$ since $\sigma^2 > 0$. But, it is

trivial to see if $\sum_{i=1}^n a_i b_j = 0$, $\text{Cov}(U_1, U_2) = 0$. So, U_1 and U_2 are orthogonal.

- b.** Given in the problem, (U_1, U_2) has a bivariate normal distribution. Note that $E(U_1) = \mu \sum_{i=1}^n a_i$, $E(U_2) = \mu \sum_{i=1}^n b_i$, $V(U_1) = \sigma^2 \sum_{i=1}^n a_i^2$, and $V(U_2) = \sigma^2 \sum_{i=1}^n b_i^2$. If they are orthogonal, $\text{Cov}(U_1, U_2) = 0$ and then $\rho_{U_1, U_2} = 0$. So, they are also independent.
- 5.131 a.** The joint distribution of Y_1 and Y_2 is simply the product of the marginals $f_1(y_1)$ and $f_2(y_2)$ since they are independent. It is trivial to show that this product of density has the form of the bivariate normal density with $\rho = 0$.
- b.** Following the result of Ex. 5.130, let $a_1 = a_2 = b_1 = 1$ and $b_2 = -1$. Thus, $\sum_{i=1}^n a_i b_i = 0$ so U_1 and U_2 are independent.
- 5.132** Following Ex. 5.130 and 5.131, U_1 is normal with mean $\mu_1 + \mu_2$ and variance $2\sigma^2$ and U_2 is normal with mean $\mu_1 - \mu_2$ and variance $2\sigma^2$.
- 5.133** From Ex. 5.27, $f(y_1 | y_2) = 1/y_2$, $0 \leq y_1 \leq y_2$ and $f_2(y_2) = 6y_2(1 - y_2)$, $0 \leq y_2 \leq 1$.
- a.** To find $E(Y_1 | Y_2 = y_2)$, note that the conditional distribution of Y_1 given Y_2 is uniform on the interval $(0, y_2)$. So, $E(Y_1 | Y_2 = y_2) = \frac{y_2}{2}$.
- b.** To find $E(E(Y_1 | Y_2))$, note that the marginal distribution is beta with $\alpha = 2$ and $\beta = 2$. So, from part a, $E(E(Y_1 | Y_2)) = E(Y_2/2) = 1/4$. This is the same answer as in Ex. 5.77.
- 5.134** The z-score is $(6 - 1.25)/\sqrt{1.5625} = 3.8$, so the value 6 is 3.8 standard deviations above the mean. This is not likely.
- 5.135** Refer to Ex. 5.41:
- a.** Since Y is binomial, $E(Y|p) = 3p$. Now p has a uniform distribution on $(0, 1)$, thus $E(Y) = E[E(Y|p)] = E(3p) = 3(1/2) = 3/2$.
- b.** Following part a, $V(Y|p) = 3p(1 - p)$. Therefore,

$$V(p) = E[3p(1 - p)] + V(3p) = 3E(p - p^2) + 9V(p)$$

$$= 3E(p) - 3[V(p) + (E(p))^2] + 9V(p) = 1.25$$
- 5.136 a.** For a given value of λ , Y has a Poisson distribution. Thus, $E(Y | \lambda) = \lambda$. Since the marginal distribution of λ is exponential with mean 1, $E(Y) = E[E(Y | \lambda)] = E(\lambda) = 1$.
- b.** From part a, $E(Y | \lambda) = \lambda$ and so $V(Y | \lambda) = \lambda$. So, $V(Y) = E[V(Y | \lambda)] + E[E(Y | \lambda)]^2 = 2$.
- c.** The value 9 is $(9 - 1)/\sqrt{2} = 5.657$ standard deviations above the mean (unlikely score).
- 5.137** Refer to Ex. 5.38: $E(Y_2 | Y_1 = y_1) = y_1/2$. For $y_1 = 3/4$, $E(Y_2 | Y_1 = 3/4) = 3/8$.
- 5.138** If $Y = \#$ of bacteria per cubic centimeter,
- a.** $E(Y) = E(Y) = E[E(Y | \lambda)] = E(\lambda) = \alpha\beta$.

b. $V(Y) = E[V(Y | \lambda)] + V[E(Y | \lambda)] = \alpha\beta + \alpha\beta^2 = \alpha\beta(1+\beta)$. Thus, $\sigma = \sqrt{\alpha\beta(1+\beta)}$.

5.139 a. $E(T | N = n) = E\left(\sum_{i=1}^n Y_i\right) = \sum_{i=1}^n E(Y_i) = n\alpha\beta$.

b. $E(T) = E[E(T | N)] = E(N\alpha\beta) = \lambda\alpha\beta$. Note that this is $E(N)E(Y)$.

5.140 Note that $V(Y_1) = E[V(Y_1 | Y_2)] + V[E(Y_1 | Y_2)]$, so $E[V(Y_1 | Y_2)] = V(Y_1) - V[E(Y_1 | Y_2)]$. Thus, $E[V(Y_1 | Y_2)] \leq V(Y_1)$.

5.141 $E(Y_2) = E(E(Y_2 | Y_1)) = E(Y_1/2) = \frac{\lambda}{2}$

$$V(Y_2) = E[V(Y_2 | Y_1)] + V[E(Y_2 | Y_1)] = E[Y_1^2/12] + V[Y_1/2] = (2\lambda^2)/12 + (\lambda^2)/2 = \frac{2\lambda^2}{3}.$$

5.142 a. $E(Y) = E[E(Y|p)] = E(np) = nE(p) = \frac{n\alpha}{\alpha + \beta}$.

b. $V(Y) = E[V(Y | p)] + V[E(Y | p)] = E[np(1 - p)] + V(np) = nE(p - p^2) + n^2V(p)$. Now:

$$nE(p - p^2) = \frac{n\alpha}{\alpha + \beta} - \frac{n\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}$$

$$n^2V(p) = \frac{n^2\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

$$\text{So, } V(Y) = \frac{n\alpha}{\alpha + \beta} - \frac{n\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)} + \frac{n^2\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{n\alpha\beta(\alpha + \beta + n)}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

5.143 Consider the random variable y_1Y_2 for the fixed value of Y_1 . It is clear that y_1Y_2 has a normal distribution with mean 0 and variance y_1^2 and the mgf for this random variable is

$$m(t) = E(e^{ty_1Y_2}) = e^{t^2y_1^2/2}.$$

$$\text{Thus, } m_U(t) = E(e^{tU}) = E(e^{tY_1Y_2}) = E[E(e^{tY_1Y_2} | Y_1)] = E(e^{t^2Y_1^2/2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{(-y_1^2/2)(1-t^2)} dy_1.$$

Note that this integral is essentially that of a normal density with mean 0 and variance $\frac{1}{1-t^2}$, so the necessary constant that makes the integral equal to 1 is the reciprocal of the standard deviation. Thus, $m_U(t) = (1-t^2)^{-1/2}$. Direct calculations give $m'_U(0) = 0$ and $m''_U(0) = 1$. To compare, note that $E(U) = E(Y_1Y_2) = E(Y_1)E(Y_2) = 0$ and $V(U) = E(U^2) = E(Y_1^2Y_2^2) = E(Y_1^2)E(Y_2^2) = (1)(1) = 1$.

$$\begin{aligned} 5.144 \quad E[g(Y_1)h(Y_2)] &= \sum_{y_1} \sum_{y_2} g(y_1)h(y_2)p(y_1, y_2) = \sum_{y_1} \sum_{y_2} g(y_1)h(y_2)p_1(y_1)p_2(y_2) = \\ &= \sum_{y_1} g(y_1)p_1(y_1) \sum_{y_2} h(y_2)p_2(y_2) = E[g(Y_1)] \times E[h(Y_2)]. \end{aligned}$$

- 5.145 The probability of interest is $P(Y_1 + Y_2 < 30)$, where Y_1 is uniform on the interval $(0, 15)$ and Y_2 is uniform on the interval $(20, 30)$. Thus, we have

$$P(Y_1 + Y_2 < 30) = \int_{20}^{30} \int_0^{30-y_2} \left(\frac{1}{15}\right) \left(\frac{1}{10}\right) dy_1 dy_2 = 1/3.$$

- 5.146 Let (Y_1, Y_2) represent the coordinates of the landing point of the bomb. Since the radius is one mile, we have that $0 \leq y_1^2 + y_2^2 \leq 1$. Now,

$P(\text{target is destroyed}) = P(\text{bomb destroys everything within } 1/2 \text{ of landing point})$
This is given by $P(Y_1^2 + Y_2^2 \leq (\frac{1}{2})^2)$. Since (Y_1, Y_2) are uniformly distributed over the unit circle, the probability in question is simply the area of a circle with radius $1/2$ divided by the area of the unit circle, or simply $1/4$.

- 5.147 Let Y_1 = arrival time for 1st friend, $0 \leq y_1 \leq 1$, Y_2 = arrival time for 2nd friend, $0 \leq y_2 \leq 1$. Thus $f(y_1, y_2) = 1$. If friend 2 arrives $1/6$ hour (10 minutes) before or after friend 1, they will meet. We can represent this event as $|Y_1 - Y_2| < 1/3$. To find the probability of this event, we must find:

$$P(|Y_1 - Y_2| < 1/3) = \int_0^{1/6} \int_0^{y_1+1/6} 1 dy_2 dy_1 + \int_{1/6}^{5/6} \int_{y_1-1/6}^{y_1+1/6} 1 dy_2 dy_1 + \int_{5/6}^1 \int_{y_1-1/6}^1 1 dy_2 dy_1 = 11/36.$$

$$5.148 \quad \text{a. } p(y_1, y_2) = \frac{\binom{4}{y_1} \binom{3}{y_2} \binom{2}{3-y_1-y_2}}{\binom{9}{3}}, y_1 = 0, 1, 2, 3, y_2 = 0, 1, 2, 3, y_1 + y_2 \leq 3.$$

b. Y_1 is hypergeometric w/ $r = 4, N = 9, n = 3$; Y_2 is hypergeometric w/ $r = 3, N = 9, n = 3$

$$\text{c. } P(Y_1 = 1 \mid Y_2 \geq 1) = [p(1, 1) + p(1, 2)] / [1 - p_2(0)] = 9/16$$

$$5.149 \quad \text{a. } f_1(y_1) = \int_0^{y_1} 3y_1 dy_2 = 3y_1^2, 0 \leq y_1 \leq 1, \quad f_1(y_1) = \int_{y_2}^1 3y_1 dy_1 = \frac{3}{2}(1 - y_2^2), 0 \leq y_2 \leq 1.$$

$$\text{b. } P(Y_1 \leq 3/4 \mid Y_2 \leq 1/2) = 23/44.$$

$$\text{c. } f(y_1 \mid y_2) = 2y_1 / (1 - y_2^2), y_2 \leq y_1 \leq 1.$$

$$\text{d. } P(Y_1 \leq 3/4 \mid Y_2 = 1/2) = 5/12.$$

- 5.150 a. Note that $f(y_2 \mid y_1) = f(y_1, y_2) / f(y_1) = 1/y_1, 0 \leq y_2 \leq y_1$. This is the same conditional density as seen in Ex. 5.38 and Ex. 5.137. So, $E(Y_2 \mid Y_1 = y_1) = y_1/2$.

$$\mathbf{b.} \ E(Y_2) = E[E(Y_2 | Y_1)] = E(Y_1/2) = \int_0^1 \frac{y_1}{2} 3y_1^2 dy_1 = 3/8.$$

$$\mathbf{c.} \ E(Y_2) = \int_0^1 y_2 \frac{3}{2}(1 - y_2^2) dy_2 = 3/8.$$

5.151 a. The joint density is the product of the marginals: $f(y_1, y_2) = \frac{1}{\beta^2} e^{-(y_1+y_2)/\beta}$, $y_1 \geq 0, y_2 \geq 0$

$$\mathbf{b.} \ P(Y_1 + Y_2 \leq a) = \int_0^a \int_0^{a-y_2} \frac{1}{\beta^2} e^{-(y_1+y_2)/\beta} dy_1 dy_2 = 1 - [1 + a/\beta] e^{-a/\beta}.$$

5.152 The joint density of (Y_1, Y_2) is $f(y_1, y_2) = 18(y_1 - y_1^2)y_2^2$, $0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1$. Thus,

$$P(Y_1 Y_2 \leq .5) = P(Y_1 \leq .5/Y_2) = 1 - P(Y_1 > .5/Y_2) = 1 - \int_{.5}^1 \int_{.5/y_2}^1 18(y_1 - y_1^2)y_2^2 dy_1 dy_2. \text{ Using straightforward integration, this is equal to } (5 - 3\ln 2)/4 = .73014.$$

5.153 This is similar to Ex. 5.139:

a. Let $N = \#$ of eggs laid by the insect and $Y = \#$ of eggs that hatch. Given $N = n$, Y has a binomial distribution with n trials and success probability p . Thus, $E(Y | N = n) = np$. Since N follows as Poisson with parameter λ , $E(Y) = E[E(Y | N)] = E(Np) = \lambda p$.

$$\mathbf{b.} \ V(Y) = E[V(Y | N)] + V[E(Y | N)] = E[Np(1 - p)] + V[Np] = \lambda p.$$

5.154 The conditional distribution of Y given p is binomial with parameter p , and note that the marginal distribution of p is beta with $\alpha = 3$ and $\beta = 2$.

a. Note that $f(y) = \int_0^1 f(y, p) dp = \int_0^1 f(y | p) f(p) dp = 12 \binom{n}{y} \int_0^1 p^{y+2} (1-p)^{n-y+1} dp$. This integral can be evaluated by relating it to a beta density w/ $\alpha = y + 3$, $\beta = n + y + 2$. Thus,

$$f(y) = 12 \binom{n}{y} \frac{\Gamma(n - y + 2) \Gamma(y + 3)}{\Gamma(n + 5)}, y = 0, 1, 2, \dots, n.$$

b. For $n = 2$, $E(Y | p) = 2p$. Thus, $E(Y) = E[E(Y | p)] = E(2p) = 2E(p) = 2(3/5) = 6/5$.

5.155 a. It is easy to show that

$$\begin{aligned} \text{Cov}(W_1, W_2) &= \text{Cov}(Y_1 + Y_2, Y_1 + Y_3) \\ &= \text{Cov}(Y_1, Y_1) + \text{Cov}(Y_1, Y_3) + \text{Cov}(Y_2, Y_1) + \text{Cov}(Y_2, Y_3) \\ &= \text{Cov}(Y_1, Y_1) = V(Y_1) = 2v_1. \end{aligned}$$

b. It follows from part a above (i.e. the variance is positive).

5.156 a. Since $E(Z) = E(W) = 0$, $\text{Cov}(Z, W) = E(ZW) = E(Z^2 Y^{-1/2}) = E(Z^2)E(Y^{-1/2}) = E(Y^{-1/2})$.

This expectation can be found by using the result Ex. 4.112 with $a = -1/2$. So,

$$\text{Cov}(Z, W) = E(Y^{-1/2}) = \frac{\Gamma(\frac{v}{2} - \frac{1}{2})}{\sqrt{2}\Gamma(\frac{v}{2})}, \text{ provided } v > 1.$$

b. Similar to part a, $\text{Cov}(Y, W) = E(YW) = E(\sqrt{Y} W) = E(\sqrt{Y})E(W) = 0$.

c. This is clear from parts (a) and (b) above.

5.157 $p(y) = \int_0^\infty p(y|\lambda) f(\lambda) d\lambda = \int_0^\infty \frac{\lambda^{y+\alpha-1} e^{-\lambda[(\beta+1)/\beta]}}{\Gamma(y+1)\Gamma(\alpha)\beta^\alpha} d\lambda = \frac{\Gamma(y+\alpha)\left(\frac{\beta}{\beta+1}\right)^{y+\alpha}}{\Gamma(y+1)\Gamma(\alpha)\beta^\alpha}, y = 0, 1, 2, \dots$. Since

it was assumed that α was an integer, this can be written as

$$p(y) = \binom{y+\alpha-1}{y} \left(\frac{\beta}{\beta+1}\right)^y \left(\frac{1}{\beta+1}\right)^\alpha, y = 0, 1, 2, \dots$$

5.158 Note that for each X_i , $E(X_i) = p$ and $V(X_i) = pq$. Then, $E(Y) = \Sigma E(X_i) = np$ and $V(Y) = npq$. The second result follows from the fact that the X_i are independent so therefore all covariance expressions are 0.

5.159 For each W_i , $E(W_i) = 1/p$ and $V(W_i) = q/p^2$. Then, $E(Y) = \Sigma E(X_i) = r/p$ and $V(Y) = rq/p^2$. The second result follows from the fact that the W_i are independent so therefore all covariance expressions are 0.

5.160 The marginal probabilities can be written directly:

$$\begin{array}{ll} P(X_1 = 1) = P(\text{select ball 1 or 2}) = .5 & P(X_1 = 0) = .5 \\ P(X_2 = 1) = P(\text{select ball 1 or 3}) = .5 & P(X_2 = 0) = .5 \\ P(X_3 = 1) = P(\text{select ball 1 or 4}) = .5 & P(X_3 = 0) = .5 \end{array}$$

Now, for $i \neq j$, X_i and X_j are clearly pairwise independent since, for example,

$$\begin{array}{l} P(X_1 = 1, X_2 = 1) = P(\text{select ball 1}) = .25 = P(X_1 = 1)P(X_2 = 1) \\ P(X_1 = 0, X_2 = 1) = P(\text{select ball 3}) = .25 = P(X_1 = 0)P(X_2 = 1) \end{array}$$

However, X_1 , X_2 , and X_3 are not mutually independent since

$$P(X_1 = 1, X_2 = 1, X_3 = 1) = P(\text{select ball 1}) = .25 \neq P(X_1 = 1)P(X_2 = 1)P(X_3 = 1).$$

$$5.161 \quad E(\bar{Y} - \bar{X}) = E(\bar{Y}) - E(\bar{X}) = \frac{1}{n} \sum E(Y_i) - \frac{1}{m} \sum E(X_i) = \mu_1 - \mu_2$$

$$V(\bar{Y} - \bar{X}) = V(\bar{Y}) + V(\bar{X}) = \frac{1}{n^2} \sum V(Y_i) + \frac{1}{m^2} \sum V(X_i) = \sigma_1^2 / n + \sigma_2^2 / m$$

5.162 Using the result from Ex. 5.65, choose two different values for α with $-1 \leq \alpha \leq 1$.

5.163 a. The distribution functions with the exponential distribution are:

$$F_1(y_1) = 1 - e^{-y_1}, y_1 \geq 0; \quad F_2(y_2) = 1 - e^{-y_2}, y_2 \geq 0.$$

Then, the joint distribution function is

$$F(y_1, y_2) = [1 - e^{-y_1}][1 - e^{-y_2}][1 - \alpha(e^{-y_1})(e^{-y_2})].$$

Finally, show that $\frac{\partial^2}{\partial y_1 \partial y_2} F(y_1, y_2)$ gives the joint density function seen in Ex. 5.162.

b. The distribution functions with the uniform distribution on $(0, 1)$ are:

$$F_1(y_1) = y_1, 0 \leq y_1 \leq 1; \quad F_2(y_2) = y_2, 0 \leq y_2 \leq 1.$$

Then, the joint distribution function is

$$F(y_1, y_2) = y_1 y_2 [1 - \alpha(1 - y_1)(1 - y_2)].$$

$$c. \frac{\partial^2}{\partial y_1 \partial y_2} F(y_1, y_2) = f(y_1, y_2) = 1 - \alpha[(1 - 2y_1)(1 - 2y_2)], 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1.$$

d. Choose two different values for α with $-1 \leq \alpha \leq 1$.

5.164 a. If $t_1 = t_2 = t_3 = t$, then $m(t, t, t) = E(e^{t(X_1 + X_2 + X_3)})$. This, by definition, is the mgf for the random variable $X_1 + X_2 + X_3$.

b. Similarly with $t_1 = t_2 = t$ and $t_3 = 0$, $m(t, t, 0) = E(e^{t(X_1 + X_2)})$.

c. We prove the continuous case here (the discrete case is similar). Let (X_1, X_2, X_3) be continuous random variables with joint density function $f(x_1, x_2, x_3)$. Then,

$$m(t_1, t_2, t_3) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x_1} e^{t_2 x_2} e^{t_3 x_3} f(x_1, x_2, x_3) dx_1 dx_2 dx_3.$$

Then,

$$\frac{\partial^{k_1 + k_2 + k_3}}{\partial t_1^{k_1} \partial t_2^{k_2} \partial t_3^{k_3}} m(t_1, t_2, t_3) \Big|_{t_1 = t_2 = t_3 = 0} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1^{k_1} x_2^{k_2} x_3^{k_3} f(x_1, x_2, x_3) dx_1 dx_2 dx_3.$$

This is easily recognized as $E(X_1^{k_1} X_2^{k_2} X_3^{k_3})$.

$$5.165 \quad a. \quad m(t_1, t_2, t_3) = \sum_{x_1} \sum_{x_2} \sum_{x_3} \frac{n!}{x_1! x_2! x_3!} e^{t_1 x_1 + t_2 x_2 + t_3 x_3} p_1^{x_1} p_2^{x_2} p_3^{x_3} \\ = \sum_{x_1} \sum_{x_2} \sum_{x_3} \frac{n!}{x_1! x_2! x_3!} (p_1 e^{t_1})^{x_1} (p_2 e^{t_2})^{x_2} (p_3 e^{t_3})^{x_3} = (p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3})^n. \quad \text{The}$$

final form follows from the multinomial theorem.

b. The mgf for X_1 can be found by evaluating $m(t, 0, 0)$. Note that $q = p_2 + p_3 = 1 - p_1$.

c. Since $\text{Cov}(X_1, X_2) = E(X_1 X_2) - E(X_1)E(X_2)$ and $E(X_1) = np_1$ and $E(X_2) = np_2$ since X_1 and X_2 have marginal binomial distributions. To find $E(X_1 X_2)$, note that

$$\frac{\partial^2}{\partial t_1 \partial t_2} m(t_1, t_2, 0) \Big|_{t_1=t_2=0} = n(n-1)p_1 p_2.$$

Thus, $\text{Cov}(X_1, X_2) = n(n-1)p_1 p_2 - (np_1)(np_2) = -np_1 p_2$.

5.166 The joint probability mass function of (Y_1, Y_2, Y_3) is given by

$$p(y_1, y_2, y_3) = \frac{\binom{N_1}{y_1} \binom{N_2}{y_2} \binom{N_3}{y_3}}{\binom{N}{n}} = \frac{\binom{Np_1}{y_1} \binom{Np_2}{y_2} \binom{Np_3}{y_3}}{\binom{N}{n}},$$

where $y_1 + y_2 + y_3 = n$. The marginal distribution of Y_1 is hypergeometric with $r = Np_1$, so $E(Y_1) = np_1$, $V(Y_1) = np_1(1-p_1)\left(\frac{N-n}{N-1}\right)$. Similarly, $E(Y_2) = np_2$, $V(Y_2) = np_2(1-p_2)\left(\frac{N-n}{N-1}\right)$. It can be shown that (using mathematical expectation and straightforward albeit messy algebra) $E(Y_1 Y_2) = n(n-1)p_1 p_2 \frac{N}{N-1}$. Using this, it is seen that

$$\text{Cov}(Y_1, Y_2) = n(n-1)p_1 p_2 \frac{N}{N-1} - (np_1)(np_2) = -np_1 p_2 \left(\frac{N-n}{N-1}\right).$$

(Note the similar expressions in Ex. 5.165.) Finally, it can be found that

$$\rho = -\sqrt{\frac{p_1 p_2}{(1-p_1)(1-p_2)}}.$$

5.167 a. For this exercise, the quadratic form of interest is

$$At^2 + Bt + C = E(Y_1^2)t^2 + [-2E(Y_1 Y_2)]t + [E(Y_2^2)]^2.$$

Since $E[(tY_1 - Y_2)^2] \geq 0$ (it is the integral of a non-negative quantity), so we must have that $At^2 + Bt + C \geq 0$. In order to satisfy this inequality, the two roots of this quadratic must either be imaginary or equal. In terms of the discriminant, we have that

$$B^2 - 4AC \leq 0, \text{ or}$$

$$[-2E(Y_1 Y_2)]^2 - 4E(Y_1^2)E(Y_2^2) \leq 0.$$

Thus, $[E(Y_1 Y_2)]^2 \leq E(Y_1^2)E(Y_2^2)$.

b. Let $\mu_1 = E(Y_1)$, $\mu_2 = E(Y_2)$, and define $Z_1 = Y_1 - \mu_1$, $Z_2 = Y_2 - \mu_2$. Then,

$$\rho^2 = \frac{[E(Y_1 - \mu_1)(Y_2 - \mu_2)]^2}{[E(Y_1 - \mu_1)^2]E[(Y_2 - \mu_2)^2]} = \frac{[E(Z_1 Z_2)]^2}{E(Z_1^2)E(Z_2^2)} \leq 1$$

by the result in part **a**.